

Golf Launch Monitor Technology Review

**Measurement Principles, Patent Landscape, and Club/Ball
Parameter Estimation**

A Comprehensive Technical Reference

August 19, 2026

Abstract

This review surveys the measurement technology behind commercial golf launch monitors — Doppler-radar systems (TrackMan, FlightScope, Garmin R10, Full Swing KIT), photometric camera systems (Foresight GCQuad/GC3, Uneekor, SkyTrak, ProTee VX, Garmin R50), and hybrid radar+camera architectures that combine these sensing families — with emphasis on *how each system measures or infers club-delivery parameters* (club speed, club path, attack angle, face angle, dynamic loft, impact location) *and ball-launch parameters* (ball speed, launch angles, spin rate, spin axis). Primary sources include selected patent disclosures (TrackMan/Tuxen radar spin and spin-axis methods, FlightScope/EDH radar-spin methods, Wintriss/Foresight photometric methods, and Acushnet optical launch-monitor designs), regulatory (FCC) filings, manufacturer technical documentation, and the peer-reviewed validation literature. The review develops the underlying physics — the D-plane impact model, oblique-impact spin generation, gear effect, and golf-ball aerodynamics — as the common mathematical framework for interpreting launch-monitor outputs, and closes with a technical map of patent disclosures and design guidance for implementers building or evaluating a system. The central organizing principle is the provenance of each reported quantity: the observations, calibration and model assumptions that support it. Combining radar and optical observations can constrain different parts of the inference problem, but does not automatically make every output directly measured or independently validated. Early optical disclosures illustrate surface-image rotation and club-marker geometry; their age does not establish implementation accuracy or design-specific rights clearance.

This document is written to be vendor- and project-neutral. Where a specific product is named it is as evidence — a published definition, a measured tolerance, a patent claim — not as an endorsement or a design target.

Contents

1	Introduction	5
1.1	Purpose and Scope	5
1.2	The Two Sensing Families	5
1.3	Reading Guide	6
2	Parameter Definitions and Conventions	7
2.1	Club-Delivery Parameters	7
2.1.1	Reference Point and Rigid-Body Velocity	8
2.1.2	Spin Loft and Contact Kinematics	9
2.2	Ball-Launch Parameters	9
2.2.1	Spin Components Need a Frame	9
2.3	The Directness Hierarchy	10
3	Impact Physics: From Club Delivery to Ball Launch	12
3.1	The D-Plane	12
3.1.1	Face/Path Weighting of Launch Direction	13
3.1.2	Spin Loft and Approximate Launch Maps	14
3.1.3	Spin-Axis Tilt	14
3.2	Ball Speed and Smash Factor	15
3.3	Spin Generation	15
3.4	Gear Effect and Impact Location	16
3.4.1	Why a Coupled Contact Solve Matters	16
3.5	Vertical Plane and the Low-Point Geometry	17
3.6	Connecting Delivery, Impact and Measurement	17
4	Doppler Radar Systems	18
4.1	Continuous-Wave Doppler Fundamentals	18
4.2	From Velocity Sensor to 3D Tracker: Phase Interferometry	19
4.3	CW + FMCW Hybrids	19
4.4	Ball Measurement	20
4.4.1	Speed and Launch Angles	20
4.4.2	Spin Rate: Harmonic Signatures and Their Ambiguities	20
4.4.3	Spin Axis: Conditional Trajectory Inversion	21
4.4.4	A Dimensional and Algebraic Check on a Delay Disclosure	22
4.4.5	Flight Observation, Extrapolation, and Ball Modes	22
4.5	Club Measurement	22
4.5.1	What the Radar Tracks on the Head	22
4.5.2	Face Angle and Dynamic Loft	23
4.5.3	Known Limitations	23
4.6	Optically Enhanced Radar: OERT and Fusion Tracking	24

5	Photometric (Camera) Systems	25
5.1	Imaging Architectures	25
5.2	Ball Measurement	25
5.2.1	Photogrammetric Position and Velocity	25
5.2.2	Why an Ellipse Center Is Not the Ball Center	26
5.2.3	Spin: Surface-Pattern Registration	26
5.3	Club Measurement	27
5.3.1	Fiducial Geometry and Reference Points	27
5.3.2	Overhead Tracking and Calibration	27
5.4	The Indoor Argument and Downrange Limits	28
5.5	DIY and Open-Source Photometric Systems	28
6	Commercial System Survey	29
6.1	Radar and Hybrid Systems	30
6.1.1	TrackMan 4 and iO	30
6.1.2	FlightScope X3, X3C and Mevo+	30
6.1.3	Garmin Approach R10	31
6.1.4	Full Swing KIT and Rapsodo MLM2PRO	31
6.2	Camera Systems	31
6.2.1	Foresight Families	31
6.2.2	Uneekor and ProTee	32
6.2.3	Garmin R50 and SkyTrak ST+	32
6.3	Open-Source Systems and Comparison Practice	32
7	Patent Landscape	33
7.1	TrackMan A/S (Interactive Sports Games A/S) — Fredrik Tuxen	33
7.2	Foresight Sports, WAWGD and Wintriss Disclosures	34
7.3	FlightScope / EDH — Henri Johnson	35
7.4	Acushnet (Titleist) — Optical Measurement Architectures	35
7.5	United States Golf Association — Aerodynamic Experiments	36
7.6	Korean Ecosystem: Creatz/Uneekor and Golfzon	36
7.7	Timing, Association and Environmental Inference	37
7.8	Technical Research Map and Rights Boundaries	37
8	Ball Flight Models and Trajectory Estimation	39
8.1	Equations of Motion and Coefficient Conventions	39
8.2	What Aerodynamic Data Establish	40
8.3	Numerical Integration and the Landing Event	40
8.4	Trajectory Estimation and Filtering	41
8.5	Indoor Extrapolation and Coupled Uncertainty	42
9	Accuracy: The Independent Evidence	43
9.1	Comparison Against an Optical Reference	43
9.2	Reliability Across Repeated Swings	44
9.3	Agreement Between Commercial Monitors	44
9.4	Robot Tests, Specifications and Causal Interpretation	45
9.5	A Validation Protocol That Supports the Intended Claim	45

10 Design Guidance for Implementers	47
10.1 Ball Data: Match Each Claim to Its Observations	47
10.2 Club Data: Define the Point, Frame and Time	48
10.3 Capability Tiers and Acceptance Conditions	49
10.4 Reporting Honesty as a Design Feature	50
11 Governed Analytics and Validation Program	51
11.1 Qualified Corpus Snapshot	51
11.2 Release A: Qualified Analytics Platform	52
11.3 Qualified Model-Comparison Results	52
11.4 Release B: Paired-Device Scientific Validation	54
11.4.1 Pairing and Reference Uncertainty	54
11.4.2 Agreement, Repeatability, and Repeated Shots	55
11.4.3 Sample Planning and the Completion Gate	55
11.5 Reproducibility and Claim Discipline	56
A Annotated Reference Library	57
A.1 Patents (Google Patents / USPTO)	57
A.2 Regulatory Records and Teardown Reports	59
A.3 Manufacturer Documentation and Product Reporting	59
A.3.1 The TrackMan Newsletter Archive	60
A.4 Research Papers, Standards, and Technical Notes	62
A.5 Engineering References	63
A.6 DIY / Open-Source	63
A.7 Hardware Datasheets, Protocols, and Standards	63
A.8 Comparative Testing and Market Analysis	64
B Detailed Implementation Guidance	66
B.1 Radar Signal Chain (OPS243-A)	66
B.1.1 Doppler Scaling and Resolution Budget	66
B.1.2 Candidate Spin-Rate Estimation	66
B.1.3 Club-Speed Extraction	67
B.2 Angle Measurement (K-LD7) and the EKF	68
B.2.1 Interferometric Angles and Hardware Limits	68
B.2.2 Measurement Functions and Observability	68
B.2.3 Filtering, Smoothing, and the Impact Event	69
B.2.4 Alignment Calibration Procedure	69
B.3 Derived Club Data, With Priors	70
B.4 Optical Module (Phase 2) Design Parameters	70
B.4.1 Shutter, Illumination, and Motion Blur	70
B.4.2 Multiple Exposures and Trigger Timing	71
B.4.3 Position, Spin, and Club Pose	71
B.5 Flight Model and Validation Protocol	71
C Sensor Hardware and Integration Reference	73
C.1 OmniPreSense OPS243-A (24 GHz CW Doppler)	73
C.1.1 Hardware and Configuration Boundaries	73
C.1.2 Sampling, Grid Spacing and Identifiability	73

C.1.3	Rolling Buffer and Trigger Timing	74
C.1.4	Interpreting the Vendor Golf Example	74
C.2	RFbeam K-LD7 (24 GHz FSK, Dual-RX Angle)	75
C.3	TI IWR6843 (60–64 GHz FMCW MIMO)	75
C.4	Raspberry Pi Global Shutter Camera (Optical Module)	76
C.5	Simulator Integration: GSPRO Open Connect	76
C.6	Equipment Rules and Model Parameters	77
C.7	Detection Theory: CFAR Selection	77
D	Patent Portfolio Compendium	79
D.1	TrackMan A/S (Incl. Interactive Sports Games A/S)	79
D.2	Topgolf Sweden AB (Toptracer / Protracer)	82
D.3	FlightScope / EDH (Henri Johnson)	83
D.4	Full Swing Golf	83
D.5	Garmin	84
D.6	Rapsodo Pte. Ltd.	84
D.7	Camera Vendors: Foresight/Wintriss, Creatz/Uneekor, Golfzon	85
D.8	Acushnet (Titleist) — Selected Optical Disclosures	86
D.9	Others and Foundational Radar Art	87
E	Radar Clubhead Kinematics: Motion and Observability	88
E.1	Context: Definitions Are Not an Estimator Specification	88
E.2	Rigid Motion With Declared Frames and Reference Points	88
E.2.1	Body and Spatial Twists for Pose Integration	89
E.3	The Measurement Model and Its Exact Ambiguity	89
E.3.1	What Least Squares and Covariance Actually Provide	90
E.4	What Representative Hardware Adds	91
E.4.1	OPS243-A: Radial-Speed Information	91
E.4.2	K-LD7: Geometry and Operating Envelope	91
E.4.3	IWR6843: A Coupled Waveform Budget	91
E.5	An Estimation Recipe With Explicit Acceptance Conditions	92
E.6	Validation That Can Falsify the Proposed Estimator	94

Chapter 1

Introduction

1.1 Purpose and Scope

A golf launch monitor answers two questions for every shot: *what did the club do through impact* (delivery), and *what did the ball do after impact* (launch and flight). Commercial systems answer these questions with strikingly different sensor architectures — continuous-wave Doppler radar, multi-camera photogrammetry, and, increasingly, fusions of the two — yet they all implement the same underlying physics: an oblique-impact collision model connecting club delivery to ball launch, and an aerodynamic model connecting ball launch to flight.

This review is a technical reference on how launch monitors work. It is written for engineers building or evaluating such a system, for researchers using one as an instrument, and for anyone who needs to know what a reported number actually represents. Its goals are:

1. catalogue *how each major commercial system works*, at the level of radar physics, imaging geometry, and signal processing;
2. establish, parameter by parameter, *what is measured directly, what is inferred through a model, and what is essentially estimated* — especially for club-delivery data such as face angle and club path;
3. survey the *patent landscape* as a source of disclosed sensor architectures, signal-processing methods and claim limitations, while distinguishing those disclosures from verified commercial implementations and design-specific rights determinations;
4. collect the *physics and algorithms* (D-plane, impact mechanics, gear effect, aerodynamic coefficients, spectral spin estimation, stereo triangulation, Kalman filtering) that any implementation needs; and
5. translate all of the above into *concrete design guidance*, organised by sensing architecture rather than by product.

Key point

A note on neutrality. Products are named throughout, and often criticised. That is because published vendor definitions, patent claims and measured tolerances are the primary evidence available in this field — the peer-reviewed literature is thin, as Chapter 9 documents. Naming a system is a citation, not a recommendation, and no architecture in this review is presented as the one to build.

1.2 The Two Sensing Families

Doppler radar systems (TrackMan, FlightScope, Garmin Approach R10, Full Swing KIT) illuminate the hitting area and downrange volume with a microwave carrier and extract target radial

velocity from the Doppler shift. Multiple receive antennas turn a velocity sensor into a 3D tracker via phase interferometry (Chapter 4). Radar excels outdoors: it tracks the entire flight, so carry and curvature are *observed*, not modeled. Its weaknesses are at the club face — radar cannot see face orientation directly — and indoors, where a screen truncates the observable flight.

Photometric (camera) systems (Foresight GC-series, Uneekor, SkyTrak, ProTee VX, Garmin R50) capture a burst of high-speed, IR-strobed stereo images over the first ~ 30 cm of ball flight and reconstruct position and orientation photogrammetrically (Chapter 5). They measure launch conditions — including 3D spin from dimple-pattern rotation — essentially perfectly for simulator purposes, then *model* the flight. Their weakness is the mirror image of radar’s: everything downrange of the capture volume is simulated, and club data requires either fiducial stickers on the face or an overhead viewing geometry.

The market’s convergent evolution is the single most instructive fact for a new design: *every* high-end vendor has concluded that neither modality suffices alone. TrackMan added cameras to its radar (OERT, Section 4.6); FlightScope added Fusion Tracking cameras; SkyTrak added radar to its camera; Rapsodo pairs radar with impact cameras; Full Swing feeds a camera into its radar ML pipeline. Chapter 6 tabulates the landscape.

1.3 Reading Guide

Chapter 2 fixes terminology and coordinate conventions (TrackMan’s definitions, the de facto industry standard). Chapter 3 develops the impact physics connecting club delivery to ball launch — the D-plane model, spin generation, and gear effect — which every monitor uses either forward (simulation) or inverse (parameter estimation). Chapters 4 and 5 treat the two sensing families in depth. Chapter 6 surveys the commercial devices. Chapter 7 compares disclosed measurement mechanisms and their evidentiary limits. Chapter 8 covers ball-flight aerodynamics and trajectory estimation. Chapter 9 reviews the independent validation literature. Chapter 10 distills the review into design guidance.

Key point

Throughout, we use a right-handed coordinate system for a right-handed golfer: x down the target line, y up, z to the golfer’s right. Angles are positive right/up. All club-delivery values are referenced to the moment of *maximum ball compression*; all ball-launch values to the moment of *separation from the face*.

Chapter 2

Parameter Definitions and Conventions

A launch-monitor number describes a chosen quantity, at a chosen point and time, through a particular estimation procedure. A disagreement can therefore reflect different definitions as well as measurement error. This chapter uses TrackMan’s published conventions where stated, without treating them as a universal standard or an independent accuracy validation [92, 87, 86]. The right-handed world frame used here and in Chapter 3 has x forward along the target line, y upward and z right. Horizontal angles are positive right of target and elevations are positive upward; right and left do not change with the golfer’s handedness. Terms such as open, closed, in-to-out and draw need an additional handedness convention.

2.1 Club-Delivery Parameters

The time assigned to an output need not be the time interval used to estimate it. TrackMan’s 2017 account describes club-motion estimates fitted from pre-impact observations and referenced to maximum compression, excluding collision-induced motion [87]. Its current parameter list explicitly defines club speed just before first contact, while path and attack angle retain the maximum-compression reference [86, 94]. Thus the latter reference does not justify interpreting every displayed club parameter as an instantaneous measurement of the colliding head. Keep the device, software version, output definition and observation window with a comparison dataset.

The delivery definitions in Table 2.1 follow the cited vendor documentation; Foresight’s lie output instead describes face-plane roll [86, 17]. A shaft inclination and a face-roll coordinate cannot be compared by name alone. Likewise, a rate about a shaft axis does not supply the other components of a clubhead’s angular velocity, and a finite-window rate need not equal its value at one instant. No exclusivity claim about which product can report these quantities follows from their definitions.

A fitted swing plane summarizes a selected trajectory segment; it does not imply that the whole swing is planar. TrackMan’s low-point convention locates the arc minimum relative to the GC at maximum compression [84]. An arc minimum ahead of that reference does not establish ball-first contact with the ground: absolute height, the sole and leading-edge geometry, ball position and the ground surface also matter. A translated copy of the same arc can have the same fore-aft low point while intersecting the ground at a different place.

Face angle and dynamic loft refer here to the normal at the contact center. Bulge and roll make that normal differ from the face-center normal on an off-center driver strike (Section 3.4). A measurement of face-center pose is not intrinsically wrong; it answers a different question unless local surface geometry and strike location are supplied. Similarly, face-to-path can help interpret a centered strike under a stated collision model, but its sign alone does not determine curvature in the presence of gear effect or differing flight conditions.

Table 2.1: Club-Delivery Quantities and Their Declared Conventions

Parameter	Quantity	Interpretation
Club speed	Norm of geometric-center (GC) velocity before first contact	Reference-point dependent
Attack angle	Elevation of GC motion at the stated reference event	Positive upward
Club path	Azimuth of GC motion at that event	Positive right of target
Face angle	Azimuth of the local face normal at the contact center	Positive right of target
Face to path	Face angle minus club path	A difference of headings
Dynamic loft	Elevation of the local contact normal	Includes delivered pose and face curvature
Spin loft	Angle between the declared motion direction and face normal	A three-dimensional angle
Swing plane / direction	Inclination / horizontal orientation of a plane fitted to a clubhead trajectory segment	Declare the fitted interval and point
Low point	Location of the swing-arc minimum relative to the impact reference	Does not alone determine turf contact
Impact height / offset	Strike coordinates relative to face center	Declare face axes and signs
Dynamic lie	TrackMan: lower-shaft inclination to horizontal at impact	Not identical to face roll
Closure rate	Foresight: heel-to-toe rotation rate about the shaft	Not the full angular-velocity norm

2.1.1 Reference Point and Rigid-Body Velocity

For two material points on an ideal rigid clubhead at the same instant,

$$\mathbf{v}_P = \mathbf{v}_O + \boldsymbol{\omega} \times \mathbf{r}, \quad \mathbf{r} = \mathbf{p}_P - \mathbf{p}_O. \quad (2.1)$$

All vectors must be expressed in the same frame. Both speed and direction can change with the reference point; geometric center, center of mass and face center are distinct definitions even when two happen to be close. TrackMan’s 2017 vendor illustration reports an approximately 3° driver path difference between center of gravity and face center, with the face-center path farther left [87]. That illustration is neither a universal correction nor evidence that every radar and optical product chooses those respective points.

For a manufactured example in SI units, choose $\mathbf{v}_O = (40, 0, 0)$, $\mathbf{r} = (0.04, 0, 0)$ and $\boldsymbol{\omega} = (0, 50, 0)$. Then $\mathbf{v}_P = (40, 0, -2)$ and its horizontal heading is $\text{atan2}(-2, 40) \simeq -2.86^\circ$. Reversing $\boldsymbol{\omega}$ reverses the heading offset; choosing $\boldsymbol{\omega} = (50, 0, 0)$ gives zero offset despite the same angular-speed norm. With horizontal speed fixed and nonzero, increasing upward velocity increases attack angle; for an already upward motion, that makes the path steeper. These examples demonstrate geometry, not typical golfer values or measured device discrepancies.

Key point

A rigid-body twist $(\boldsymbol{\omega}, \mathbf{v}_O)$ and known $\mathbf{r}$ permit conversion between reference points (Chapter E). The scalar $\|\boldsymbol{\omega}\|$ alone does not. The velocity difference obeys $\|\Delta \mathbf{v}\| \leq \|\boldsymbol{\omega}\| \|\mathbf{r}\|$, but converting that bound to a heading difference also requires the original velocity and a nonvanishing horizontal projection. Two devices can be compared only after reconciling their reference points, frames and times, with the conversion uncertainty retained.

2.1.2 Spin Loft and Contact Kinematics

Write attack angle as a , club path as p , dynamic loft as ℓ and face angle as f . For nonvertical directions, the unit motion vector is $(\cos a \cos p, \sin a, \cos a \sin p)$ and the unit face normal has the analogous form with (ℓ, f) . Their dot product gives the spin-loft angle $\sigma \in [0, \pi]$:

$$\cos \sigma = \sin a \sin \ell + \cos a \cos \ell \cos(f - p). \quad (2.2)$$

This implements the vendor's three-dimensional angular definition [91]. When $f = p$ and the elevation difference lies in $[-\pi, \pi]$, $\sigma = |\ell - a|$; using a signed difference silently assumes its sign. For $a = -5^\circ$, $\ell = 15^\circ$, changing $f - p$ from zero to 10° increases spin loft from 20° to approximately 22.32° . Direct vector evaluation still works when an azimuth is undefined at a vertical direction, provided both vectors are known.

Reported face-to-path and spin loft are legitimate declared quantities even when face orientation is evaluated at contact and velocity at GC. They are not automatically the local relative contact kinematics required by an impact law. For ideal rigid club and ball kinematics immediately before contact, the club-point velocity relative to the prospective ball contact point is

$$\mathbf{v}_{\text{rel}} = \mathbf{v}_P - [\mathbf{v}_B + \boldsymbol{\omega}_B \times (\mathbf{p}_Q - \mathbf{p}_B)], \quad (2.3)$$

where P and Q are the respective surface points and B is the ball center. During a deformable impact, local surface-deformation velocities must also be included; the displayed rigid-body expression alone does not represent them. Its normal and tangential components enter a contact model; compliance, friction and impact location then influence the outgoing ball state (Chapter 3). Knowing two headings does not identify all of those quantities.

2.2 Ball-Launch Parameters

TrackMan's carry convention uses the descending crossing of the launch elevation, not necessarily the first contact with actual terrain [85]. Its parameter documentation distinguishes this flight descriptor from initial launch quantities and describes total using a calculated resting position [86]. A launch-only indoor measurement therefore cannot by itself observe carry, landing or roll. Comparisons must state whether flight is tracked, extrapolated or normalized to chosen environmental conditions (Chapter 8). Smash factor is a speed ratio, not a fraction of kinetic energy transferred: it omits the relevant masses and rotational and deformational energy.

2.2.1 Spin Components Need a Frame

An ideal rigid ball has one instantaneous angular-velocity vector, not independent physical "backspin" and "sidespin" rotations. Use the orthonormal launch frame in Equation (3.6): outgoing direction $\hat{\mathbf{v}}_b$, transverse backspin axis $\hat{\mathbf{e}}_b$ and transverse up axis $\hat{\mathbf{e}}_u$. Express the spin vector $\mathbf{S}$ in rpm, so

Table 2.2: Launch State and Separately Derived Flight Descriptors

Parameter	Meaning and Scope
Ball speed	Norm of center velocity immediately after separation
Launch angle / direction	Elevation / azimuth of that outgoing velocity
Spin rate	Angular-speed magnitude, usually expressed in rpm
Spin axis	Direction of the nonzero spin vector; one tilt angle requires an additional convention
Smash factor	Ball-speed / club-speed ratio, using their respective events and reference points
Carry / carry side	Distance / lateral position at a declared flight endpoint, not an instantaneous launch state
Total	Distance to a resting position, requiring bounce and roll treatment
Apex / landing angle	Maximum flight height / velocity elevation at a declared landing event

$\mathbf{S} = 60\boldsymbol{\omega}_b/(2\pi)$ when angular velocity is in radians per second. Define $S_{\text{back}} = \mathbf{S} \cdot \hat{\mathbf{e}}_b$, $S_{\text{side}} = -\mathbf{S} \cdot \hat{\mathbf{e}}_u$ and $S_{\parallel} = \mathbf{S} \cdot \hat{\mathbf{v}}_b$. Then

$$\begin{aligned}
S_{\perp} &= \sqrt{S_{\text{back}}^2 + S_{\text{side}}^2}, & \theta &= \text{atan2}(S_{\text{side}}, S_{\text{back}}), \\
S_{\text{side}} &= S_{\perp} \sin \theta, & S_{\text{back}} &= S_{\perp} \cos \theta, \\
S^2 &= S_{\perp}^2 + S_{\parallel}^2, & S &= \|\mathbf{S}\|.
\end{aligned} \tag{2.4}$$

Replacing $S_{\perp}$ by total spin S is valid only when the axial component vanishes. For horizontal forward flight and $\mathbf{S} = (1000, -300, 3000)$ rpm in the world frame, $S_{\perp} \simeq 3014.96$ rpm, $S \simeq 3176.48$ rpm and $\theta \simeq 5.71^\circ$. Thus total spin and one tilt angle do not determine the general three-dimensional state. The tilt is undefined if both transverse components vanish, and zero spin has no axis; the chosen launch-frame construction also needs a nonvertical launch. Device-specific sidespin and tilt conventions must be converted explicitly before applying these equations.

In the usual spin-induced lift model, the transverse spin relative to air velocity contributes to lateral and vertical lift; wind can make that velocity differ from ground-relative launch velocity. A fixed lateral-distance-per-degree rule cannot replace the flight calculation: speed, launch direction, spin magnitude, wind and aerodynamic coefficients also matter (Chapter 8). Left/right curvature labels need not be draw/fade labels for both handednesses.

2.3 The Directness Hierarchy

Here, *measured* means estimated from observations that constrain the quantity, with calibration and a stated observation model; it does not mean algorithm-free or error-free. *Derived* denotes a transformation of reported quantities, and *estimated from a physical model* denotes an inference or prediction that additionally depends on collision, flight or ground-interaction assumptions. These are provenance descriptions rather than mutually exclusive product classes. Radar records signals and cameras record images; neither records a calibrated three-dimensional club path as raw data. Table 2.3 therefore lists the evidence needed for an output rather than assigning every device of one architecture an identical capability.

The radar and camera chapters develop these conditional observation relationships (Chapters 4 and 5). An outdoor radar track can support trajectory-based spin inference, while a short indoor

Table 2.3: Output Provenance and Conditions to Check

Output	Possible Provenance	Necessary Qualification
Ball speed / launch	Measured through calibrated radar or image observations	Geometry, timing, track association and launch-event fit
Spin rate / axis	Measured from rotation-sensitive observations; or model-inferred from other quantities	Visible features or spin signatures, ambiguity, axial component and flight window
Club motion	Measured by tracking a declared point or fitting rigid motion	Reference point, observation interval and collision exclusion
Face pose / strike	Measured from resolved geometry; or inferred through an impact model	Surface normal, strike location, markers if required, and model identifiability
Face to path / smash	Derived from the constituent outputs	Shared uncertainty and definition compatibility
Carry / landing	Derived from a sufficiently observed trajectory, or predicted from launch	Observed endpoint versus extrapolation and normalization
Total	Estimated with flight, bounce and roll models when rest is not observed	Ground slope, restitution, friction and endpoint definition

track may require different information; neither statement identifies every product’s algorithm. For example, TrackMan now describes iO as using cameras for full three-dimensional spin, including axial spin [101]. That product-specific vendor claim is evidence against a universal “indoor radar means estimated axis” label, not independent proof of its advertised accuracy. Similarly, a marker requirement belongs to the particular club-tracking implementation, not to optical sensing as a physical necessity.

Design implication

To use launch-monitor data to investigate the golf swing, retain the measurement chain: club motion and contact geometry constrain impact; impact produces launch; launch and environment determine flight. A face angle inferred from ball launch is not independent corroboration of a proposed relationship between those same outputs. Report definitions, uncertainty and shared inputs alongside the numbers before attributing a difference to technique or to one sensor family.

Chapter 3

Impact Physics: From Club Delivery to Ball Launch

The collision connects a golfer's delivered club state to the ball's launch velocity and spin. A forward model predicts those outcomes from specified incident states, geometry and contact properties; an inverse model estimates some of those inputs from observations. Either can be used with optical, radar or combined measurements. The sensing modality alone does not establish a product's internal algorithm or which reported quantities are independently measured. This chapter separates exact geometry, conditional impact mechanics and empirical approximations so that each can support an appropriate inference.

3.1 The D-Plane

Use a right-handed basis with x forward along the target, y upward and z to the right when viewed toward the target. Horizontal angles remain positive to the right and elevations positive upward, consistent with the right-handed-player signs in Chapter 2. A unit direction with azimuth φ and elevation λ is

$$\mathbf{d}(\varphi, \lambda) = \begin{pmatrix} \cos \lambda \cos \varphi \\ \sin \lambda \\ \cos \lambda \sin \varphi \end{pmatrix}. \quad (3.1)$$

The local face normal is $\hat{\mathbf{n}} = \mathbf{d}(\varphi_f, \lambda_d)$. For collision mechanics, define $\hat{\mathbf{p}}_c = \mathbf{d}(\varphi_c, \alpha_c)$ from the incident velocity of the *contact point*, with $V_c = \|\mathbf{v}_c\| > 0$ and all states evaluated at a declared common instant. A monitor's reported club path and attack angle may instead describe another reference point or maximum compression. They cannot be substituted silently: for a rigid head, $\mathbf{v}_c = \mathbf{v}_O + \boldsymbol{\omega}_h \times (\mathbf{r}_c - \mathbf{r}_O)$, and the state can change during impact.

In the centered, symmetric oblique-impact idealization associated with the D-plane [44], the nonparallel directions $\hat{\mathbf{n}}$ and $\hat{\mathbf{p}}_c$ span a plane. Assume an initially stationary, nonspinning spherical ball, an isotropic contact response and a face normal treated as fixed during the impulse. If the net impulse lies in that plane, so does the outgoing ball velocity. Its location between face and path, and its closeness to either, additionally depend on the contact law; geometry alone supplies no universal weighting. For tangential impulse directed along the contact-plane component of club motion, the ball's contact lever arm is $-R\hat{\mathbf{n}}$, giving

$$\Delta\boldsymbol{\omega}_b = I_b^{-1}(-R\hat{\mathbf{n}}) \times \mathbf{J}_t, \quad \Delta\boldsymbol{\omega}_b \parallel \hat{\mathbf{p}}_c \times \hat{\mathbf{n}}. \quad (3.2)$$

Here R is the ball radius and I_b its scalar inertia in this spherical model; the signed tangential impulse sets the orientation along that axis and its magnitude. Parallel directions do not define a unique D-plane or a normalized cross-product axis. Initial spin, off-center head rotation, anisotropy and changing contact geometry require an expanded model.

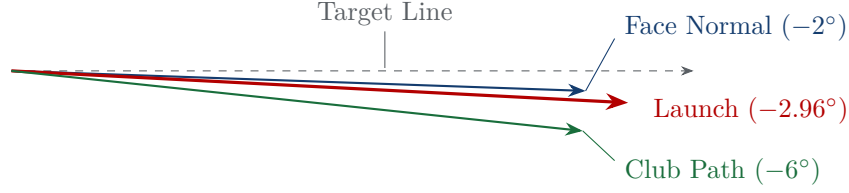


Figure 3.1: Manufactured Horizontal Launch Example. The declared 0.76/0.24 weighting gives -2.96° from face angle -2° and path -6° . This top-view projection illustrates the arithmetic, not a measured shot or a complete three-dimensional spin prediction.

3.1.1 Face/Path Weighting of Launch Direction

The filtered player data of Wood et al. [112] give the following study-specific horizontal weights in an approximate model of launch direction between face angle and club path:

$$\varphi_{\text{launch}} \approx w_f \varphi_f + (1 - w_f) \varphi_p, \quad w_f \approx \begin{cases} 0.76 \pm 0.08 & \text{driver} \\ 0.69 & \text{7-iron} \\ 0.61 & \text{wedge.} \end{cases} \quad (3.3)$$

Caution

The values above are the *horizontal* weights measured by Wood et al. [112]. Their player data contain 1,575 shots before filtering, with 157 golfers testing driver/7-iron and ten also testing a wedge. Club motion was measured optically at 720 fps and ball launch with a GC2. The horizontal means use strikes within 0.25 in of face centre in each axis and at least 1.5° face-to-path separation: 89 driver, 172 iron and 28 wedge shots remained. A separate 15-shot PING Man robot test returned 0.63 for the 7-iron.

The familiar 0.85/0.15 figure comes from TrackMan's *Ten Fundamentals*, which gives driver rules of thumb in both planes (and 75% for irons). Wood's driver means are 0.83 ± 0.08 vertically and 0.76 ± 0.08 horizontally, reported with 95% confidence intervals. The difference from 0.85 is a reason to retain the model's conditions, not a universal calibration correction. These are estimates from filtered samples, not individual-shot tolerance bands; protocol and measurand differences preclude treating the comparison as a definitive test of every application of the vendor rule.

For any estimator using this weighting model, inverting Eq. (3.3) to recover face angle amplifies any bias in measured launch direction by $1/w_f$ when path and weight are fixed. At $w_f = 0.76$ the factor is 1.32, versus 1.18 at $w_f = 0.85$: about 12% greater sensitivity, not a factor of two. Path error and uncertainty in the weight contribute separately, so this derivative is not a complete uncertainty budget for inferred face angle.

Henrikson and colleagues [37] compare a tangentially compliant contact model with plate impacts and player data. Their model's launch ratio varies with incidence even at fixed friction, and changing friction has different effects across incidence regimes. It uses a fixed impact surface and an adjustment for normal inelasticity; agreement in those comparisons is not a universal validation for moving, rotating clubheads. The findings support examining the contact-force history rather than attributing all club-dependent weighting to surface smoothness.

3.1.2 Spin Loft and Approximate Launch Maps

The geometric angle Φ between the local normal and contact-point incident velocity satisfies the full dot product

$$\cos \Phi = \hat{\mathbf{n}} \cdot \hat{\mathbf{p}}_c = \sin \lambda_d \sin \alpha_c + \cos \lambda_d \cos \alpha_c \cos(\varphi_f - \varphi_c). \quad (3.4)$$

This contact obliquity equals a reported spin loft only when point, time and angle definitions match. For coplanar forward directions it reduces to $|\lambda_d - \alpha_c|$. Writing $A_c = \varphi_f - \varphi_c$, neither $\cos A_c \cos \lambda_d$ nor $\cos A_c \cos(\lambda_d - \alpha_c)$ is a general replacement when horizontal separation and both elevations are nonzero.

Tutelman's historical model [97] uses its own loft/relative-angle construction and an approximate launch map,

$$f(\Phi) = 0.96 - 0.0071\Phi, \quad \text{DA}_{\text{vert}} = Lf(\Phi), \quad \text{DA}_{\text{horiz}} = Af(\Phi), \quad (3.5)$$

with angles in degrees and A, L interpreted according to that model. Its historical loft terminology is not interchangeable with target-frame dynamic loft, and its later comparison with TrackMan does not establish that the original model was fitted to that comparison. Treat the map as an author approximation with a declared domain, not an exact spherical-coordinate transformation or a general friction identification. For a three-dimensional implementation, obtain launch angles from the predicted velocity vector using `atan2`; adding component angles is not a general rotation composition.

3.1.3 Spin-Axis Tilt

A spin axis needs a reference frame as well as a direction. For outgoing unit velocity $\hat{\mathbf{v}}_b$ that is not vertical, let $\hat{\mathbf{e}}_y$ be world up and define

$$\hat{\mathbf{e}}_b = \frac{\hat{\mathbf{v}}_b \times \hat{\mathbf{e}}_y}{\|\hat{\mathbf{v}}_b \times \hat{\mathbf{e}}_y\|}, \quad \hat{\mathbf{e}}_u = \hat{\mathbf{e}}_b \times \hat{\mathbf{v}}_b, \quad \theta_{\text{axis}} = \text{atan2}(-\boldsymbol{\omega}_b \cdot \hat{\mathbf{e}}_u, \boldsymbol{\omega}_b \cdot \hat{\mathbf{e}}_b). \quad (3.6)$$

This convention makes positive backspin point along $\hat{\mathbf{e}}_b$ and gives positive tilt for a right-curving spin component in ordinary forward flight. Compare a device's sign and reference-axis definition before comparing its number. A vertical launch needs a different frame construction; a zero spin vector has no spin-axis direction. The tilt is also undefined when both transverse projections vanish, even if an axial spin component is nonzero. A general spin vector can also have a component along $\hat{\mathbf{v}}_b$, so total spin and one tilt angle do not specify every three-dimensional spin state.

The often-used ratio $\tan(\text{face-to-path})/\tan(\text{spin loft})$ is not an exact tilt identity for general target-frame angles. Calculate the cross product and its projections under the stated collision assumptions instead. Likewise, a fixed yards-per-degree or return-to-target multiplier cannot replace a flight calculation: speed, spin magnitude, launch, wind and aerodynamic coefficients also affect curvature.

Design implication

If an estimator adopts the launch-weight model, it may infer $\varphi_f = [\varphi_{\text{launch}} - (1 - w_f)\varphi_p]/w_f$. Its local launch sensitivity is $1/w_f$; path, weight, reference-frame covariance and model discrepancy contribute separately. Agreement between an inferred face angle and the launch measurement used to infer it is not an independent validation of that face angle.

3.2 Ball Speed and Smash Factor

Let J_n be the normal impulse on an initially stationary ball of mass m . For a free rigid head of mass M , centroidal inertia tensor I_h and contact lever arm $\mathbf{r}_h$, its normal inverse effective mass is

$$\frac{1}{M_{\text{eff},n}} = \frac{1}{M} + (\mathbf{r}_h \times \hat{\mathbf{n}})^\top I_h^{-1} (\mathbf{r}_h \times \hat{\mathbf{n}}). \quad (3.7)$$

In the scalar limit with no normal–tangential coupling, normal restitution e_n and closing speed $V_c \cos \Phi > 0$ give [9, 61]

$$J_n = \frac{(1 + e_n)V_c \cos \Phi}{1/m + 1/M_{\text{eff},n}}, \quad \mathbf{v}_b^+ \cdot \hat{\mathbf{n}} = \frac{1 + e_n}{1 + m/M_{\text{eff},n}} V_c \cos \Phi. \quad (3.8)$$

This predicts the *normal component* of ball velocity. If there is tangential impulse, $\mathbf{v}_b^+ = (J_n \hat{\mathbf{n}} + \mathbf{J}_t)/m$ and the speed norm also contains that component. Normal–tangential coupling requires a coupled contact solve rather than inserting an effective mass into an otherwise unrelated scalar fit.

For illustration only, taking $m = 0.04593$ kg, $M_{\text{eff},n} = 0.200$ kg and $e_n = 0.83$ gives the following normal-component ratios to V_c :

Obliquity	0°	10°	20°	30°
Normal Ratio	1.488	1.466	1.398	1.289

These are manufactured values, not a current equipment-conformance test or a loft-dependent physical ceiling. The assumed restitution is a model parameter; it does not identify every real ball/face combination or impact speed.

Key point

A reported smash factor divides measured ball speed by a specified reported club speed, which need not be the contact-point incident speed V_c . Exceeding one row of this illustrative table therefore does not by itself prove measurement error. Before rejecting a shot, reconcile reference point and time, effective inertia, contact model and uncertainty. A discrepancy is a diagnostic lead, not a universal impossibility verdict.

3.3 Spin Generation

The same tangential impulse changes both ball translation and angular momentum. For a spherical ball with contact arm $-R\hat{\mathbf{n}}$, $\Delta\boldsymbol{\omega}_b = I_b^{-1}(-R\hat{\mathbf{n}}) \times \mathbf{J}_t$. For an impulse in the obliquity plane, the launch deflection from the normal satisfies $\tan \delta_b = J_t/J_n$. Thus independently chosen launch and spin fits must still be checked for compatibility with one contact solution. Tangential compliance can reverse force during contact; a single friction coefficient does not specify the entire slip–stick history [37, 10].

When its club-speed input is identified with V_c , an empirical approximation in Tutelman’s model is [97]

$$S \text{ [rpm]} \approx 160 V_c \text{ [mph]} \sin \Phi. \quad (3.9)$$

The coefficient carries the stated units and is not a dimensional identity or a universal spin law. Retain the model’s input conventions and calibration limits; use the predicted spin vector for component decomposition rather than imposing a separate trigonometric tilt rule. Rules based only on iron loft, or a universal spin-loft saturation angle, omit contact material, speed and force history. Changing surface conditions changes the contact problem; this chapter supplies no universal contamination correction or threshold.

3.4 Gear Effect and Impact Location

An off-center impulse rotates the head and changes contact-point velocity during collision. In a simplified horizontal construction, let x be the signed lateral offset from the head CG line, C the normal CG-to-face distance and I_h the head inertia about the relevant vertical axis. Tutelman's angular-impulse heuristic [95] estimates $\Delta\omega_h = xmV_b/I_h$ and equates a face tangential speed to the ball's rotational surface speed at separation, giving

$$s \text{ [rpm]} \approx 58,830 \frac{V_b[\text{mph}] C[\text{in}] x[\text{in}]}{I_h[\text{g cm}^2]} \approx 16.4 V_b[\text{mph}] x[\text{in}]. \quad (3.10)$$

The last coefficient uses the author's representative $C = 1.42$ in and $I_h = 5100 \text{ g cm}^2$; there is no extra factor of 100. For $V_b = 150$ mph and $x = 0.25$ in, the simplified estimate is 615 rpm, not 61,500 rpm. The cited 2.5% describes the relative sample standard deviation of a historical, filtered inertia/depth ratio, not a guaranteed spin-prediction error or all current drivers. The contact-speed assumption neglects ball tangential translation and coupled recoil, so the heuristic is not a complete no-slip impact solution.

3.4.1 Why a Coupled Contact Solve Matters

The missing terms can be seen without fitting a golfer or product. Consider an initially nonspinning free head moving normally at speed V toward a stationary spherical ball, with head contact arm $C\hat{n} + x\hat{t}$ in a planar model. The unit tangent $\hat{t}$ is perpendicular to $\hat{n}$. The head has mass M and scalar inertia I_h ; the ball has mass m , radius R and inertia I_b . Take positive inertial parameters and $V > 0$, with $0 \leq e_n \leq 1$ and an impulsive friction coefficient $\mu \geq 0$. With impulse components (J_n, J_t) on the ball, the ball-minus-head contact-velocity increment is $K(J_n, J_t)^\top$, where

$$K = \begin{pmatrix} \frac{1}{m} + \frac{1}{M} + \frac{x^2}{I_h} & -\frac{Cx}{I_h} \\ -\frac{Cx}{I_h} & \frac{1}{m} + \frac{1}{M} + \frac{R^2}{I_b} + \frac{C^2}{I_h} \end{pmatrix}. \quad (3.11)$$

Normal restitution and zero final tangential relative velocity then give the sticking candidate

$$K \begin{pmatrix} J_n \\ J_t \end{pmatrix} = \begin{pmatrix} (1 + e_n)V \\ 0 \end{pmatrix}, \quad |J_t| \leq \mu J_n. \quad (3.12)$$

The inequality must hold for the candidate to be admissible under this impulsive friction model; otherwise a sliding branch is needed. The term $1/m$ in the tangential entry accounts for ball translation, R^2/I_b for ball spin and the head terms for recoil. Consequently, matching contact velocities does not mean that face speed equals R times ball spin. This idealization still omits contact compliance, shaft impulses and changing face geometry; it is a consistency calculation, not a universally identified gear-effect law.

Bulge and roll alter the local normal as strike location changes, while gear effect alters the impulse through head motion. Their contributions must be evaluated together; changing the rotation axis also changes the relevant head inertia. An optical strike measurement can constrain the geometry, but uncertainty in pose, CG, inertia and contact response remains. Neither a quoted yardage example nor Equation (3.10) provides a universal bound on face-angle inference error.

Design implication

Impact location is valuable when it resolves an important uncertainty for the intended outcome. Its priority relative to face pose, local velocity or spin requires a sensitivity and uncertainty analysis. Off-center launch/spin cannot in general be inverted with a centered D-plane model alone, and adding a strike sensor does not remove collision-model discrepancy.

3.5 Vertical Plane and the Low-Point Geometry

A fixed inclined-plane construction explains one conditional coupling between attack and horizontal path. Set the plane's horizontal base along the target and its inclination to $0 < \beta < 90^\circ$, with unit normal $(0, \cos \beta, \sin \beta)$ in the forward/up/right frame. A velocity direction with elevation α_c and horizontal offset ψ lies in that plane only if

$$\sin \alpha_c \cos \beta + \cos \alpha_c \sin \psi \sin \beta = 0, \quad \sin \psi = -\frac{\tan \alpha_c}{\tan \beta}. \quad (3.13)$$

On the forward-moving branch, $\alpha_c = -5^\circ$ and $\beta = 60^\circ$ give $\psi \approx 2.8953^\circ$ to the right. The result requires $|\tan \alpha_c / \tan \beta| \leq 1$ and the declared side of plane inclination; a mirrored plane changes the sign. A measured swing need not follow one fixed plane, and this derivation does not establish any particular product's processing algorithm.

3.6 Connecting Delivery, Impact and Measurement

Swing dynamics determines the delivered local velocity, orientation and retained elastic state; contact mechanics converts that state into an impulse and outgoing ball state; the instrument estimates selected quantities from its observations. An explanation of technique must keep those stages connected while testing each with sufficiently independent evidence. For example, changing wrist motion may alter both face pose and contact-point velocity, and changing strike location may alter both the local normal and effective inertia. A single fitted launch percentage cannot identify which change caused an observed ball flight. Report the retained states, reference points, contact assumptions and uncertainty alongside the conclusion, and compare joint predictions of launch direction, speed and spin rather than validating each with incompatible scalar rules.

Chapter 4

Doppler Radar Systems

Radar observations constrain a moving target through frequency, phase, amplitude and timing. A launch monitor combines those observations with geometry, signal processing and, for some outputs, a collision or flight model. The useful question is which quantities the available observations identify, under which assumptions, and with what uncertainty. Those distinctions matter when launch-monitor outputs are used to explain a golfer's swing: two reported parameters can share the same measured input and therefore share its errors.

Key point

What This Shows: The measurement equations below explain radial velocity, angular ambiguity, spin signatures and conditional trajectory-based spin-axis inference. They also show how an inferred face angle can inherit uncertainty from ball launch and club path. **What It Does Not Show:** A patent disclosure, vendor specification or manufactured calculation does not establish the accuracy or internal implementation of every commercial product. Device generation, firmware, ball type, geometry and operating mode must accompany any performance claim.

4.1 Continuous-Wave Doppler Fundamentals

A continuous-wave (CW) radar transmits a carrier at frequency f_c , with wavelength $\lambda = c/f_c$. For monostatic transmission and reception, target speeds small compared with c , and a sign convention assigning positive Doppler to increasing range, the instantaneous range rate $v_r = \dot{r}$ gives

$$f_d = \frac{2v_r}{\lambda} = \frac{2f_c v_r}{c}. \quad (4.1)$$

The factor of two accounts for the outgoing-plus-return path; the displayed Doppler sign depends on the receiver's mixing convention. At an illustrative carrier of 24.125 GHz, one mph of radial speed gives approximately 71.95 Hz; at 10.5 GHz, it gives 31.31 Hz, using $c = 299\,792\,458$ m/s and $1 \text{ mph} = 0.447\,04$ m/s. These conversions do not identify a particular product's carrier or detection range.

Carrier frequency alone does not set a flight-tracking limit. For a coherent observation of duration T , Fourier-bin spacing of order $1/T$ corresponds to radial-velocity spacing of order $\lambda/(2T)$. That is spectral spacing, not a universal velocity-error bound: windowing, acceleration, estimator design and signal-to-noise ratio matter. For an ideal wideband ranging waveform, range resolution instead scales with transmitted bandwidth B , approximately $c/(2B)$; angular performance depends on aperture and geometry. Detection range additionally depends on transmitted power, antenna gain, target scattering, propagation, clutter and detection thresholds. A blanket thirty-yard limit for consumer 24 GHz radar cannot follow from the carrier frequency; the hybrid disclosure discussed below itself proposes much longer ranges [11].

4.2 From Velocity Sensor to 3D Tracker: Phase Interferometry

A single unmodulated CW channel supplies radial-velocity information without independently resolving absolute range and two angular coordinates. Multiple calibrated receive antennas can add direction information. Let $\hat{\mathbf{u}}$ denote the unit direction from the array toward the target. For a locally planar wavefront in a narrowband, far-field approximation, receivers separated by baseline $\mathbf{d}$ have a relative delay $\tau = (\mathbf{d} \cdot \hat{\mathbf{u}})/c$, with direction and phase signs chosen consistently. For baseline length d , direction cosine $u = \mathbf{d} \cdot \hat{\mathbf{u}}/d$ along it and a calibrated measured phase $\Delta\varphi$, the possible directions are

$$\Delta\varphi \equiv \frac{2\pi d}{\lambda} u \pmod{2\pi}, \quad u_k = \frac{\lambda(\Delta\varphi + 2\pi k)}{2\pi d}, \quad k \in \mathbb{Z}, \quad |u_k| \leq 1. \quad (4.2)$$

The integer branch is part of the inverse problem. For $d = \lambda$, direction cosines 0.2 and -0.8 give the same complex phase. Half-wavelength spacing avoids interior aliases over the full direction-cosine interval, but its opposite endpoints still coincide in phase. Larger baselines may be usable over a restricted field of view or with additional, appropriately chosen baselines and joint ambiguity resolution; an antenna count alone does not guarantee uniqueness [99].

Two independent baselines constrain two direction cosines, with a physical hemisphere choice needed to reconstruct a direction. Near endfire, converting a direction cosine to an angle is poorly conditioned: for an angle θ measured from broadside so that $u = \sin \theta$, small unwrapped phase errors give $\delta\theta \simeq \lambda\delta\varphi/(2\pi d \cos \theta)$. This local approximation excludes branch errors and becomes unreliable near $\cos \theta = 0$. Near-field wavefront curvature requires a range-dependent propagation model rather than the plane-wave formula. Combining a resolved direction with independently estimated range and range rate can yield a three-dimensional track, provided the channels observe the same target and share calibrated coordinates and timing. A per-Doppler-bin direction is a direction of contributing scattering, not automatically the ball center or a labeled point on a clubface [100].

Design implication

One speed module and two angle modules form an investigational sensing arrangement, not a validated launch monitor merely by supplying three outputs. RFbeam's K-LD7 exposes raw I/Q data: its RADC packet contains Rx1 and Rx2 at one carrier setting and Rx1 at a second setting used for FSK ranging. The March 2021 Revision B datasheet specifies a maximum configured speed of 100 km/h, a typical 29 ms frame at that setting, and warnings about out-of-range returns and serial throughput [72]. Its person-detection example and configurable range do not establish golf-ball performance. An implementer must establish speed and signal bandwidth, target association, synchronization, ambiguity handling and error propagation before treating these modules as a three-dimensional ball tracker. A rigid mount and surveyed calibration help, but do not remove the need to model drift, multipath and processing errors.

4.3 CW + FMCW Hybrids

Frequency-modulated continuous-wave (FMCW) sensing adds delay information through a known transmitted frequency history. For a moving target, beat frequency contains range and Doppler contributions, so separating them depends on the waveform and estimator; a chirp does not make range an assumption-free observable. Full Swing's US 11,311,789 describes example 24 GHz CW and linear-FMCW channels, multiplexing alternatives and several antenna layouts [11]. The disclosure associates Doppler signatures with speed and spin and modulated signals with range, and discusses

ranges exceeding 250 m. Its examples include a ball swung on a line at 75 m; that demonstration is not an independent accuracy study of full golf shots. The array drawings include both symmetric and nonuniform arrangements, so one example should not be presented as the definitive commercial KIT layout. The disclosure establishes proposed mechanisms; which mechanisms are deployed, and their error distributions, require separate product-specific evidence.

4.4 Ball Measurement

4.4.1 Speed and Launch Angles

A processor must associate the post-impact Doppler return with the ball before interpreting it as ball radial velocity. Three-dimensional launch velocity also requires direction information from a fitted track or another sensing channel. If α is the angle between target velocity and the radar line of sight, $v_r = \|\mathbf{v}\| \cos \alpha$; it is not generally enough to substitute the launch angle relative to the ground for α . Dividing by $\cos \alpha$ becomes poorly conditioned near a tangential trajectory, and uncertainty in direction then strongly affects inferred speed. The reported impact state must also distinguish a sample taken after impact from a value extrapolated back to the defined event.

Leach et al. compared TrackMan Pro IIIe and Foresight GC2+HMT with a benchmark optical system using 240 shots across a driver, 7-iron and utility wedge [47]. They found generally stronger agreement for ball parameters than clubhead parameters, with several results dependent on club. That evidence supports a device- and study-specific distinction; it does not establish best agreement for the same parameters in every study or newer product.

4.4.2 Spin Rate: Harmonic Signatures and Their Ambiguities

Tuxen's US 8,845,442 describes extracting spin information from Doppler spectral structure [98]. In a rigid-ball point-feature model, let $\mathbf{v}_C$ be center velocity, $\boldsymbol{\rho}(t)$ the feature's offset from the center, and $\boldsymbol{\omega}$ the angular velocity, all expressed in one frame. Using a common line-of-sight unit vector $\hat{\mathbf{n}}$ across the small target gives

$$f_{d,\text{feature}}(t) = \frac{2}{\lambda} \hat{\mathbf{n}} \cdot [\mathbf{v}_C + \boldsymbol{\omega} \times \boldsymbol{\rho}(t)]. \quad (4.3)$$

The rotational term vanishes for spin parallel to that line of sight. For constant spin and approximately fixed viewing direction, a feature at perpendicular distance $\rho_\perp$ from the spin axis has a sinusoidal projected velocity with amplitude $\|\boldsymbol{\omega}\| \rho_\perp \sin \beta$, where β is the spin-axis/viewing angle. The simpler amplitude $\rho_\perp \|\boldsymbol{\omega}\|$ requires a view perpendicular to the spin axis; $\rho_\perp$ equals the ball radius only for an equatorial surface feature. Actual returns also depend on coherent scattering, visibility and rotating reflectivity, so this point-feature model is not a complete electromagnetic ball model.

Let S denote spin rate in revolutions per minute. Periodic modulation at rotation frequency $f_{\text{spin}} = \|\boldsymbol{\omega}\|/(2\pi)$ can produce spectral lines

$$f_n = f_d + n f_{\text{spin}}, \quad n \in \mathbb{Z}, \quad S [\text{rpm}] = 60 f_{\text{spin}} [\text{Hz}]. \quad (4.4)$$

Observed peaks need not include every integer order, symmetric peaks need not have equal detectable amplitude, and some rotation aspects may yield weak modulation. For example, peaks offset by 200 and 400 Hz can represent the first two orders of 200 Hz or the second and fourth orders of 100 Hz. A reflector pattern that repeats after half a revolution can suppress odd harmonics relative to the

mechanical rotation frequency. Consequently, multiplying the spacing of any two visible peaks by sixty is not a generally valid spin estimator.

The patent describes tracking candidate traces, qualifying harmonics and assigning their orders before estimating rotation frequency [98]. Comb or cepstral processing can organize candidates, but cannot by itself resolve information absent from the observation. Window duration, frequency drift, noise, clutter, scattering asymmetry and ball markings affect which candidates are distinguishable. A harmonic spacing yields a candidate spin magnitude; it does not alone give a three-dimensional spin axis, though additional phase, amplitude, geometric or trajectory observations may supply axis information.

4.4.3 Spin Axis: Conditional Trajectory Inversion

Trajectory inversion combines observations with a force model. Let $\mathbf{v}_a = \mathbf{v} - \mathbf{w}$ be air-relative velocity for wind $\mathbf{w}$; let $\mathbf{A}$ denote ball acceleration in the ground-fixed inertial approximation, $\mathbf{g}$ gravitational acceleration, and $\mathbf{D}$ and $\mathbf{L}$ drag and lift accelerations, not forces. Assume aerodynamic acceleration separates into drag parallel to $\mathbf{v}_a$ and lift perpendicular to it, with no omitted transverse force. For nonzero airspeed,

$$\mathbf{D} = \frac{(\mathbf{A} - \mathbf{g}) \cdot \mathbf{v}_a}{\|\mathbf{v}_a\|^2} \mathbf{v}_a, \quad \mathbf{L} = \mathbf{A} - \mathbf{g} - \mathbf{D}. \quad (4.5)$$

Physical drag additionally requires the fitted parallel component to oppose airspeed; a projection alone does not enforce that sign. Unknown wind, omitted forces and differentiation errors can contaminate the inferred lift. Acceleration estimated from position is especially sensitive to noise and the chosen temporal fit.

For a fixed spin-axis direction over the fitted interval and a Magnus model with $\mathbf{L}$ perpendicular to that axis, stacking the constraints at N sample times gives

$$\mathbf{M}\hat{\boldsymbol{\omega}} = \mathbf{0}, \quad \mathbf{M} = \begin{bmatrix} \mathbf{L}_1^\top \\ \vdots \\ \mathbf{L}_N^\top \end{bmatrix}, \quad \|\hat{\boldsymbol{\omega}}\| = 1. \quad (4.6)$$

In exact consistent data, rank two leaves one axis line, with two possible signs; rank one leaves a plane of candidate vectors and therefore a continuum of unit axes. No lift leaves no directional information through this constraint. The zero vector is always a solution of the unconstrained homogeneous system, which is why unit normalization is essential. Noisy rows generally do not have an exact common nullspace: a weighted unit-norm fit and its conditioning are needed, rather than declaring uniqueness from a numerical rank alone. The separation between the smallest singular values matters to axis stability.

A signed Magnus relation, such as $\mathbf{L} = \kappa \boldsymbol{\omega} \times \mathbf{v}_a$ with a stated positive effective aerodynamic coefficient κ , can help select the orientation and estimate spin components. At one instant, adding spin parallel to $\mathbf{v}_a$ leaves this cross product unchanged. Changing air-relative velocity over time can add information if the same spin vector and force model remain applicable; instantaneous insensitivity does not prove non-identifiability over every trajectory. The patent's trajectory route must be interpreted with those model and geometry assumptions [98]. A longer observed flight often supplies more useful curvature, but does not guarantee accurate inference. A short indoor segment can be poorly conditioned, especially with low lift or uncertain wind; both outdoor and indoor fitted axes remain estimates with different information and uncertainty.

4.4.4 A Dimensional and Algebraic Check on a Delay Disclosure

FlightScope’s US 10,775,492 describes a different spin-axis approach using spatially separated receiver pairs [41]. In that disclosure, S_V and S_H are vertical and horizontal receiver separations, D is target distance, and Φ is the spin-axis angle inferred from those pairs. Its displayed relations give $T_V = (S_V/D) \cos \Phi$ and $T_H = (S_H/D) \sin \Phi$, while calling the T quantities time delays, and subsequently give $\arctan[S_H T_H / (S_V T_V)]$ for the angle. With S_V, S_H, D all lengths, those right-hand sides are dimensionless and require an additional time scale to represent delays. Moreover, even treating the printed T quantities as normalized values, elimination gives

$$\Phi = \operatorname{atan2}\left(\frac{T_H}{S_H}, \frac{T_V}{S_V}\right), \quad \tan \Phi = \frac{S_V T_H}{S_H T_V}, \quad (4.7)$$

where the tangent form applies only when its denominator is nonzero and loses quadrant information. For manufactured values $S_V = 0.1$, $S_H = 0.2$, $D = 3$ in consistent length units and $\Phi = 30^\circ$, the displayed patent ratio returns approximately 66.59° ; the algebraic elimination returns 30° . This identifies an inconsistency in the displayed relationships, not a validated replacement for the full measurement method or an interpretation of patent scope. A working delay estimator still needs timing definitions, feature correspondence, rotational geometry and experimental validation.

4.4.5 Flight Observation, Extrapolation, and Ball Modes

A trajectory tracked to landing can support measured-flight outcomes; a trajectory truncated by a screen requires a flight model for the unobserved part. A normalized trajectory, computed for selected weather or ball conditions, must also be distinguished from the actual observed shot. These are output modes, not an absolute division between radar brands. TrackMan’s published TM4 minimum indoor setup of 4.7 m is *device-to-net* distance, not 4.7 m of ball flight [88]. FlightScope’s X3C instructions separately specify sensor-to-tee and ball-flight distances and identify marking/RCT requirements for indoor spin capture [15]. Instructions for that generation should not silently be applied to every earlier X3 or Mevo.

Garmin’s general R10 guidance describes standard-ball conditions under which spin is modeled rather than captured, including ball flight below 20 m or ball speed below 90 mph [25]. It must be read alongside the RCT mode: Garmin describes indoor spin capture with compatible Titleist RCT balls and current software, with at least eight feet tee-to-net and about two ball rotations for its most reliable results [29]. The support text allows successful capture with fewer rotations but also describes fallback to estimated spin when capture is inadequate; the app distinguishes that estimate typographically. Thus the standard-ball twenty-meter/ninety-mph guidance is not a universal requirement to measure spin in every supported mode. Marked radar-reflective balls and optically printed balls are different sensing aids and are not interchangeable across devices.

4.5 Club Measurement

4.5.1 What the Radar Tracks on the Head

A rotating clubhead has a velocity field: for a point P offset by ρ from reference point C , $\mathbf{v}_P = \mathbf{v}_C + \boldsymbol{\omega} \times \rho$. Here $\boldsymbol{\omega}$ is clubhead angular velocity, rather than the ball angular velocity in the preceding section. The fastest or strongest radar return need not correspond to the reference point used for reported club speed. Its bias is not universally positive; it depends on geometry and which part of the head dominates the return. TrackMan describes a reconstructed clubhead silhouette

and defines reported parameters using specific points and event times [93, 94]. Club path and attack angle concern a declared velocity direction; face angle and dynamic loft concern surface orientation, potentially at the impact location rather than the geometric center. For curved faces, those orientations differ. Point, frame, timing and surface definitions must therefore be reconciled before attributing a disagreement between instruments to sensor error.

4.5.2 Face Angle and Dynamic Loft

Radial-velocity sensing alone does not uniquely determine every clubface orientation, but it is too strong to say radar can never provide orientation information. Resolved scattering geometry, multiple views, temporal constraints and other sensors can add information; the available model and measurements determine observability (Chapter E). One possible estimation route is inversion of a local collision approximation. Let φ_f be face angle, φ_p club path angle and w_f the effective face weighting in a local collision model. For small horizontal angles about the same reference direction, suppose launch direction satisfies $\varphi_{\text{launch}} \simeq w_f \varphi_f + (1 - w_f) \varphi_p$ with known nonzero w_f . Then

$$\varphi_f \simeq \frac{\varphi_{\text{launch}} - (1 - w_f) \varphi_p}{w_f}. \quad (4.8)$$

This is an inverse of the stated local model, not a description of every radar product. The effective weighting depends on collision conditions and must not be transferred unchanged between clubs, lofts, ball types and impact locations (Section 3.1.1). A vertical-angle approximation likewise requires its own justified weighting and operating domain; it is not automatically obtained by substituting names into a horizontal fit.

For small perturbations within this model,

$$\delta\varphi_f \simeq \frac{\delta\varphi_{\text{launch}}}{w_f} - \frac{1 - w_f}{w_f} \delta\varphi_p + \frac{\varphi_p - \varphi_{\text{launch}}}{w_f^2} \delta w_f. \quad (4.9)$$

The launch-error gain is $1/w_f$ and therefore depends on the chosen collision weighting. Correlated measurement errors and uncertainty in the collision model also contribute. Off-center impact and gear effect can invalidate a fit that omits impact location; a richer estimator may use additional observations to address them. A common horizontal reference offset moves face and path together and cancels in their difference in this approximation, whereas independent misalignments need not cancel.

Key point

An inferred face angle can be statistically coupled to the measured launch direction because the estimator uses that direction as an input. Agreement or correlation between those outputs is therefore not independent confirmation of a physical face-to-launch relationship. For swing analysis, document the measurement and inference dependencies, propagate their covariance, and compare against an independent criterion before drawing conclusions about technique.

4.5.3 Known Limitations

The Leach et al. comparison motivates caution about clubhead outputs for research, but does not rank all modern radar and optical systems [47]. A pickup rate describes availability, not the error distribution of successful readings. TrackMan's stated club-data pickup rate in its 2020 OERT description should be read in that sense [93]. Validation should retain missed shots, state the tested clubs and speeds, and distinguish bias, dispersion and repeatability from agreement with an independent reference.

4.6 Optically Enhanced Radar: OERT and Fusion Tracking

TrackMan’s OERT description presents synchronized dual radar and camera observations, including clubhead silhouette and impact-location information [93]. Its current specification table describes TM4 and the indoor iO separately; the latter combines 24 GHz radar with cameras and describes optical three-dimensional spin measurement [88]. Those descriptions support the stated product distinctions without exposing every proprietary processing step. FlightScope describes Fusion Tracking as synchronized image processing combined with radar, and US 10,338,209 discloses mechanisms for combining the modalities [15, 40]. The presence of a patent does not establish independent accuracy or prove that every disclosed option is deployed.

Rapsodo explicitly requires the printed RPT pattern for MLM2PRO optical spin measurement; its support distinguishes RPT from RCT radar-reflective balls [69]. Full Swing markets machine-learning radar and video capabilities, but that description alone does not establish that its camera contributes to a fused spin/face estimator [24]. A camera used for swing video is not automatically a measurement camera.

Design implication

Choose an additional sensor to reduce a specified ambiguity or uncertainty, then test whether the combined estimator does so. Optical features can constrain orientation or impact location; radar can contribute radial velocity and flight tracking; ball markings can improve identifiable rotational structure. Their value depends on visibility, timing, calibration and the joint model. Combining modalities can improve inference, but neither sensor count nor a vendor’s architectural choice proves a universal limit of radar or the accuracy of a new implementation.

Chapter 5

Photometric (Camera) Systems

5.1 Imaging Architectures

Photometric launch monitors estimate motion from calibrated images taken near impact. Their useful capture volume depends on camera placement, illumination, exposure timing, resolution and visibility; it is not a universal length of ball flight. Multiple cameras can supply simultaneous geometric constraints, while strobed exposures can supply several time-separated observations within one image. Neither camera count nor a nominal frame rate alone specifies measurement uncertainty.

Commercial examples illustrate different observation geometries. Foresight’s GC3 uses three cameras, while the GCQuad family uses four; supported club outputs depend on the model and enabled features [19, 20]. Uneekor’s overhead EYE XO2 uses three cameras and club stickers for the documented club-face data; overhead placement does not itself make club measurement markerless [103]. ProTee VX instead advertises two overhead cameras and sticker-free club and ball outputs [65]. Garmin’s R50 uses three cameras and a club-face sticker for club tracking [28, 27, 26]. These are manufacturer descriptions of capabilities, not a common independent accuracy specification. The model-specific distinctions in Chapter 6 matter more than a camera-versus-radar label, especially for hybrid instruments.

5.2 Ball Measurement

5.2.1 Photogrammetric Position and Velocity

A calibrated camera maps a 3D point to an image through its intrinsics, pose and lens-distortion model. For matched observations of the *same physical point*, corrected image coordinates can be triangulated from several views, for example by minimizing reprojection error. A ball silhouette needs additional care: its apparent contour is not a set of fixed material points shared between viewpoints. Estimating the sphere center from its outline should therefore use a sphere-projection model before treating the result as a point observation.

Let $\mathbf{p}_k$ and $\mathbf{p}_{k+1}$ be estimated centers in one calibrated target frame at times separated by $\Delta t > 0$. Then

$$\bar{\mathbf{v}}_k = \frac{\mathbf{p}_{k+1} - \mathbf{p}_k}{\Delta t} \quad (5.1)$$

is an interval velocity estimate; its norm is speed. Displacement alone is not speed, and an interval estimate need not equal the instantaneous velocity at separation from the club. A local trajectory fit can estimate velocity at a declared event time, with timing uncertainty and any extrapolation included in the error budget. Launch angles follow by resolving that velocity in the target frame, not by reading image-axis slopes as physical launch angles.

For an ideal rectified stereo pair, let B be the baseline, f the common focal length in pixels, and $d = u_L - u_R > 0$ the disparity in pixels. Depth along the camera optical axis is $Z = fB/d$. Holding

calibration fixed, first-order propagation gives

$$\sigma_Z \simeq \frac{Z^2}{fB} \sigma_d, \quad \sigma_d^2 = \text{Var}(u_L) + \text{Var}(u_R) - 2 \text{Cov}(u_L, u_R). \quad (5.2)$$

Thus two independent image-coordinate errors of standard deviation σ_{px} give $\sigma_d = \sqrt{2}\sigma_{px}$, not σ_{px} . The approximation becomes unreliable when disparity uncertainty is large relative to disparity. Baseline, focal-length, distortion and alignment uncertainty add further contributions; Equation (5.2) is not a bound for every multi-camera layout. Camera depth Z must also be transformed into the book's target frame before comparison with launch-monitor coordinates.

5.2.2 Why an Ellipse Center Is Not the Ball Center

The distinction can be derived directly for an ideal pinhole camera. Let a sphere of calibrated radius r have center $\mathbf{c} = (X, Y, Z)^\top$ in camera coordinates, with $Z > r$. For a normalized image ray $\mathbf{q} = (u, v, 1)^\top$, tangency requires

$$(\mathbf{c}^\top \mathbf{q})^2 = (\mathbf{c}^\top \mathbf{c} - r^2) \mathbf{q}^\top \mathbf{q}. \quad (5.3)$$

The center of this image conic is

$$(u_e, v_e) = \frac{Z}{Z^2 - r^2} (X, Y), \quad (u_c, v_c) = \frac{1}{Z} (X, Y) \quad (5.4)$$

for the ellipse center and projected sphere center, respectively. Their difference is $r^2/(Z^2 - r^2)$ times (u_c, v_c) . It vanishes on the optical axis but generally does not vanish merely because an ellipse was fitted accurately. These are normalized coordinates; the intrinsics convert them to pixels. A sphere model, calibration and known or estimated radius supply the correction, while partial occlusion, ball deformation near contact and nonuniform illumination require separate treatment. The familiar ball-size range cue is therefore conditional on size and projection assumptions, not an independent exact depth measurement.

5.2.3 Spin: Surface-Pattern Registration

Visible dimples, logos or printed marks can constrain rotation by matching their surface locations between exposures. A geometric estimator jointly needs the ball center, radius, camera calibration, exposure times and feature correspondences. Changes in lighting, specular reflections, occlusion and repeated texture can produce matches that are photometrically plausible but geometrically wrong. Describing this as spherical registration explains a possible observation model; it does not establish the proprietary algorithm inside every camera product.

Let $Q_k \in SO(3)$ map ball-fixed coordinates into a fixed world frame. The spatial rotation increment is $R = Q_{k+1} Q_k^\top$. Under constant spatial angular velocity over Δt , and on the branch $\|\boldsymbol{\omega}\| \Delta t < \pi$,

$$\boldsymbol{\omega} = \frac{\text{vee}(\text{Log } R)}{\Delta t}, \quad \theta = \arccos\left(\frac{\text{tr } R - 1}{2}\right) \in [0, \pi]. \quad (5.5)$$

Here vee converts a skew matrix to its axial vector and Log denotes the principal rotation logarithm on that branch. The trace angle alone is unsigned; an eigenvector with eigenvalue one also has an arbitrary sign. A consistent rotation-vector construction supplies the orientation information away from the branch boundary. Near zero rotation the axis direction is poorly determined; near π the

principal branch needs particular care. If angular velocity changes direction during the interval, the logarithm describes an equivalent finite rotation and is not generally its arithmetic time average.

Rotations separated by complete revolutions can generate the same endpoint orientation, and repeated surface patterns can introduce further ambiguity. Additional exposures, independent observations or a justified speed prior may select a candidate; they do not guarantee that an otherwise ambiguous rotation becomes observable. Three noncollinear matched 3D surface points are sufficient for a noiseless proper rigid-transform fit, but three arbitrary image detections are not automatically three known 3D correspondences. A marked ball can improve correspondence without removing timing or calibration error. Rapsodo's RPT pattern is an optical spin aid; Titleist RCT's embedded radar-reflective structure serves a different observation mechanism [68, 29]. No universal percentage spin accuracy, or independence from ball speed, follows from the registration equations.

5.3 Club Measurement

5.3.1 Fiducial Geometry and Reference Points

Reflective club markers provide identifiable image features. Foresight's documented single-marker mode supplies a limited club-data set, while supported four-marker modes add orientation and impact information [23, 21]. The actual marker layout matters: four collinear points still leave rotation about their line unconstrained. A suitable noncollinear constellation can define a rigid pose when its geometry and correspondences are known and visible. For a curved driver face, the relationship between that constellation and the selected face-center tangent plane also needs calibration; a plane fitted to arbitrary surface points is not automatically the desired face plane.

A single marker trajectory provides that point's position and velocity, not a unique full club pose. For two points P and O fixed in a rigid clubhead,

$$\mathbf{v}_P = \mathbf{v}_O + \boldsymbol{\omega} \times (\mathbf{p}_P - \mathbf{p}_O), \quad (5.6)$$

with every quantity expressed in the same frame and at the same time. Consequently, differentiating a pose origin or marker position does not automatically yield the manufacturer's club-speed reference point. Path and attack angle inherit this point and timing dependence. Impact coordinates additionally require a registered face frame and a defined ball-contact event; a replay image alone is not a calibrated numerical impact-location measurement. These are the same measurand distinctions that govern the radar comparison in Chapter 9.

The GC3's published club list contains speed, path, attack angle and smash factor; it should not be credited with the GCQuad's face-angle and impact-location feature set [19]. Nor is a ratio such as smash factor a separate directly observed primitive simply because it appears in a measured-data list. Feature entitlement, marker mode and metric definition should accompany any comparison between units.

5.3.2 Overhead Tracking and Calibration

Overhead placement offers useful views but does not guarantee an unobstructed face, a marker-free algorithm or a uniquely determined pose. The EYE XO2 manual specifies club stickers, whereas ProTee VX advertises sticker-free image-based club outputs [103, 65]. Those different designs should not be collapsed into one generic overhead measurement method. Learned shape models can help interpret images; their training domain and failure behavior remain part of the measurement system.

Factory calibration concerns internal camera geometry; setup also establishes the target line, ground plane, working distance and hitting zone. Moving or tilting a unit, changing its support or

obscuring calibration features can change that relationship. The EYE XO2, for example, specifies a mounting envelope and a calibration procedure [103]; other products require their own instructions. A DIY build should verify residuals on independent calibration poses and assess timing, distortion, blur and occlusion across its intended volume. Fitting the calibration images well is not a held-out accuracy test.

5.4 The Indoor Argument and Downrange Limits

A camera system that obtains the necessary launch and spin observations near impact can estimate launch conditions before the ball reaches a screen. This is an important indoor advantage, conditional on the capture window and visibility actually being adequate. It does not observe the subsequent aerodynamic trajectory, environmental variation, landing or bounce beyond that screen. Those outputs are predictions from the estimated state and the flight/ground models of Chapter 8. Short-flight radar systems likewise combine observations with estimation, while hybrid systems can add complementary image information. The useful comparison is which quantities are identifiable under the installed geometry and which uncertainty remains, rather than a claim that one modality measures everything the other models.

5.5 DIY and Open-Source Photometric Systems

PiTrac documents Raspberry Pi cameras, infrared strobes and image processing for launch speed, angles and three-axis spin [63]. Strobed multi-exposure capture can separate effective observation times from the camera's frame rate: pulse duration controls motion blur, while pulse spacing controls displacement, overlap and rotational ambiguity. Its evolving build documentation should be consulted for the actual camera configuration, calibration and software version; a computer-and-camera subtotal is not a complete installed-system cost. Neither a parts list nor agreement between two frames establishes a validated error distribution across balls, speeds and lighting conditions. The shared imaging principles discussed in Section 7.2 establish neither direct product lineage nor rights clearance.

Design implication

Optical launch and club measurements depend on a connected chain: visible features, calibrated geometry, synchronized exposures, defined reference points and justified estimation assumptions. A change anywhere in that chain can look like a change in the golfer's swing. Preserving those definitions and validating across the intended operating conditions makes the resulting motion estimates interpretable; adding a camera alone does not establish accuracy or all club-face capabilities.

Chapter 6

Commercial System Survey

A product survey is useful only if its categories preserve what the instrument observes, what its estimator assumes and what the purchased configuration reports. Table 6.1 summarizes selected architectures using manufacturer documentation checked on October 4, 2026. It is neither an accuracy ranking nor an exhaustive list of current products. Firmware, enabled software, ball type, club markers and installation geometry can change the available outputs.

The measurement hierarchy in Table 2.3 applies to every row. Camera images and radar signals are observations; launch and club states are estimates from those observations; ratios and coordinate transformations are derived quantities; simulated downrange flight is a model prediction. An output advertised as “measured” still requires a definition, an estimator and uncertainty assessment. Conversely, a radar-based face-angle output cannot be assigned to a particular D-plane inverse merely from its sensor category. Where public documentation does not disclose that algorithm, the table does not invent one.

Table 6.1: Documented Launch-Monitor Architectures and Interpretation Limits

System	Observation Architecture	Configuration and Interpretation
TrackMan 4	Synchronized short- and long-range radar with a camera	Optical/radar integration; indoor distance is specified from unit to net, not wholly as ball flight.
TrackMan iO	Radar, high-speed camera and alignment camera	Overhead indoor geometry; club outputs depend on the software package.
FlightScope X3 / X3C	Radar and camera Fusion Tracking	Model-specific setup; X3 documentation includes face impact, and current X3C documentation also lists it.
FlightScope Mevo+	Radar with a built-in camera	Pro and Face Impact Location are software options; neither means adding a previously absent camera.
Garmin R10	Radar	Standard-ball spin requirements differ from RCT-assisted indoor requirements; calculated face angle is not a direct image observation.
Garmin R50	Three cameras	Club sticker required; an impact replay is not by itself evidence of calibrated numerical impact coordinates.
Full Swing KIT	Radar with a video camera	Manufacturer describes ML-enhanced radar; the public account does not establish a camera contribution to every estimate.
Rapsodo MLM2PRO	Radar and two cameras	RPT optical-pattern balls required for reported spin; RCT is a different technology.

System	Observation Architecture	Configuration and Interpretation
Foresight GCQuad / QuadMAX / GCHawk	Four-camera family	Club-marker mode and enabled features matter; do not transfer every model's feature list to the entire family.
Foresight GC3 / Bushnell Launch Pro	Three-camera family	GC3's documented club set is speed, path, attack and smash; face orientation and impact location are not established by that list.
Uneekor EYE XO2	Three overhead cameras	The manual requires club stickers for club data; overhead placement does not imply marker-free operation.
Uneekor EYE MINI	Two cameras	Club Optix supplies impact video; manual/support and product-page feature descriptions need model/version reconciliation.
ProTee VX	Two overhead cameras	Vendor lists sticker-free ball and club estimates, including numerical impact coordinates; these are capability claims.
SkyTrak ST+	Photometric ball tracking with dual-radar club tracking	Manufacturer describes learned club estimation; do not assume every ST MAX specification is identical.
OpenFlight	Radar development architecture	Hardware interfaces and experimental processing are discussed in Chapter B; no universal spin-detection rate follows.
PiTrac	Camera and infrared-strobe development architecture	Configuration and calibration are version-dependent; low-cost parts do not establish validated accuracy.

6.1 Radar and Hybrid Systems

6.1.1 TrackMan 4 and iO

TrackMan describes two synchronized radar subsystems and optical assistance for TrackMan 4 [88, 93]. Its published minimum indoor distance of 4.7 m is measured from the device to the net; part of that distance lies behind the ball. Treating the entire distance as a required ball-flight segment changes the installation claim. The overhead iO has a different geometry and lists 24 GHz radar, a high-speed camera capable of up to 4,600 frames/s and an alignment camera [88]. It is not simply a rear-mounted TrackMan 4 repackaged above the golfer. The listed Home package club output is narrower than the Home Complete/Commercial set, which includes face orientation and impact location. Room-clearance and capture requirements still apply even when a product does not require a long radar flight segment. These specifications establish advertised configuration, not common error bounds between packages or measurement equivalence with other brands.

6.1.2 FlightScope X3, X3C and Mevo+

FlightScope's Fusion Tracking combines radar and image information; Mevo+ already contains a camera used for alignment, video and tracking [16]. The Pro Package enables additional software outputs, rather than converting a camera-free unit into a camera-equipped one. The separate Face Impact Location option is available for Mevo+ without requiring Pro; FlightScope's X3 account describes impact location as included, and its X3C page also lists that feature [14, 15]. Consequently,

a categorical “no impact location” entry for these systems is incorrect. Short-flight spin operation depends on the prescribed ball treatment, spacing and environment; face-impact imaging also has its own lighting requirements. Published patents explain possible sensing methods, but they do not prove that each current model implements every claimed embodiment. The distinction between patent evidence and deployed-product evidence is developed in Chapter 7.

6.1.3 Garmin Approach R10

Garmin lists club-head speed, ball speed, launch angle and launch direction as radar-measured quantities and identifies face angle as calculated [25]. That statement does not identify all details of the internal estimator, nor does it make the remaining displayed metrics independent observations. Its standard-ball guidance ties measured spin to substantial flight distance and ball speed; those conditions should not be imposed unchanged on its separate RCT-assisted indoor mode. The RCT guidance specifies a minimum ball-to-screen distance and sufficient observed rotation, with longer flight sometimes required; an italic spin result indicates estimation [29]. Titleist’s Radar Capture Technology (RCT) uses an embedded radar-reflective structure, unlike the optical RPT pattern. A vendor tolerance or signal-detection percentage also cannot substitute for an independent, condition-specific error distribution.

6.1.4 Full Swing KIT and Rapsodo MLM2PRO

Full Swing describes KIT as an ML-enhanced radar system with 16 reported data points and video capability [24]. The presence of a camera does not establish that its images enter every kinematic estimate. The cited manufacturer technology page does not substantiate a general third-party finding of 1–2% agreement with other launch monitors, so no such accuracy claim is made here.

Rapsodo’s MLM2PRO combines radar with two cameras and requires its RPT-pattern balls to report spin rate and spin axis [68, 69]. RPT denotes Rapsodo Precision Technology; it is not the Titleist RCT radar treatment. The current FAQ lists 15 metrics, including club path and attack angle, but does not list face angle. That feature list therefore cannot support calling the device a demonstrated face-angle estimator. Its impact video, launch estimates and simulated flight remain distinct outputs. Neither a marketing comparison percentage nor an unsupported camera frame rate establishes accuracy across balls, clubs and capture conditions.

6.2 Camera Systems

6.2.1 Foresight Families

The GCQuad manual distinguishes a single-marker club mode from a four-marker mode that adds orientation information; its driver and iron diagrams show a noncollinear layout [21]. Supported GCQuad-family features include face and impact information, but QuadMAX and overhead GCHawk capabilities should be checked against their own model documentation and enabled features [23]. No generic four-camera description establishes the exact spin algorithm, image count per shot or every club metric. The robot comparisons in Chapter 9 concern the tested conditions; they do not establish a universal class-leading accuracy ranking.

GC3 and Bushnell Launch Pro belong to the three-camera family, but branding and subscriptions should not replace model-specific entitlement checks [19, 5]. The GC3 product list specifies club speed, path, attack angle and smash factor. Those outputs do not include the GCQuad’s full

face-orientation and impact-location set. The point-velocity and marker-registration distinctions in Section 5.3 apply even when two units display the same metric name.

6.2.2 Uneekor and ProTee

The EYE XO2 manual describes three cameras and club stickers; its mounting and calibration instructions are part of the measurement configuration [103]. Its optical ball tracking does not make its club tracking sticker-free. EYE MINI is a different, portable two-camera model with impact video. Uneekor’s model-specific support requires club stickers, whereas its current product page calls them optional or recommended and displays a broader club-metric list [106, 102]. That documentation discrepancy should be resolved for the installed hardware/software before relying on numerical face or impact outputs; video replay alone is insufficient evidence.

ProTee VX’s current product documentation explicitly advertises two cameras, sticker-free ball and club tracking, and numerical horizontal and vertical impact coordinates [65]. That is evidence of an advertised feature set, not an independent validation of the underlying image estimator. Overhead placement, learned image interpretation and a large hitting zone each solve different parts of the observation problem. None removes the need for calibration, adequate visibility and a declared face-coordinate convention.

6.2.3 Garmin R50 and SkyTrak ST+

Garmin documents three cameras for the R50 and requires a club-face sticker for club tracking [28, 27, 26]. The manual’s impact-viewing functions should be distinguished from a verified specification of numerical impact coordinates. Its use of cameras instead of the R10’s radar is an architectural difference, not by itself proof of a quantified accuracy improvement.

SkyTrak describes photometric ball tracking and dual-radar club estimation with machine learning for ST+ [74]. This is another reason to avoid a strict camera-versus-radar taxonomy. The existence of another product, ST MAX, does not justify pooling its hardware, software and feature specifications with ST+ without separate documentation.

6.3 Open-Source Systems and Comparison Practice

The OpenFlight-oriented hardware example and PiTrac illustrate radar and strobed-camera development routes [57, 63]. The sensor interfaces, sample cadence and observability limits in Chapter B determine what can be inferred from available data; an experimental spin-detection rate is not a transferable product specification. PiTrac’s evolving documentation likewise needs a pinned build and calibration record before numerical performance can be reproduced. A subtotal for computers and cameras omits other hardware, assembly and validation costs. Neither project should be assigned a universal price advantage or a proven future club-face capability from a parts list or a patent analogy.

Design implication

Preserve device, firmware, enabled features, ball and club preparation, installation, reference points, frames and event times. Assess changes against repeatability, reference uncertainty and model discrepancy. A changed face angle, spin or carry may otherwise reflect the measurement or prediction process rather than the golfer’s movement.

Chapter 7

Patent Landscape

Patent publications expose proposed measurement architectures that can be hard to reconstruct from product brochures. Their technical value lies in the signal being observed, the calibration and inference steps, and the conditions under which an output is identifiable. A disclosure, a granted claim, a commercial implementation and an accuracy experiment are four different kinds of evidence. This chapter compares selected disclosures; Chapter D provides a wider catalog of document identifiers.

Read an abstract as an overview, a description as a set of disclosed embodiments, and each claim as its own combination of limitations. An example camera count, exposure time or receiver arrangement in the specification is not automatically required by every claim. Conversely, a broad title such as “spin measurement” does not mean that a patent covers every method producing that output. Company groupings below aid navigation; they do not replace assignment records or establish which product implements a particular embodiment.

7.1 TrackMan A/S (Interactive Sports Games A/S) — Fredrik Tuxen

TrackMan’s virtual-marking notice contains 47 distinct US patent numbers at the October 1, 2026 check [90]. That dated list is a route into its disclosures, not an exhaustive search of applications, foreign publications or the company’s ownership history.

Spin Frequency and Trajectory-Based Axis. US8845442B2. The disclosure describes radar spectral spacings associated with spin and a separate trajectory-based approach to spin axis [98]. Claim 1 concerns estimating spin frequency from a frequency distance in the modulated radar signal; it should not be paraphrased as requiring all the axis-inversion steps in the description. Spectral structure still needs a defensible mapping from the observed periodicity to ball rotation. Trajectory-based axis inference also depends on the aerodynamic model and the information present in the observed flight segment (Section 4.4.2 and Equation (4.5)).

Target-Referenced Measurements. US8085188B2 / US9857459B2 / US10473778B2. US8085188B2 determines direction deviation relative to an image-designated target; US9857459B2 adds a co-rotated camera/radar arrangement in its first claim. US10473778B2 claim 1 instead correlates radar trajectory data with an image feature and displays the resulting trajectory information. Related coordinate transformations appear in US8912945B2 and US10690764B2: a launch position and image-identified reference define a target-relative frame. These operations change how an observation is related to the scene; they do not by themselves add an independent observation of projectile motion. Sensor and image coordinates must be registered before a target-relative angle has a physical meaning.

A Radar View of Spin Axis. US10850179B2. Claim 1 separates the Doppler spectrum into components, determines their angular positions using a non-collinear receiver arrangement and

identifies a *projection* of the spin axis onto a plane perpendicular to the radar’s line of sight [100]. That projection is not, by itself, an unrestricted three-dimensional axis measurement. Motion compensation, receiver geometry and the distinction between a projected axis and a full axis belong in any explanation of the method. US11446546B2 claim 1 likewise identifies a projected axis; US11938375B2 claim 1 identifies an axis component from phase differences for radar frequency components. The related publications US11446546B2 and US11938375B2 therefore require their own account of the output and receiver geometry.

Tracking, Calibration and Impact. US10989791B2 and US11828867B2 concern radar/imager tracking; US11619708B2 and US12517218B2 address inter-sensor calibration; US10953303B2 describes estimating club impact information from image features. These are distinct inference problems, even when a product presents their outputs together. The similarly titled radar/image documents US10596416B2, US11697046B2 and US12128275B2 identify Topgolf Sweden AB, not TrackMan, in their publication records. In US11619708B2 claim 1, the initial camera track already uses radar range together with image angles. Agreement between that track and the radar track is therefore not a comparison of two wholly independent measurements. US12517218B2 claim 1 selects corrections by minimizing quality-weighted three-dimensional track residuals. Improved correspondence is a calibration objective; agreement alone does not establish absolute accuracy against an independent reference.

Display and Attribution. The abstract of US10315093B2 describes trajectory illustration, while its first claim concerns calibrated simultaneous camera/radar tracking. The related documents show why a shared abstract is not enough to specify a method: US10441863B2 claim 1 recalibrates after a camera field-of-view change, US11135495B2 claim 1 derives part of a striking implement’s trajectory, and US11291902B2 claim 1 includes a three-dimensional launch-vector overlay. Associating tracked balls with launch locations is described in US11086005B2. A convincing video overlay and a correct association with a player are separate requirements from accurate trajectory measurement.

Terrain-Constrained Position. US12186643B2 intersects a calibrated camera-to-ball ray with a three-dimensional terrain model and outputs that intersection as a ball position. The terrain constraint supplies depth information that the image alone lacks. When the observed ball point is above the modeled surface, the ray–surface intersection is a different point; interpreting that intersection as the ball’s unconstrained spatial position introduces a geometric error. An output labeled three-dimensional therefore still needs an account of the geometric assumptions that determined it.

7.2 Foresight Sports, WAWGD and Wintriss Disclosures

The Wintriss/WAWGD documents provide examples of image-based launch measurement and integrated monitor operation. Uneekor’s product marking notice lists several of the same numbers, and the companies announced a license agreement in 2024 [104, 6]. That evidence should not be expanded into an assertion that every Foresight or Uneekor device implements every disclosed feature.

Surface Images. US7292711B2 / US7324663B2. The disclosures describe recovering ball motion from images of surface features, including dimples and non-precision marks [45]. US7292711B2 claim 1 includes compensation for sensor misalignment relative to a calibrated orientation. US7324663B2 claim 1 describes a search over rotations, scaling and correlation scores, together

with coordinate and alignment calculations. Here, “markerless” means that prescribed fiducials need not be applied; it does not mean that the measurement needs no resolvable texture, lighting control or calibration. Correlation quality and rotation ambiguity must be evaluated on the actual images.

Trigger and Placement. US7497780B2 / US7641565B2. The former integrates trigger-zone monitoring, capture, parameter calculation and output. The latter detects ball placement using images and physical attributes. Reliable triggering determines which frames reach a spin estimator; it is part of the measurement chain, rather than merely a display convenience.

Club Motion. US8951138B2. This disclosure addresses club path and face orientation around impact. Its inclusion supports discussion of club measurement methods, not a blanket assertion that four cameras or reflective face dots are universally protected, or that one patent identifies a particular commercial club-data system.

7.3 FlightScope / EDH — Henri Johnson

Phase-Modulated Spin Signals. US9868044B2. Claim 1 derives spin from the period of a demodulated radar component whose portions correspond to apparent speeds above and below a nominal speed [42]. The disclosure’s dielectric-lens explanation and surface-feature examples should be distinguished from that claim’s signal-processing requirements. Symmetry can cause a feature pattern to repeat more than once per revolution; a measured modulation frequency therefore needs interpretation before it becomes a spin rate.

Receiver-Pair Timing. US10775492B2. Claim 1 uses demodulated Doppler signals from receiver pairs in perpendicular planes, their relative time delays and receiver separations to calculate an axis relative to a receiver plane [41]. This supplies rotation-related observations without first fitting a long curved trajectory. It does not establish unconditional indoor accuracy: signal quality, delay estimation, geometry and the stated axis convention remain part of the result. A ratio of delays also needs careful treatment near a vanishing denominator and when resolving angle quadrants; a bare one-argument arctangent is not a complete implementation.

Camera/Radar Tracking. US10338209B2. The document describes calibrated simultaneous tracking and combining camera and radar information [40]. Claim 1 specifically includes using radar data to interpolate a three-dimensional camera trajectory when one sensor fails to supply usable measurements. Recovery through such a gap is an inference step whose uncertainty must be tested, not a second independent observation of the missing interval.

7.4 Acushnet (Titleist) — Optical Measurement Architectures

These documents illustrate several ways to assemble an optical launch monitor [31, 30, 46]. They do not reduce to one fixed camera-and-marker recipe.

Marked Club and Ball. US5501463A. Claim 1 uses at least two cameras and repeated exposures of contrasting areas on both the striking instrument and the object. Image patterns support estimation of club path and orientation. The conversion from marker positions to a club-face quantity requires a defined relationship between those markers and the club geometry.

Launch Conditions to Trajectory. US6500073B1. Claim 1 obtains positions and orientations from images at different times, calculates launch conditions, then calculates a trajectory. Measured initial conditions and the resulting modeled flight must be distinguished: a predicted landing point is not an observed landing point.

Separate Club and Ball Capture. US6758759B2. The first claim places club and ball cameras in vertically separated planes and derives their motion data from repeated images. The minimum in that claim is one camera for each task; describing the entire patent as requiring two stereo pairs would confuse an embodiment with the claim.

Integrated Optical Hardware. US7143639B2. Claim 1 specifies a housing layout with a club camera, at least two ball cameras, illumination, triggering and a teeing aid, arranged for use by both handednesses through rotation of the apparatus. The first claim does not require four cameras.

Three-Dimensional Image Generation. US10668350B2. The disclosure discusses stereo and light-field approaches. Claim 1 specifically includes a light-field camera that captures light intensity and ray direction, followed by reconstruction and kinematic processing. Example frame rates or exposures in a disclosure are design proposals; they do not demonstrate a product's achieved measurement error.

7.5 United States Golf Association — Aerodynamic Experiments

The United States Golf Association's US6186002B1 belongs outside the Acushnet grouping. It describes controlled launches and timed light-screen crossings for aerodynamic coefficient fitting (Chapter D) [51]. Its experimental setup matters: an arbitrary short trajectory need not contain enough information to separate initial spin, wind and uncertain aerodynamic coefficients.

7.6 Korean Ecosystem: Creatz/Uneekor and Golfzon

Uneekor's marking notice connects specific products with listed patent numbers [104]. The Creatz disclosures include US10247553B2, which describes a start sensor for identifying ball placement and starting movement, and US12008770B2, which concerns dimple-based spin estimation. A product's markerless-spin label does not specify its entire algorithm or establish performance across every ball surface.

Creatz's US11191998B2 makes the distinction between observation and inference particularly clear. Claim 1 uses speed- or angle-related quantities to estimate spin when image marks cannot supply it; when mark-based calculation is possible, it also uses such an estimate to correct that calculation. This is more than a fallback for missing observations. The reported spin can depend on both image evidence and a relationship inferred from other ball or club quantities. Validation should distinguish these operating modes and test their errors separately; a plausible corrected value does not establish an independent measurement of rotation.

Golfzon's US12002222B2 provides a different example of this dependence. Claim 1 obtains dynamic loft and backspin through a reference database using observed club/ball motion parameters, then calculates spin-axis and side-spin quantities from motion vectors and specified relationships. These inferred outputs must not be used as independent evidence that the assumed impact relationships hold for a particular shot.

Golfzon's US9242158B2 describes beginning the displayed flight with a first set of ball information and updating it when a second set arrives [111]. This separates feedback latency from completion of the measurement pipeline. Timing examples in a description are not measured latency guarantees. Readers comparing monitors should ask both when an output appears and when the final parameters become available.

7.7 Timing, Association and Environmental Inference

Several disclosures show why a measurement pipeline needs more than a list of sensors. Topgolf's US11335013B2 interpolates timestamped image positions to a common time before combining them with another sensor. Golfzon's US10045008B2 describes alternating camera capture with estimated intermediate images. These methods synthesize information at a requested time; they do not create an additional exposure at that time. Their interpretation therefore depends on the motion between samples, particularly across impact or bounce, where an interpolation rule must account for a rapid change in velocity.

Object association is a separate inference. Topgolf's US11644562B2 extrapolates a trajectory backward and combines position and angular error estimates before assigning a launch origin; when its criterion is unmet, it waits for more observations. Foresight's US10639537B2 uses launch-monitor predictions to choose among observed flight and landing records. A correctly reconstructed flight assigned to the wrong strike still produces incorrect feedback to the golfer. Moreover, agreement with a prediction used to select that flight is partly a selection effect, rather than wholly independent validation of the prediction.

Environmental quantities can also be inferred rather than directly sensed. Topgolf's US12298326B2 compares model and observed accelerations and combines wind estimates from multiple ball trajectories. Because the comparison includes a flight model, aerodynamic-model error can affect the inferred wind; adding trajectories does not automatically remove a shared model error. FlightScope's US11573082B2 instead maps temperature, pressure and humidity to an effective altitude by matching air density to a reference environment. Equal density does not establish equal wind or every other aerodynamic condition, so this equivalent altitude is a defined environmental summary rather than a complete replacement for the atmosphere in a trajectory calculation.

These dependencies matter when a launch monitor is used to test a swing hypothesis. Record which data were observed, which were interpolated, how the shot was associated with a player and which environmental assumptions entered the calculation. An apparent change in inferred spin or predicted distance can otherwise be attributed to the golfer when it partly reflects a change in capture, association or modeling.

7.8 Technical Research Map and Rights Boundaries

The following map identifies engineering questions to investigate. It replaces a technique-wide “free to use” classification: the age or expiration of one document cannot establish clearance for every implementation bearing the same technique name. The USPTO describes a patent as a right to exclude, not permission to practice an invention. Term cannot be computed indiscriminately as twenty years from any date labeled “priority”; the applicable filing basis and other term conditions matter.¹ A design-specific rights review requires the relevant claims, jurisdictions and current records. This technical catalog does not supply that determination.

¹USPTO, *Patent Essentials*, <https://www.uspto.gov/patents/basics/essentials>.

Table 7.1: Selected Measurement Methods and Their Validation Questions

Method	Example Documents	Question to Test
Surface-image spin	US7292711B2; US7324663B2	Are texture, exposure and rotation matching sufficient to resolve ambiguity?
Marked club/ball images	US5501463A; US6758759B2	Are calibration and marker-to-face geometry known at the reported time?
Radar spin periodicity	US8845442B2; US9868044B2	Which spectral or temporal repeat corresponds to one revolution?
Radar axis information	US10850179B2; US10775492B2	Is the result a projection or a full axis, and in which coordinate frame?
Camera/radar fusion	US10338209B2; US10989791B2	Are clocks, coordinates, association and missing-data uncertainty controlled?
Trajectory-informed spin	US12544624B2	How strongly does the result depend on club probabilities and the flight model?
Placement and triggering	US7641565B2; US10247553B2	Does detection reliably capture the interval needed for the requested output?
Staged display	US9242158B2	Which values are provisional, and when does the displayed trajectory change?

Key point

A launch monitor is a chain of observations and inferences. Radial speed, image displacement, surface rotation, launch conditions and predicted flight occupy different places in that chain. Combining sensors can add information, but only after timing, coordinates and object identity agree; it does not turn repeated use of the same data into independent evidence.

For example, Topgolf's US12544624B2 uses trajectory observations, spin candidates and probabilities associated with a likely club. That approach differs from tracking rotating surface features. A plausible spin value may partly reflect the prior model, so evaluating both methods requires experiments that expose their different ambiguities.

When using these outputs to investigate a swing, retain the measurement frame, time interval, calibration and model assumptions. Ball launch conditions and club-delivery kinematics constrain terminal states; they alone do not identify the golfer's joint torques, force history or unique movement strategy. Technical rigor means carrying those information limits through the argument, rather than letting a precise display imply a uniquely determined cause.

Chapter 8

Ball Flight Models and Trajectory Estimation

A launch monitor connects three different problems: observing motion, estimating a launch state, and predicting the flight that follows. An indoor carry value is principally a prediction from a short observed segment; an outdoor system may also observe much of the trajectory. The same aerodynamic model can be used forward to predict carry or inversely to infer launch conditions from later observations. Neither direction makes the model exact, and using one model in both directions does not provide independent validation.

8.1 Equations of Motion and Coefficient Conventions

Let $\mathbf{p}$ and $\mathbf{v}$ denote ball-centre position and velocity in a ground-fixed frame, and let $\mathbf{w}(\mathbf{p}, t)$ be the local wind velocity. Define air-relative velocity $\mathbf{v}_a = \mathbf{v} - \mathbf{w}$, airspeed $V = \|\mathbf{v}_a\|$, cross-sectional area $A = \pi R^2$, and dynamic pressure $q = \rho V^2/2$. Radius R and mass m should describe the ball being modeled; values near 21.34 mm and 45.93 g are illustrative inputs, not identities for every ball. A quasi-steady drag-and-Magnus model is

$$\dot{\mathbf{p}} = \mathbf{v}, \quad m\dot{\mathbf{v}} = -qAC_D \frac{\mathbf{v}_a}{V} + qAC_L \mathbf{e}_L + m\mathbf{g}, \quad \mathbf{e}_L = \frac{\boldsymbol{\omega} \times \mathbf{v}_a}{\|\boldsymbol{\omega} \times \mathbf{v}_a\|}. \quad (8.1)$$

Here $\boldsymbol{\omega}$ is the physical spin vector and C_L is a *signed* coefficient in the displayed direction. This convention assigns lift magnitude $qA|C_L|$; a formula using $\hat{\boldsymbol{\omega}} \times \hat{\mathbf{v}}_a$ instead has an additional factor equal to the sine of the angle between spin and airflow. Coefficients fitted under one convention cannot be moved to the other without that conversion. With the book's right-handed axes (x forward, y up, z right), positive z backspin and forward air-relative motion give upward lift when $C_L > 0$. The force law is an approximation to the mean aerodynamic force; it does not represent every unsteady or orientation-dependent force on a real dimpled ball.

The unit vectors in Equation (8.1) require explicit limiting rules. At zero airspeed, use the model's zero-force limit rather than divide by V . At zero spin or parallel spin and airflow, the cross-product direction is undefined. A symmetric Magnus model sets that lift contribution to zero and requires its coefficient to approach zero consistently; an empirical fit measured away from these limits does not establish its own continuation to them. For drag force $\mathbf{F}_D = -qAC_D \mathbf{v}_a/V$ with $C_D \geq 0$, the air-relative power is nonpositive: $\mathbf{F}_D \cdot \mathbf{v}_a = -qAC_D V$. It can nevertheless do positive work in the ground frame when a sufficiently fast tailwind accelerates the ball. Lift in this model is perpendicular to air-relative velocity, not necessarily to ground-relative velocity.

Useful dimensionless inputs are Reynolds number $\text{Re} = 2RV/\nu$ and spin ratio $S = R\|\boldsymbol{\omega}\|/V$, with kinematic viscosity ν . The angle α between spin and airflow, ball construction and surface condition may also matter. A model interface should therefore declare its dependencies and calibration domain, schematically

$$C_D = f_D(\text{Re}, S, \alpha; \mathbf{b}), \quad C_L = f_L(\text{Re}, S, \alpha; \mathbf{b}), \quad (8.2)$$

where $\mathbf{b}$ denotes a versioned set of ball-specific parameters. This is an interface definition, not a numerical fit attributed to a particular author. Fits obtained with spin perpendicular to the airflow need an additional justified assumption before use at general α .

Spin evolution needs its own equation, for example $I\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_a$ for an idealized ball with constant scalar moment of inertia I and aerodynamic torque $\boldsymbol{\tau}_a$. An empirical decay law $\dot{\boldsymbol{\omega}} = -k\boldsymbol{\omega}$ further assumes no change in axis direction from torque, and k may depend on speed and spin. Even when physical spin decays, spin ratio can rise as the ball slows. For fixed R , nonzero spin and positive V ,

$$\frac{\dot{S}}{S} = \frac{\boldsymbol{\omega} \cdot \dot{\boldsymbol{\omega}}}{\|\boldsymbol{\omega}\|^2} - \frac{\dot{V}}{V}. \quad (8.3)$$

Thus integrating S with a physical-spin decay constant is generally inconsistent unless the airspeed term is included. For a varying wind field, $\dot{V} = \mathbf{v}_a \cdot \dot{\mathbf{v}}_a / V$ uses $\dot{\mathbf{v}}_a = \dot{\mathbf{v}} - \partial_t \mathbf{w} - (\mathbf{v} \cdot \nabla) \mathbf{w}$, including the wind variation along the ball's path.

8.2 What Aerodynamic Data Establish

Bearman and Harvey's wind-tunnel study compared particular spinning, dimpled ball models [1]. It motivates accounting for Reynolds number, spin and surface geometry; its tested dimple comparison is not a universal ranking of modern balls. Smits and Smith [75] and Quintavalla [66] provide further modeling approaches. Quintavalla's author description identifies six-term lift and drag models fitted to Indoor Test Range measurements over driver trajectories. That is a calibration strategy, not evidence that any six-term polynomial is accurate outside its fitted conditions or intrinsically superior to wind-tunnel data. This review does not prescribe numerical coefficients from an unverified transcription of those models; an implementation must establish the original equations and their domain before adopting them.

An implementer needs the actual coefficient data, normalization, units, fitted ball, validity range and spin-torque law, not just a model name. Preserve source tables or reproducible digitization and report extrapolation beyond the calibration domain. Fit aerodynamic parameters using independent launch observations and multiple trajectories with varied speeds and spins. Otherwise a biased launch angle, unknown wind or a spin error can be absorbed into apparently plausible lift and drag coefficients. US6186002 describes determination of coefficients from measured trajectories [51]; a patent's disclosure supplies a method description, not independent evidence of accuracy or a legal clearance determination.

Design implication

Version the coefficient data, spin law, atmosphere, wind, ground definition and numerical solver together. Validate predicted trajectories on shots excluded from fitting. Agreement with another simulator is a comparison of predictions; it does not establish that either prediction matches the physical flight.

8.3 Numerical Integration and the Landing Event

Fourth-order Runge–Kutta is one possible integrator, but its order alone does not establish adequate carry accuracy. Check convergence as the step is reduced, resolve coefficient transitions, and locate landing between integration steps. Define landing as a descending intersection of the ball-centre trajectory with a declared height surface, excluding the launch-time root. Returning to launch

elevation is only one convention; tee height, uneven ground and a surface referenced to the ball's contact point require explicit treatment. Bounce and roll require additional contact and terrain models and are not part of airborne carry.

Landing-time uncertainty matters even if the trajectory integrator is exact. Let a be a scalar uncertain input and let the landing event satisfy $G(\mathbf{p}(t_*, a), t_*, a) = 0$. For fixed-time sensitivity $Y = \partial \mathbf{p} / \partial a$, implicit differentiation gives

$$\frac{dt_*}{da} = -\frac{G_{\mathbf{p}}Y + G_a}{G_{\mathbf{p}}\mathbf{v} + G_t}, \quad \frac{d\mathbf{p}_*}{da} = Y + \mathbf{v} \frac{dt_*}{da}. \quad (8.4)$$

All terms are evaluated at the same landing event. The formula assumes differentiability and a transverse crossing, $G_{\mathbf{p}}\mathbf{v} + G_t \neq 0$; a nearly tangential crossing is poorly conditioned. For a vector of uncertain inputs, differentiate with respect to each input component in turn. For an output $c(\mathbf{p}_*, t_*, a)$, also include the explicit terms $c_t dt_*/da + c_a$ along with $c_{\mathbf{p}} d\mathbf{p}_*/da$. For downrange position $c = x_*$ with a fixed origin, those explicit terms vanish; the landing-time contribution is already included in $d\mathbf{p}_*/da$.

A vacuum trajectory makes the missing term easy to see. For level launch and landing, $x = v_x t$, $y = v_y t - g_0 t^2/2$, and $v_y > 0$, the descending root is $t_* = 2v_y/g_0$. Downrange carry is $c = 2v_x v_y/g_0$, so $\partial c / \partial v_y = 2v_x/g_0$. At fixed time, however, $\partial x / \partial v_y = 0$. Ignoring the change in landing time would therefore erase the entire vertical-launch-speed contribution to carry uncertainty in this simple example.

8.4 Trajectory Estimation and Filtering

A useful physical state is $\mathbf{x} = (\mathbf{p}, \mathbf{v}, \boldsymbol{\omega})$, augmented only with additional parameters that the data can constrain. For a fixed radar at position $\mathbf{s}$, range is $r = \|\mathbf{p} - \mathbf{s}\|$ and radial velocity is $\dot{r} = \mathbf{n} \cdot \mathbf{v}$, with $\mathbf{n} = (\mathbf{p} - \mathbf{s})/r$. Angles require the radar's own axes; camera pixels require calibrated projection, pose and timing. Different sensors therefore need different measurement functions, Jacobians and noise models, even when they share one state estimator. Observation windows, processing delays, association of returns to ball or club, and correlated errors belong in that model; treating every output as a simultaneous independent point measurement overstates information.

Extended or unscented Kalman filtering and subsequent smoothing provide approximate state estimates conditional on the process and observation models [73]. Backward estimation to ball separation can use the observed arc, but its uncertainty grows with uncertain timing, model error and weak geometric information. Including a spin-axis state does not make that axis observable. A short arc can be consistent with multiple combinations of launch velocity, wind, lift coefficients and spin; a tight prior may select one combination without the observations independently identifying it. Check sensitivity rank and conditioning, residual structure and performance on withheld data, rather than equating a small filter covariance with demonstrated accuracy.

For the K-LD7 example, use the speed-dependent measurement cadence documented in Chapter B; a roughly 29 ms cycle at the stated 100 km/h example does not supply 10–20 independent detections in 50 ms. Processing a complete burst can help only to the extent that it contains useful, correctly associated observations. Neither a smoother nor a confidence weight repairs an incorrect timestamp or a missing observation direction.

8.5 Indoor Extrapolation and Coupled Uncertainty

Let $\mathbf{a}$ collect launch state, aerodynamic parameters, atmospheric inputs and ground geometry, and let $c(\mathbf{a})$ be a carry or lateral-distance prediction evaluated at landing. For locally smooth behavior and sufficiently small uncertainty,

$$\text{Var}(c) \approx J_c \Sigma_a J_c^\top, \quad J_c = \frac{dc}{d\mathbf{a}}. \quad (8.5)$$

The covariance Σ_a must retain correlations, including those introduced when launch and aerodynamic parameters are fitted together. This expression covers uncertainty in the declared inputs, not every discrepancy between the chosen force law and reality. Model discrepancy needs separate evidence from held-out trajectories; combine it probabilistically only after stating its dependence on the input errors. For nonlinear, multimodal or threshold-sensitive predictions, propagate representative joint samples and report the resulting distribution rather than only a local Gaussian approximation.

There is consequently no universal conversion from 300 rpm of spin error, two degrees of axis error, or a 0.01 change in drag coefficient to a fixed number of yards. Report the baseline launch, ball model, atmosphere, perturbation and landing convention with any numerical sensitivity. Likewise, tilting spin changes lift direction only under stated geometric and coefficient assumptions; lateral displacement integrates that changing force over the flight and is not a universal percentage of carry per degree. An optical spin observation can constrain an indoor extrapolation, but its calibration and rotation ambiguity still need validation. The evidence question is which uncertain inputs control this shot's prediction and how well they were observed, not which sensor family wins by definition.

Chapter 9

Accuracy: The Independent Evidence

Accuracy is a property of a specified quantity, instrument configuration and operating condition. Ball speed at separation, club speed at a face point before impact, and predicted carry are different measurands with different validation requirements. A study can establish useful agreement for some of these without validating every number on a display. The evidence below separates comparisons against an independently characterized reference, repeated-swing reliability, and agreement between commercial devices.

9.1 Comparison Against an Optical Reference

Leach, Forrester, Mears and Roberts [47] recorded 240 shots from eight golfers using a driver, 7-iron and utility wedge, comparing TrackMan Pro IIIe and Foresight GC2+HMT with a four-camera 5,400 Hz GOM system. The experiment used the then-current software in January 2015. The authors characterized the reference with a calibration artefact, position tests, simulated noise propagation and a separate spinning-sphere experiment. Their simulated ball-parameter uncertainties were 0.06 mph for speed, 0.01° for launch angle and direction, and 14 rpm for spin; the spinning-sphere comparison gave a 9 rpm uncertainty. These are setup-specific estimates under the paper’s stated uncertainty convention, not proof of an error-free reference. The same propagation experiment was not performed for the club quantities.

Ball parameters generally agreed more closely than club parameters, but the details resist a simple radar-versus-camera ranking. Table 4 of the paper gives pooled club-speed median differences (device minus reference) of -1.1 mph for TrackMan and $+2.8$ mph for Foresight, with interquartile intervals of $[-1.9, 0.1]$ and $[1.4, 4.5]$ mph. TrackMan’s often-quoted -0.4 mph club-speed median is the *driver-only* result in Table 5. Its 0.0° launch-angle median is also driver-only; the pooled median is 0.1° . Pooling or switching clubs changes the question being answered.

The paper’s “research grade” fractions also need their original definition. The tolerance interval was centered on each sample’s *median difference*, rather than on zero difference from the reference. Thus the reported 54% of TrackMan and 29% of Foresight club-speed values within the ± 1 mph research band describe spread after centering. They are not the fractions of all attempted shots with absolute club-speed error below 1 mph. The thresholds were selected by the authors for research and coaching purposes; they are not a universal instrument standard. Report bias, spread, tolerance definition and denominator together.

Availability was a separate limitation. TrackMan returned ball and club-speed data for about 98% of trials, but the other club parameters for 62% overall and only 19% of utility-wedge trials. Foresight tracked the ball in 90% and the club in 75% of trials. The simultaneous laboratory arrangement required compromises in lighting, markers and nearby equipment that the authors identified as possible interference sources. These return rates characterize that experiment, not every current model or installation. TrackMan spin values flagged as calculated were excluded from the analysis.

Several disagreements also concern definitions. The reference club velocity used a manufacturer-

supplied centre-of-gravity location, while a launch monitor may track a different point on a rotating head. The study compared different face-normal definitions for the two devices and could not exactly reproduce all vendors' impact-time conventions. Such differences are physically consequential, not merely interchangeable labels. In this experiment, some impact-model-derived orientation results agreed more closely than optical orientation results. The lesson is to validate each estimator and measurand; “measured” does not automatically mean accurate and “derived” does not automatically mean poor.

9.2 Reliability Across Repeated Swings

Bishop et al. [2], in a 2023 journal volume published online in 2024, studied 13 male golfers performing ten shots with their own driver in each of two indoor sessions. The paper reported high within-session intraclass correlation coefficients (ICCs) for club speed (0.99) and ball speed (0.97–0.99), while spin ICCs were 0.60 and 0.02; the between-session spin ICC was 0.15. These findings concern repeatability of the combined golfer, protocol and instrument. Without a simultaneous independent spin reference, they do not isolate radar spin error from actual variation in impact and ball launch. Nor does a high speed ICC establish a small absolute speed bias.

An ICC compares variance components and depends on the variation among the subjects or shots being compared. In a simple random-intercept model, it has the form $\sigma_b^2/(\sigma_b^2 + \sigma_w^2)$, where σ_b^2 is between-subject variance and σ_w^2 includes within-subject variation and measurement error. This schematic formula is not a substitute for specifying the fitted ICC model, agreement definition and single- or average-measurement target. A more heterogeneous group can produce a higher ICC without any improvement in the instrument. For monitoring one golfer, absolute uncertainty and the repeatability of the chosen session summary matter alongside relative reliability.

The Bishop tables label spin dispersion in degrees per second and the abstract mislabels carry dispersion in mph, while its tables use yards for carry. Do not silently reinterpret the spin units as rpm or propagate the carry typo into a device specification. The source supports the reported ICC comparison, but the ambiguous spin units limit reuse of its dimensional error figures.

The uncertainty of a change also involves both measurements. For two session estimates with error standard deviations s_1, s_2 and error covariance c_{12} ,

$$s_{\Delta}^2 = s_1^2 + s_2^2 - 2c_{12}. \quad (9.1)$$

Under an independent, equal-precision, approximately normal error model with known error standard deviation s , a two-sided 95% noise threshold is about $1.96\sqrt{2}s$, not $2s$. Estimating that standard deviation from limited data introduces additional uncertainty and requires a corresponding interval method. This derivation assumes that s describes the actual session estimator being compared and excludes systematic drift. It does not make every larger change practically important or prove that training caused the change. Repeated-swings variability, alignment changes and instrument error all need consideration when using such a threshold.

9.3 Agreement Between Commercial Monitors

Bliss and Langdown's full paper [4] is available, replacing the earlier abstract-only account. Published online in December 2025 in a 2026 volume, it compares Mevo+ with TrackMan 4 indoors using one golfer and three clubs. Of 118 driver, 118 seven-iron and 174 pitching-wedge attempts, 58, 80 and 69 were retained after exclusions for missing fields, estimated spin or severe mishits: approximately

49%, 68% and 40%. These are retained analysis fractions, not either device’s standalone detection rate. Table 2 reports a seven-iron ball-speed consistency ICC of 0.996 alongside a mean TrackMan-minus-Mevo+ difference of -3.03 mph; its absolute-agreement ICC is 0.90 with a wide interval from -0.02 to 0.97. The protocol used mean-rating ICCs and reported both consistency and absolute agreement. Results describe the selected paired observations from that golfer and setup; they do not establish error against an independent physical reference or performance on the excluded attempts. The paper’s criterion of 95% of differences inside computed 95% agreement limits does not specify a practically acceptable error width.

More generally, consistency can remain high despite an additive calibration offset. An average-measures ICC targets an average of ratings, rather than the reliability of one instrument reading; its value should not be quoted as a single-reading accuracy specification. Limits of agreement and confidence intervals answer different questions: the former describe the spread of paired differences, while the latter quantify uncertainty in estimated statistics. For approximately normal differences, $\bar{d} \pm 1.96s_d$ estimates population agreement limits; their width must be compared with a tolerance chosen for the intended use. Merely finding most observations inside those fitted limits cannot establish interchangeability.

If two devices measure the same scalar measurand with errors e_A and e_B , the variance of their difference is

$$\text{Var}(e_A - e_B) = \text{Var}(e_A) + \text{Var}(e_B) - 2\text{Cov}(e_A, e_B). \quad (9.2)$$

Agreement alone cannot identify both devices’ error variances or expose a bias they share. This reasoning presumes matched reference points, timing and definitions; otherwise the physical measurand difference contributes too. Comparing two indoor carry predictions adds another layer because both outputs depend on launch estimates and flight models.

9.4 Robot Tests, Specifications and Causal Interpretation

Robot and industry tests can be useful when the protocol and raw paired observations are available. A smaller standard deviation of displayed spin over robot shots is not, by itself, evidence of smaller spin-measurement error. True impact variation, ball differences, rejection rules, averaging and output smoothing can all change that standard deviation. A constant but biased output would have excellent repeatability and poor accuracy. Comparing these output standard deviations cannot establish a universal photometric advantage without an independent reference and an uncertainty analysis of the test. The industry reports cited elsewhere in this review [34, 56] should be read as protocol-specific comparisons, not replacements for that analysis. Likewise, a manufacturer’s tolerance requires its conditions, confidence convention and parameter definition before it can be compared with a study’s spread statistic.

9.5 A Validation Protocol That Supports the Intended Claim

1. **Define the measurand.** Specify units, coordinate frame, reference point, event time and whether carry is observed or simulated. For club speed, a constant per-club offset is not a substitute for transforming velocities between points on a rotating rigid body (Section 2.1.1 and Chapter E).
2. **Characterize the reference.** Calibrate space and time and document reference uncertainty over the tested range. A consumer monitor can be a useful comparator, but agreement with it does not certify physical accuracy. RPT optical markings and RCT radar-reflective

construction serve different measurement routes; choose balls supported by each instrument's actual spin mode (Chapter B).

3. **Retain every attempt.** Report attempted shots, returned values, rejected values and estimated fallbacks separately for each parameter and condition. Analyze conditional accuracy among returned observations alongside total availability. Missingness may depend on club, speed, strike location and spin; neither dropping failures nor assigning them zero error is defensible.
4. **Separate calibration from validation.** Freeze calibration, model versions and rejection criteria before testing held-out shots, golfers or sessions appropriate to the deployment claim. If a bias correction is fitted, report its uncertainty and validate it independently.
5. **Report effect sizes and uncertainty.** Show paired differences, bias, spread, outliers and uncertainty intervals in physical units, stratified by relevant conditions. Account for repeated shots within golfers and sessions when estimating uncertainty; hundreds of shots from one person are not hundreds of independent golfers. Use predeclared tolerances tied to the coaching, fitting or research decision.

Design implication

Validation follows the whole chain from observation to decision. Timing and geometry determine what is observed; the estimator determines what is inferred; the flight model determines what is predicted. Log those distinctions with the uncertainty and version of every reported quantity, and preserve replayable observations where feasible. That makes a disagreement investigable without assuming that the more expensive device, the smaller variance or the label “measured” identifies the correct answer.

Chapter 10

Design Guidance for Implementers

A useful design starts with the quantity the user needs and works backward to the observations required to estimate it. Ball launch, club delivery and simulated carry share data, but they are not interchangeable validation targets. This chapter connects the sensing geometry of the earlier chapters, the implementation conditions of Chapter B, and the evidence standard of Chapter 9. The resulting capability tiers are proposed development stages with acceptance conditions, not claims that a parts list already delivers commercial performance.

A possible starting architecture combines a continuous-wave Doppler module, one or more angle-sensitive radar modules, a synchronized impact trigger and an embedded host. The OPS243-A and K-LD7 are examples discussed elsewhere in the review, not interchangeable building blocks: their observables, latency, sampling and access to raw data differ. Demonstrating a valid radial-speed channel is a useful first milestone. Calling this arrangement the cheapest accurate monitor, or equivalent to a commercial radar architecture, would require evidence about costs and performance that the component list does not supply.

10.1 Ball Data: Match Each Claim to Its Observations

1. **Ball speed.** A Doppler observation constrains radial velocity, not automatically total ball speed. Separate ball returns from club, shaft and multipath returns, and establish calibration, aliasing limits and timing before claiming an accuracy tolerance. For a correctly associated target with radial velocity $v_r = V \cos \beta$, where β is the angle between velocity and the sensor's line of sight, the local correction is

$$V = v_r \sec \beta, \quad \delta V \approx \sec \beta \delta v_r + V \tan \beta \delta \beta. \quad (10.1)$$

Use radians for the angle perturbation and retain covariance between the inputs when propagating uncertainty. The correction becomes ill-conditioned near perpendicular motion. The relevant β depends on sensor position and the instantaneous trajectory; it is not simply the vertical launch angle in every installation.

2. **Launch angles.** Fit a trajectory only when the burst contains enough correctly timed and associated observations for the proposed state. Filtering and smoothing can combine those observations, but cannot create information that the cadence or viewing geometry did not provide (Section 8.4). Compare the resulting launch angles against independent observations at a declared separation time, and test the benefit over a simpler estimator on held-out shots.
3. **Spin rate.** Treat harmonic selection, surface-pattern ambiguity, observation duration and signal-to-noise ratio as explicit parts of the estimator (Section 4.4.2). No universal 50–60% detection ceiling follows from radar physics. Longer windows can improve spectral resolution while also admitting changing geometry, acceleration and additional targets; comb or cepstral methods need tests for missed fundamentals and integer-multiple aliases. A fallback from club, speed or loft priors is an impact-model prediction and must remain labeled as such.

Report its validation separately from successful rotation-sensitive observations. *Disclosure reference:* US8845442 describes radar spin processing; the technical map in Section 7.8 does not determine clearance or a technique-wide expiry date.

4. **Spin axis.** Short flight segments may provide too little trajectory information to separate axis, aerodynamic coefficients and launch-condition errors reliably. Rotation-related observations from receiver geometry provide another disclosed route (Section 7.3), but require suitable hardware and validated signal processing. Optical rotation estimates and explicitly labeled impact-model estimates are further options; their observation requirements and error sources differ. Retain competing rotation hypotheses or withhold an axis when the observations do not resolve them.

10.2 Club Data: Define the Point, Frame and Time

1. **Club speed.** Define the reference point and event time before processing a pre-impact spectrum (Section 2.1.1). For a rigid head, velocities at two points satisfy $\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times (\mathbf{r}_B - \mathbf{r}_A)$. A fixed spectral percentile may be a repeatable signal feature, but it does not by itself identify either point or remove the rotational contribution. Reflectivity, viewing direction and shaft returns can change the percentile without the same change in the declared head-point speed. Validate the mapping against tracked head motion over varied clubs and strikes. Smash factor is a useful diagnostic only with compatible ball and club definitions; an idealized impact-model bound is not a universal hard rejection threshold (Section 3.2).
2. **Club path and attack angle.** These describe the direction of velocity at the chosen point and time, derived from sufficient trajectory information in a calibrated frame. A scattering-centre angle is not itself either quantity. Alignment, association, differentiation, reference-point motion and timing all contribute uncertainty. Check them separately rather than assuming alignment always dominates or that adding a radar makes path and attack angle direct observations.
3. **Face angle.** The conditional D-plane inverse in Equation (4.8) is a possible estimator, not a verified description of every commercial algorithm. For a planar model $\ell = wf + (1 - w)p$ with nonzero weight w , launch direction ℓ , face angle f and path p , first-order propagation gives

$$\delta f \approx \frac{\delta \ell}{w} - \frac{1 - w}{w} \delta p - \frac{\ell - p}{w^2} \delta w. \quad (10.2)$$

The launch-to-face sensitivity $1/w$ is approximately 1.15 for $w = 0.87$ and 1.32 for $w = 0.76$; neither is a universal amplification. Include covariance, weight uncertainty and model discrepancy, especially when strike location and gear effect violate the centered-impact approximation (Section 3.1.1). If those conditions are unknown, enlarging uncertainty or withholding the result is more defensible than presenting the inverse as a face observation. A shared forward/inverse software module still requires independent validation; algebraic consistency alone is insufficient.

4. **Impact location and orientation.** Club and ball imaging offers a concrete route to locating contact relative to the face. The Acushnet and Wintriss disclosures (Sections 7.2 and 7.4) illustrate measurement architectures, rather than complete validated implementations or blanket rights clearance. Face fiducials can constrain orientation through calibrated image geometry and a known marker-to-face relationship. Camera count alone does not establish

three-dimensional observability, visibility at impact or achieved accuracy. Account for lens distortion, exposure blur, strobe timing, marker placement and the relation between the reconstructed surface and the face normal being reported.

10.3 Capability Tiers and Acceptance Conditions

Each stage adds a testable capability to the preceding system. An unsuccessful estimate at one stage must not be silently replaced by a more restrictive prior and then relabeled as a new measurement.

1. **Tier 1 — validated radar observations.** Establish calibrated radial velocity and whatever directional information the selected hardware actually provides. Synchronize the trigger and observation windows, define ball/club association and preserve raw or sufficiently detailed intermediate data for replay. Add spin or impact-model outputs only with provenance and separately tested failure criteria. Acceptance requires a declared operating range, paired comparison results and per-parameter availability, rather than merely reproducing a consumer monitor’s list of display fields.
2. **Tier 2 — optical rotation and impact constraints.** Add synchronized imaging where it addresses an identified ambiguity or missing observable. Calibrated image sequences can constrain launch direction and surface rotation; impact location also requires an adequate view of the ball and club near contact. A global shutter avoids row-to-row exposure timing differences but does not eliminate motion blur; illumination and exposure duration remain design variables. Radar can contribute speed and triggering, while optical observations can distinguish rotation hypotheses. Acceptance includes occlusion, marker symmetry, timing and calibration tests across the intended shot range, with costs measured for the actual implementation.
3. **Tier 3 — validated joint estimation and prediction.** Fuse complementary observations with sensor-specific measurement functions, timestamps and covariance models. Shared calibration or processing can correlate their errors; counting the same information twice produces unjustifiably narrow uncertainty. Version the flight model and propagate launch, aerodynamic and environmental uncertainty to the landing event (Chapter 8). Acceptance requires checks that fusion improves held-out performance and uncertainty calibration, including prediction error on independently observed trajectories. There is no universal yardage equivalent for a fixed coefficient change.
4. **Tier 4 — observable club motion.** Recover the club’s pose and velocity from an observation geometry that supports the requested rigid-body state, then evaluate speed, path and attack angle at the declared point and time. The screw-theoretic formulation in Chapter E provides the kinematic framework, but its rank and ambiguity analysis is part of the design requirement. For a single monostatic radar phase centre, resolved Doppler rows alone do not independently determine all six instantaneous twist components; additional viewpoints, temporal position information or other observations are needed under stated motion assumptions. Custom chirps and an $SE(3)$ smoother do not remove that ambiguity by themselves. An instantaneous screw axis also becomes poorly conditioned near zero angular speed and is not automatically a global swing plane. Optical face and impact measurements can be combined with the recovered motion, provided their frames and event times are aligned. Acceptance requires reference-point, pose, twist and derived-axis validation, including the configurations in which the output must be withheld.

10.4 Reporting Honesty as a Design Feature

A reported quantity should carry its definition, provenance, uncertainty and relevant model version. The hierarchy in Table 2.3 organizes dependence on observations and assumptions; it is not an accuracy ranking. Both a camera pose and a radar trajectory are estimates from sensor observations, and each can be wrong for different reasons. The independent evidence in Chapter 9 shows why return rate and agreement must be reported together, and why selected-shot precision cannot stand in for accuracy on all attempts.

A useful shot record distinguishes observed channels, model-conditioned estimates, simulated outcomes and prior-based fallbacks. Preserve the coordinate frame, reference point, event time, calibration identifier, ball configuration and rejection reason alongside the value. If an uncertainty interval is conditional on a fixed aerodynamic or impact model, state that condition. Do not manufacture an uncertainty number where its calibration has not been tested.

These distinctions help the golfer interpret changes in the swing. A change in predicted carry can originate in ball launch, atmosphere or software; a change in club speed can originate in a different reference point; an apparent face change can originate in the assumptions of an impact inverse. The design objective is to make those alternatives testable with synchronized observations and a documented chain of inference. That is what turns a plausible display into an instrument that can support research and coaching decisions.

Chapter 11

Governed Analytics and Validation Program

The distinction between a useful analytics platform and a scientific vendor comparison is the evidence design. A large collection of launch records can support exploratory model checking without supporting the stronger claim that one device disagrees with another. At the revisions pinned below, the repositories associated with this review use a two-release program with a versioned data contract, explicit use eligibility, and fail-closed statistics. The program is tracked by the public Tools program epic. UpstreamDrift is the canonical generic contract and evidence-envelope authority. Tools owns the current specialized grouped estimators: Python is the desktop calculation authority and React is a contract-tested browser twin. Those grouped operations are not yet UpstreamDrift API operations, so that boundary is published rather than obscured by a cross-service parity claim.

11.1 Qualified Corpus Snapshot

The controlled authority snapshot contains 261,666 rows from 27 traceable sources. Size alone is misleading. Only 13,855 rows contain finite, non-imputed values for all five primary launch inputs (ball speed, launch angle, launch direction, spin rate, and spin axis). Before the per-source minimum-size gate, 13,647 rows also contain carry, 13,532 lateral outcome, 9,850 descent angle, and 9,283 flight time. The strict model cohort contains 13,843 rows from 15 source families with complete launch inputs and at least one supported observed outcome. Each outcome comparison then uses its own eligible subset; 13,843 is not the sample size for every metric. Eligibility is evaluated by source, metric, provenance, and intended use; a blank field is represented as *unavailable*, never silently filled merely to increase sample size.

The largest source is a 236,590-row ShotLink-derived table. It has no complete five-input launch rows and no observed carry values. Its rights and transformation lineage also require separate approval. It is therefore quarantined: the source is **not eligible for vendor-model training**, public redistribution, or vendor-performance claims. It may be retained internally for provenance review and for analyses that are independently shown to match its permitted use. This rule prevents a large but unsuitable table from dominating the apparent evidence base.

At the present qualification gate, no vendor-labelled surrogate is currently eligible for production training. The corpus contains 11,699 TrackMan-labelled rows and 9,298 strict launch-input rows, but it lacks a trusted repeating player or session grouping. Foresight has four rows (two strict); FlightScope has 2,794 rows but no strict five-input rows. The historical TrackMan surrogate used unique shot identifiers as partitions, which cannot protect against player, session, or source leakage; it is retained for audit with status `retired_non_group_safe` and cannot be loaded, queried, or retrained. A future model that passes the group-safe gate would be described as *vendor-comparable on its qualified source*, not as a reproduction or certification of proprietary vendor software.

Key point

Every displayed statistic carries its source identity, row-selection rule, units, method version, and backing row references. An unavailable result is a valid result: it records why the requested inference is not supported.

11.2 Release A: Qualified Analytics Platform

Release A delivers the professional analysis surface on data that passes the declared gate. Its scope includes unit-aware scatter and distribution views, model residuals, arbitrary variable relationships, within-player and between-player covariance, session trends, uncertainty, exportable plots, and reproducible project files. Player-level analysis requires an explicit player identity column. Identity is never inferred from session, club, row order, filename, or directory layout; missing and whitespace-only identities are excluded and reported.

UpstreamDrift's versioned contract returns estimates, uncertainty, warnings, source lineage, backing-row references, and machine-readable unavailable states for its registered generic analyses. Tools supplies PyQt and React/Vite views with parity in variable choices, units, formulas, backing-data export, persistence, and explanatory text. The specialized grouped estimators for covariation and longitudinal synthesis are currently released within Tools, with Python as the desktop authority and TypeScript as a tested browser twin; they are not yet UpstreamDrift API operations. This bounded duplication is a tracked architectural seam, not evidence that two independent implementations are automatically equivalent.

Exploratory all-pairs scans report multiplicity warnings; a warning does not control a false-positive rate. Within-player analysis separates raw pooled association, player-mean-centred association, between-player association, and per-player estimates; these quantities answer different questions and must not be substituted for one another. Correlation and principal-component loadings are descriptive, not causal. Predictive work uses grouped splits whenever a trusted player or session identity exists and otherwise states that leakage-resistant grouping is unavailable. The split must match the intended generalization: holding out sessions from known players does not test prediction for previously unseen players or sources.

Strokes gained is similarly gated. A result labelled *strokes gained* requires a cited baseline table, a declared lie and distance state, and a reproducible mapping from shot outcome to expected strokes. A distance-only or target-error score without that backing is labelled a *performance proxy*; it must not borrow the strokes-gained name.

The canonical scoring contract requires exact lie, context, target, and distance states before and after the shot. Missing states, incomplete retained rows, and queries outside baseline support fail closed. No expected-strokes baseline is bundled with the public application. A baseline's declared source URL, licence, version, and digest preserve provenance; that declaration does not independently endorse its methodology, legal suitability, or relevance to a particular player population. The distance/target performance proxy remains a separate yard-based dispersion measure whose machine-readable flag is `is_strokes_gained=false`.

11.3 Qualified Model-Comparison Results

The Release A campaign evaluated seven open flight-model configurations on the 13,843-row strict cohort, producing 96,901 predictions with no execution failures. Its primary estimands are source-stratified error, bias, association, and residual relationships. Pooled metrics are retained only as

descriptive summaries because source composition, device, venue, ball, and acquisition conditions are confounded.

For the Blackmore TrackMan-labelled carry cohort ($n = 8,860$), the best tested configuration was the Waterloo/Penner implementation: carry RMSE 12.256 m, mean error -4.093 m, and strong linear association with the reported outcome. These are agreement statistics against the source’s reported carry, not an absolute-accuracy result and not evidence that the proprietary TrackMan equations have been identified. For context, the best tested carry configuration on one Garmin R10 source ($n = 1,541$) had RMSE 7.150 m and bias -4.722 m, while one Garmin R50 source ($n = 255$) had RMSE 10.238 m and bias $+6.554$ m. The sign change is one reason a pooled vendor ranking would be scientifically unsafe. Here “best tested” means the lowest carry RMSE among the seven configurations on that source’s evaluation rows. Selecting that minimum is not a new held-out evaluation, and comparisons between these unpaired sources cannot isolate a vendor effect.

The content-addressed calculation record is the private campaign’s `results/v2/model/model_campaign_manifest.json`. It binds the cohort, predictions, source metrics, pooled descriptive metrics, residual-correlation table, configuration, and source qualification outputs by SHA-256. The companion `multisource_v2_model_report.md` states row selection and limitations in human-readable form. Restricted rows remain in the private authority; the public paper reports only qualified aggregates.

The qualified snapshot is pinned to a private authority commit and three digests verifiable against the corresponding files by an authorized reader:

```
private merge: 8cd8ad04bf2904eedd0d8f10d2aa202e437e91fd
qualification:
679856edb290111a0106c0cbcbdb2b673f581fd6b0026409c4d643565b330709a
model campaign:
b70935fb4209b3f3aa18c3129ae948c4e20da74536af4d88433e5f0875960ca7
capability manifest digest:
1906d19fcace3284ae99d9dd8de213a0e2dabe8c062e4d63edc06c98f7eaf92e
```

These identifiers bind the reported aggregates without exposing restricted shot rows. A matching digest establishes file identity; it does not validate the measurements, selection rules, or statistical conclusions.

The associated software and public-boundary revisions are pinned separately:

```
public data facade: 70feeb4889ef4d030bdb426bd7ecb465c36ba41b
generic UpstreamDrift contract v2: 5ea0d08821ee8a1cacb93d22239abddee4813fca
Tools grouped-analytics release: 08b4b013d8f58feb0a788fc67add29f64750da23
canonical source-backed SG contract: fb4e6e41ac4980309e32dacceb957f4da02616b3
Tools canonical SG integration: a7b6dc437b45857e27cd07cccb56ff665826e342
Tools visual approval: 10d7f6fc38d27dd04324160090a7290f46d0cfd4
approved launch-monitor PNG SHA-256:
6699129037d0adf8152e73bdae3fc2b75a2e15bfb16e25a4b3cfcec2448ac00e
```

The public facade contains capability metadata and locks, not restricted rows. The older Tools revision remains the authority for the specialized grouped estimators, while the older UpstreamDrift revision remains the generic v2 envelope and claim-discipline authority. The newer UpstreamDrift revision owns the canonical source-backed scoring contract, and the newer Tools revision is its PyQt/React consumer. The visual approval covers presentation and accessibility, not statistical correctness or baseline validity. It also does not convert an unbundled or user-selected expected-strokes table into endorsed scientific evidence.

11.4 Release B: Paired-Device Scientific Validation

Release B’s scientific claims require a protocol fixed before confirmatory data collection, followed by execution and analysis under that protocol. The pinned bundle is a *structural template*, not a completed preregistration. The ball specification, numeric speed bands, device/reference set, placement, and calibration procedures still require a documented freeze before the pilot. Pilot-informed changes must be distinguished from confirmatory decisions, with the final analysis plan registered before inspecting confirmatory differences. The software bundle and its synthetic fixture establish an executable workflow, not completion of those scientific decisions.

The broader intended program covers speed, loft, spin, launch direction, impact location, representative mishits, and indoor/outdoor conditions. Record device model and firmware, ball and club, environmental state, alignment, player or robot identity, shot identity, and measured-versus-estimated status. Firmware changes form new strata rather than being pooled invisibly. The current template covers only three robot-center, outdoor club/speed cells: driver, 7-iron, and wedge. Even a completed study in those cells would not establish indoor performance, mishit robustness, or generalization to human players.

11.4.1 Pairing and Reference Uncertainty

Each paired comparison must concern the same physical shot and the same *measurand*, the quantity intended to be measured. For example, horizontal carry to first ground contact in a declared frame differs from a model’s carry normalized to standard weather, even if both outputs are labelled carry. A traceable reference requires an uncertainty statement, not merely a passed-calibration flag; calibration and repeatability have specific metrological meanings [43]. Alignment, time registration, environmental corrections, and the distinction between measured and estimated outputs remain part of the comparison. A common trigger timestamp associates records; it does not demonstrate zero acquisition lag or negligible synchronization uncertainty.

An additive model shows what pairing can and cannot remove. For shot s , let X_s be the shared physical quantity, Y_{ds} the output of device d , and R_s the reference output. Within a declared condition, write constant biases as b_d, b_R and zero-mean random errors as e_{ds}, e_{Rs} :

$$Y_{ds} = X_s + b_d + e_{ds}, \quad R_s = X_s + b_R + e_{Rs}, \quad (11.1)$$

$$D_{ds} := Y_{ds} - R_s = b_d - b_R + e_{ds} - e_{Rs}. \quad (11.2)$$

The common X_s cancels, but reference bias and uncertainty do not. Suppressing the shot index for the random variables across shots within this condition,

$$\text{Var}(D_d) = \text{Var}(e_d) + \text{Var}(e_R) - 2\text{Cov}(e_d, e_R). \quad (11.3)$$

Shared calibration or environmental corrections can therefore change apparent agreement through error covariance. As a constructed counterexample, two devices with the same 5 m offset agree perfectly while both differ from the physical quantity by 5 m. This is a mathematical example, not a measured vendor error. Across golf shots, dispersion of Y_{ds} also includes variation in X_s and any covariance between the physical outcome and measurement error. It cannot automatically be attributed to golfer inconsistency, sensor noise, or a motor-control mechanism.

11.4.2 Agreement, Repeatability, and Repeated Shots

For independent paired differences from an approximately normal distribution with constant mean and variance, descriptive 95% limits of agreement are

$$\bar{D} \pm 1.96s_D, \quad (11.4)$$

where $\bar{D}$ and s_D are the sample mean and sample standard deviation of the differences. These limits concern individual paired differences, not a confidence interval for mean bias; each estimated limit also has sampling uncertainty. Repeated observations of changing quantities and replicates of a constant quantity require different variance decompositions [3]. Condition-dependent bias or spread calls for a prespecified conditional model or justified transformation and examination of its residual assumptions. Correlation alone cannot establish agreement, and a nonsignificant mean difference cannot establish acceptable error: practical bounds and precision requirements must be specified for the intended use.

Within-device repeatability concerns repeated measurements under specified conditions [43]. For two independent normal errors from one instrument with within-device standard deviation σ_w , measured at the same underlying quantity, the standard deviation of their difference is $\sqrt{2}\sigma_w$. The corresponding 95% absolute-difference threshold is $1.96\sqrt{2}\sigma_w$. A device-reference difference standard deviation does not separately identify either instrument's σ_w . Repeated robot shots can themselves have different physical outcomes, so their raw dispersion is not automatically instrument repeatability.

The pinned descriptive analyzer computes $1.96s_D$ under the field name `repeatability_coefficient`; here it is an *agreement half-width*, not an identified within-device repeatability coefficient. The analyzer's bootstrap resamples individual pairs to estimate uncertainty in mean bias; it does not implement the proposed hierarchical analysis or confidence intervals for both agreement limits. Its pooled device summaries must not be mistaken for the completed condition-specific confirmatory analysis.

To illustrate the effect of clustering, suppose difference j in block b follows $D_{bj} = \mu + U_b + E_{bj}$. Let μ be the population mean difference, independent zero-mean block effects U_b have variance τ^2 , and mutually independent zero-mean residuals E_{bj} have variance σ^2 , also independent of all block effects. For B independent blocks of m shots each,

$$\text{Var}(\bar{D}) = \frac{\tau^2}{B} + \frac{\sigma^2}{Bm}. \quad (11.5)$$

Treating all Bm shots as independent would instead give $(\tau^2 + \sigma^2)/(Bm)$, understating uncertainty when $m > 1$ and $\tau^2 > 0$. This illustrative model is not a fit to the campaign. Unequal blocks, player effects, serial dependence, and comparisons sharing a reference require the model and resampling scheme appropriate to that design, preserving the shot pairing and relevant clusters.

11.4.3 Sample Planning and the Completion Gate

The template's planning arithmetic uses one two-sided paired-mean test with independent, identically distributed normal differences:

$$n \approx \left[\frac{(z_{1-\alpha/2} + z_{1-\beta})\sigma_D}{\delta} \right]^2. \quad (11.6)$$

Here z_p is the standard-normal p quantile, α the test's type-I error probability, $1 - \beta$ its power, σ_D the assumed difference standard deviation, and δ the mean difference to detect. With $\alpha = 0.05$,

power 0.80, $\sigma_D = 3$ m and $\delta = 1$ m, this approximation gives 70.6399, rounded up to 71 complete independent pairs. Allowing an expected 15% unusable fraction gives $\lceil 71/0.85 \rceil = 84$ *attempts*; that expected-yield calculation does not guarantee a quota. It neither establishes power for the protocol's multiplicity-adjusted family of endpoints and comparisons nor specifies precision for agreement limits. Clustering and the intended interaction estimates also require separate planning.

The existing software gate nevertheless requires **84 accepted pairs per cell**, or 252 across the three cells; that operational threshold is retained. The checked-in schedule has exactly 252 slots, so it supplies no attrition reserve for achieving 252 accepted pairs. Before collection, specify and version an attempt budget, stopping rule, and replacement policy consistent with that gate. Retain every failed attempt and its reason, rather than overwriting a failure or silently removing a device that struggles with difficult shots. Report missingness by device and condition alongside agreement among retained pairs; complete-pair results alone need not represent performance over all attempted shots.

The software's "pair" identifier actually denotes one same-shot observation set containing at least two distinct commercial devices and exactly one reference. Each device-reference contrast forms a statistical pair; contrasts within the same set share the reference, and do not create additional independent shots toward the quota. The validator checks the recorded calibration status, unique device rows, hashed serial identifiers, and a common trigger timestamp; these schema checks do not establish physical independence or reference accuracy. The versioned protocol template, capture-row JSON Schema, validator, deterministic schedule, collection ledger, command-line procedure, and synthetic fixture are present. The generated status record reports `confirmatory_ready=false`, 252 `not_collected` exclusions, and zero analyzed pairs. The schedule and ledger are bound by SHA-256:

```
schedule: 72b06c034e73751dbacce8468505517a8500754d4d769cc89fed846611790173
ledger:   ebc1a24100b0e09ea39978127d6f1bcad3bd26bf6c4f3ae042680a833281d9c9
```

No paired observations are represented as collected in this snapshot. The remaining protocol decisions and physical acquisition are hold points; neither software readiness nor a synthetic fixture is device-validation evidence.

Neural-network vendor emulators remain internal research artifacts. Their training manifests must pin the eligible sources, feature and target units, splits, random seeds, software version, and evaluation cohort. They are released only with held-out and out-of-distribution performance, uncertainty or applicability bounds, and a model card that forbids device-emulation or certification claims. Until the paired-device gate is satisfied, Release B comparisons remain unavailable rather than being approximated from unrelated shots.

11.5 Reproducibility and Claim Discipline

Release manifests bind software, interface, paper, and data revisions to their checksums. Normalization and exclusions have their own lineage; exports retain the request and backing rows needed to reproduce a displayed calculation independently of interface state, subject to the source's access restrictions.

Unpaired public records support source-specific model comparisons. Same-shot observations with characterized reference uncertainty support the proposed device comparisons. Neither design alone identifies proprietary algorithms or uniquely separates sensor and aerodynamic-model errors: those inferences require additional assumptions and identifiable experiments. Reproducible calculations make these limits inspectable; they do not remove them.

Appendix A

Annotated Reference Library

This appendix organizes sources used in the review by their evidential role. The research dossiers in `articles/Launch_Monitor_Technology_Review/research/` record the working claim trails; they are research notes, not independent validation. A source listed here does not support every claim about the associated product. Manufacturer descriptions, measured agreement, and physical models answer different questions. Read a numerical result together with its device version, measurand, reference point, event time, test conditions, and missing-data rate. Access checks on 1 October 2026 are observations, not availability guarantees: a failed retrieval does not establish permanent absence.

A.1 Patents (Google Patents / USPTO)

These records disclose proposed methods and embodiments. They do not establish the architecture shipped in a particular device, independent performance, or freedom to implement. The annotations identify technical topics; they do not determine claim scope or current legal status. Inventor and company groupings are navigation aids, not a complete ownership or portfolio history.

TrackMan A/S (Fredrik Tuxen)

- Inventor index and manufacturer patent list:
<https://patents.justia.com/inventor/fredrik-tuxen> and
<https://www.trackman.com/legal/patents>
- US8845442 — Determination of spin parameters of a sports ball (harmonic-sideband spin; Magnus spin-axis inversion): <https://patents.google.com/patent/US8845442B2/en>
- US10393870 — continuation of the above:
<https://patents.google.com/patent/US10393870B2/en>
- US10850179 — spin axis via receiver-pair interferometry:
<https://patents.google.com/patent/US10850179B2/en>
- US8085188 / US9857459 / US10473778 — target-deviation (tap-a-target) family:
<https://patents.google.com/patent/US8085188B2/en>,
<https://patents.google.com/patent/US9857459B2/en>,
<https://patents.google.com/patent/US10473778B2/en>
- US10315093 — trajectory tracer overlay:
<https://patents.google.com/patent/US10315093B2/en>
- US11086005 — multi-bay range tracking:
<https://patents.google.com/patent/US11086005B2/en>
- Historical TrackMan spin-patent explainer address (redirected to the general blog during this review; use the patent records above for content):
<https://blog.trackmangolf.com/patent-measuring-the-spin-of-a-sports-ball/>

FlightScope / EDH (Henri Johnson)

- US9868044 — spin from periodic features in a phase-demodulated signal:
<https://patents.google.com/patent/US9868044B2/en>
- US10775492 — spin-axis estimation using receiver-pair time delays:
<https://patents.google.com/patent/US10775492B2/en>
- US10338209 — combined radar and camera tracking:
<https://patents.google.com/patent/US10338209B2/en>
- Company history: <https://athlonsports.com/golf/the-flightscope-story-tracking-the-history-of-flight-in-golf>

Foresight Sports / Wintriss Engineering (Kiraly)

- US7292711 — flight parameters from imaged surface features, including markings, blemishes, or dimples: <https://patents.google.com/patent/US7292711B2/en>
- US7324663 — standalone smart-camera launch monitor:
<https://patents.google.com/patent/US7324663B2/en>
- US7497780 — integrated launch monitor UX:
<https://patents.google.com/patent/US7497780B2/en>
- US7641565 — ball-placement detection: <https://patents.google.com/patent/US7641565B2/en>
- WAWGD (Foresight) filings index:
<https://patents.justia.com/inventor/wawgd-inc-dba-foresight-sports>
- Foresight–Uneekor 2024 license (Business Wire):
<https://www.businesswire.com/news/home/20240930838753/en/>

Acushnet / Titleist (Gobush et al.)

- US5501463 — optical club/ball measurement with reflective markers:
<https://patents.google.com/patent/US5501463A/en>
- US6500073 — camera measurements and trajectory computation:
<https://patents.google.com/patent/US6500073B1/en>
- US6758759 — separate club and ball monitors before/after impact:
<https://patents.google.com/patent/US6758759B2/en>
- US7143639 — portable monitor with one or more cameras:
<https://patents.google.com/patent/US7143639B2/en>
- US8500568 / US8556267 — portable continuations:
<https://patents.google.com/patent/US8500568B2/en>,
<https://patents.google.com/patent/US8556267B2/en>
- US10668350 — stereographic or light-field camera embodiments:
<https://patents.google.com/patent/US10668350B2/en>

Others

- USGA US6186002 — aerodynamic characteristics from measured trajectories (original assignee: United States Golf Association): <https://patents.google.com/patent/US6186002B1/en>
- Full Swing US2020/0147470 — CW+FMCW dual-mode launch monitor: <https://patents.google.com/patent/US20200147470A1/en>
- Creatz US10247553 — start-position sensor: <https://patents.google.com/patent/US10247553B2/en>
- Golfzon US9242158 — two-stage latency-hiding simulation: <https://patents.google.com/patent/US9242158B2/en>
- Uneekor patent/license list: <https://uneekor.com/legal/patents>
- Pulse/exposure timing correlation US12401909: <https://patents.google.com/patent/US12401909B2/en>
- TrackMan EP1698380 litigation coverage: <https://thegolfwire.com/322771-2/>
- FlightScope 2022 BGH win vs TrackMan: <https://golfbusinessnews.com/news/innovation-centre/flightscope-wins-patent-infringement-case-against-trackman-in-germany/>

A.2 Regulatory Records and Teardown Reports

FCC filing mirrors identify equipment records and tested configurations; they do not independently validate launch parameters. Forum teardown reports are secondary observations requiring confirmation against the actual hardware.

- TrackMan 4 FCC ID SFX-TMAN4 (X-band + 24 GHz): <https://fccid.io/SFX-TMAN4>
- TrackMan iO FCC ID SFX-TMB0201 (24 GHz): <https://fccid.io/SFX-TMB0201>
- FlightScope Mevo FCC ID QXP-A7310 (24.125 GHz): <https://fccid.io/QXP-A7310>
- Mevo+ teardown / QXP-LJ361 (10.5–10.55 GHz) discussion: <https://golfsimulatorforum.com/forum/flightscope/277362-mevo-teardown>

A.3 Manufacturer Documentation and Product Reporting

Official pages describe vendor definitions and specifications. Media, retailer, and community pages below are labeled separately; their inclusion does not turn a product claim into an independent measurement.

TrackMan

- 40+ parameters explained: <https://www.trackman.com/blog/golf/40-trackman-parameters>
- Club data definitions: <https://www.trackman.com/blog/golf/club-data-definitions>
- Club speed (geometric-center definition): <https://www.trackman.com/blog/golf/what-is-club-speed>
- Spin loft / spin rate: <https://www.trackman.com/blog/golf/spin-loft>, <https://www.trackman.com/blog/golf/spin-rate>

- OERT (“Two radars, one camera, zero doubt”):
<https://www.trackman.com/blog/golf/two-radars-one-camera-zero-doubt>
- Tech specs (TM4, iO): <https://www.trackman.com/golf/launch-monitors/tech-specs>
- Ball Flight Laws newsletter (Jan. 2009; historical D-plane and launch-direction teaching material). Compare the study-specific weights in §3.1.1; 0.76 is not a universal constant:
<https://www.yumpu.com/en/document/view/11652756/trackman-ball-flight-laws>

A.3.1 The TrackMan Newsletter Archive

Historical newsletters are indexed below through a third-party mirror. Issues #7–#9 were retrieved for the present annotation review; this is not a claim that the collection is complete or that official and archival copies are unavailable. The mirror directory is

<https://www.gregsmithgolfcoach.com/wp-content/uploads/2014/04/>:

- `newsletter1.pdf` (Nov 2007) through `newsletter9.pdf` (Jan 2013), sequentially numbered.
- `TrackMan_Newsletter_2014.pdf` is issue #10 (Jan 2014); the filename differs from the numbered examples. This does not establish an end date for the series.
- **#7 (Oct 2010), “Ten Fundamentals”** presents historical vendor rules of thumb: driver launch weights of 85% toward the face in both planes, 75% for irons, and nominal spin decay of 4% per second. These are not universal material or aerodynamic constants. Its scalar loft-minus-attack definition requires a common-plane interpretation; spatial spin loft is the angle between the face normal and velocity.
- **#9 (Jan 2013), pp. 3 and 5** specifies a maximum-compression reporting reference with averaging qualifications. It lists TrackMan III/IIIe *absolute* specifications, described by the manufacturer as 95% confidence: speed ± 1.5 mph, attack $\pm 1.0^\circ$, path $\pm 1.0^\circ$, loft $\pm 0.8^\circ$, face $\pm 0.6^\circ$. Its separate repeatability column is not interchangeable with absolute error. Conditions include a centre of gravity within 10 mm of the geometric centre for attack/path, impact entirely on the face for loft/face, and successful measurement of the required club, ball-direction and spin data. A reporting reference does not mean every output is an instantaneous direct measurement; see also the dated club-data definitions above.
- **#8 (Jun 2011), p. 4** gives a 12.7 mm heel offset and a 2° face-angle change toward closed. Its universal-driver wording should not be adopted as a geometric law: the change depends on local face curvature. For an ideal circular horizontal section, $\Delta\theta = \arcsin(x/R)$ for lateral offset x and radius R , relative to the central normal. At $x = 12.7$ mm, radii of 250 and 500 mm give about 2.91° and 1.46° , respectively. These are geometric examples, not measured driver specifications; the sign depends on the heel/toe convention.

Caution

Those accuracy figures are manufacturer specifications. Leach et al. tested TrackMan Pro IIIe and GC2+HMT against an optical reference in an indoor, eight-golfer, 240-shot study (Leach et al. 2017, <https://doi.org/10.1016/j.measurement.2017.08.009>). In Table 6, 45% of TrackMan path differences and 66% of face-angle differences fall within $\pm 1^\circ$ of the *median difference*, not of zero error. Such a band describes dispersion after centering; a systematic bias can remain. It is not a direct pass/fail test of the vendor’s absolute limits.

Data availability is another quantity: TrackMan returned club velocity on 98% of shots, but

the other four club parameters on 62% overall and 19% for the utility wedge. The authors also describe installation compromises, possible radar interference, markers, and differences in reference point or impact timing. Reference calibration alone does not make all compared outputs the same measurand. Report bias, dispersion, uncertainty and return rate separately, and retain these device/setup limits when applying the result to a golfer's swing or to a newer monitor.

Foresight Sports

- GCQuad product/technology:
<https://www.foresightsports.com/products/gcquad-launch-monitor>
- Ball and club data (spherical correlation, fiducials):
<https://foresightsports.eu/ball-club-data/>
- HMT head measurement (dot-count data tiers):
<https://www.foresightsports.com/hmt-head-measurement>
- Club marker application guide:
<https://help.foresightsports.com/hc/en-us/articles/4408197030035>
- Marker troubleshooting:
<https://help.foresightsports.com/hc/en-us/articles/4405916455443>
- Design history (Popular Science): <https://www.popsci.com/gear/foresight-sports-quadmax-photometric-golf-launch-monitor-development/>

Other Vendors

- FlightScope X3 announcement (The Golf Wire): <https://thegolfwire.com/326713-2/>
- Garmin R10 third-party explainers (My Golf Simulator):
<https://mygolfsimulator.com/garmin-r10-data/>,
<https://mygolfsimulator.com/garmin-r10-accuracy/>
- Garmin R50 (three-camera): <https://www.garmin.com/en-US/p/736810/>
- Full Swing KIT technology and specs:
<https://www.fullswinggolf.com/kit-launch-monitor-technology/>,
<https://www.fullswinggolf.com/kit-specs/>
- Rapsodo MLM2PRO spin (MyGolfSpy): <https://mygolfspy.com/news-opinion/rapsodo-mlm2pro-takes-spin-measurement-to-pro-level/>
- SkyTrak+ / ST+ vendor product and measurement description:
<https://www.skytrakgolf.com/products/skytrak-plus-launch-monitor>
- Uneekor EYE XO / Dimple Optix vendor description (Canadian site):
<https://uneekor.com/en-ca/products/eye-xo-standalone>
- Uneekor calibration guide (Carl's Place):
<https://www.carlofet.com/blog/calibrating-uneekor-launch-monitors>; official PDF:
https://download.uneekor.com/docs/EYEXO2_Calibration_Guide.pdf
- ProTee VX retailer review (PlayBetter):
<https://www.playbetter.com/blogs/golf-simulator-reviews/protee-vx-review>

A.4 Research Papers, Standards, and Technical Notes

Journal and proceedings papers, preprints, theses, author notes and test standards have different review and validation roles. Agreement with another monitor is not automatically accuracy against an independent reference; repeatability is not trueness. A cross-sport method needs golf-specific validation before its reported performance can be transferred.

- Penner, “The physics of golf,” Rep. Prog. Phys. 66 (2003):
<https://iopscience.iop.org/article/10.1088/0034-4885/66/2/202>
- Leach et al. 2017 launch-monitor validation (*Measurement*):
<https://www.sciencedirect.com/science/article/abs/pii/S0263224117305079>; open-access:
https://repository.lboro.ac.uk/articles/journal_contribution/9562799
- TrackMan 4 indoor reliability (J. Sports Sci. 2024):
<https://www.tandfonline.com/doi/full/10.1080/02640414.2024.2314864>
- Mevo+ vs TrackMan 4 agreement (2025):
<https://www.sciencedirect.com/science/article/pii/S2772696725000420>
- Bearman & Harvey 1976 golf ball aerodynamics:
<https://www.cambridge.org/core/journals/aeronautical-quarterly/article/abs/golf-ball-aerodynamics/67FE0903DB1CC12001F1ED1B1261C4B9>
- Smits & Smith aerodynamic model: <https://www.researchgate.net/publication/284037213>
- USGA Indoor Test Range conditions: <https://www.usga.org/content/dam/usga/images/equipment-standards/ITR-test-conditions-2028-ODS.pdf>
- Friction/tangential compliance and launch angle (MDPI 2020):
<https://doi.org/10.3390/proceedings2020049027>
- Wood, Henrikson and Broadie (2018), face/path influence study:
<https://doi.org/10.3390/proceedings2060249>. The player-test horizontal means (driver 0.76, 7-iron 0.69, wedge 0.61) use centre-strike and face-to-path filters; they are sample-specific, not universal coefficients or individual-shot error bounds.
- Cross oblique-impact golf experiments:
<https://www.physics.usyd.edu.au/~cross/GOLF/GOLF.htm>
- Cross & Nathan oblique collision formalism: <https://arxiv.org/pdf/1610.03464>
- Dynamic models review, Sports Engineering 25 (2022):
<https://link.springer.com/article/10.1007/s12283-022-00387-0>
- Stanford LES of golf ball flow (Sports Eng. 2019):
http://aero-comlab.stanford.edu/Papers/golf_ball_sports_engineering_2019.pdf
- Golf ball aerodynamics in still air (MDPI): <https://www.mdpi.com/2504-3900/2/6/238>
- Micro-Doppler of spinning projectiles on CW radar:
<https://www.researchgate.net/publication/317119893>
- SpinDOE camera spin estimation (transferable method): <https://arxiv.org/pdf/2303.03879>
- “Measuring Ball Spin by Image Registration”:
<https://www.researchgate.net/publication/2881136>

- Doppler ball-tracking thesis (WSU, TrackMan platform):
<https://baseball.physics.illinois.edu/trackman/JasonMartinThesisWSU.pdf>
- Alan Nathan's TrackMan (baseball) technical notes:
<https://baseball.physics.illinois.edu/trackman.html>
- Cochran & Stobbs, *Search for the Perfect Swing* (1968):
<https://archive.org/details/searchforperfect0000coch>

A.5 Engineering References

- 3D launch conditions from impact conditions:
<https://www.tutelman.com/golf/ballflight/3dlaunch.php>
- Smash factor: <https://www.tutelman.com/golf/ballflight/smashfactor.php>
- Gear effect: <https://www.tutelman.com/golf/ballflight/gearEffect1.php>
- Spin decay: <https://www.tutelman.com/golf/ballflight/spinDecay.php>
- Ball flight laws summary: <https://www.perfectgolfswingreview.net/ballflight.htm>
- TrackMan face-angle derivation discussion (Manzella forum): <https://forum.brianmanzellagolf.com/threads/how-does-trackman-flightscope-measure-the-club-face-angle.16591/>

A.6 DIY / Open-Source

- PiTrac GitHub (GPL-2.0): <https://github.com/PiTracLM/PiTrac>
- PiTrac documentation: <https://docs.pitrac.org/>
- PiTrac Hackaday.io project (design logs):
<https://hackaday.io/project/195042-pitrac-the-diy-golf-launch-monitor>
- OpenFlight upstream: <https://github.com/jewbetcha/openflight>
- OpenFlight Grafana write-up: <https://medium.com/grafana-labs/openflight-building-an-open-source-golf-launch-monitor-with-raspberry-pi-grafana-cloud-and-84d68ad40fc5>
- GC2-imitating OpenCV experiments: <https://github.com/ronheywood/opencv>
- GSA Golf simulator theory (marked-ball DIY methods):
<https://www.golf-simulators.com/GolfSimulatorTheory.html>

A.7 Hardware Datasheets, Protocols, and Standards

- OPS243 product brief:
https://omnipresense.com/wp-content/uploads/2024/01/OPS243-Product-Brief_004-F.pdf
- OPS243 AN-010 API interface:
https://omnipresense.com/wp-content/uploads/2025/10/AN-010-AD_API_Interface.pdf
- OPS243 AN-027 rolling buffer:
https://omnipresense.com/wp-content/uploads/2025/06/AN-027-A_Rolling-Buffer-1.pdf

- OPS243 AN-029 sports applications: https://omnipresense.com/wp-content/uploads/2026/06/AN-029-A_OPS243-for-Sports_260621.pdf
- RFbeam K-LD7 datasheet: <https://rfbeam.ch/product/k-ld7-radar-transceiver/>
- TI IWR6843 datasheet: <https://www.ti.com/lit/ds/symlink/iwr6843.pdf>; ISK EVM: <https://www.ti.com/tool/IWR6843ISK>; placement app note: <https://www.ti.com/document-viewer/lit/html/SWRA758>
- TI mmWave golf-swing forum thread: <https://e2e.ti.com/support/sensors-group/sensors/forums/sensors-forum/1116424/iwr6843aop-golf-swing-analyzer-with-mmwave-radar>
- Raspberry Pi camera documentation (Global Shutter, XTR trigger): <https://www.raspberrypi.com/documentation/accessories/camera.html>
- GSPro Open Connect v1: <https://gsprogolf.com/GSProConnectV1.html>; community feedback: <https://github.com/tnbozman/gspro-interface/blob/main/OpenAPI-Documentation-Feedback.MD>; reference client: <https://github.com/travislang/gspro-garmin-connect-v2>
- USGA equipment standards: initial velocity TPX3007: <https://www.usga.org/content/dam/usga/pdf/Equipment/TPX3007-initial-velocity-test-procedure.pdf>; ODS TPX3006: <https://www.usga.org/content/dam/usga/pdf/Equipment/TPX3006-overall-distance-and-symmetry-test-procedure.pdf>; clubhead CT TPX3004 and MOI TPX3005 via <https://www.usga.org/equipment-standards.html>; 2028 ALC revision: <https://www.usga.org/content/usga/home-page/articles/2023/12/revised-golf-ball-testing-conditions-to-take-effect-in-2028.html>
- CFAR tutorial (CA/GO/SO/OS): <https://www.mathworks.com/help/phased/ug/constant-false-alarm-rate-cfar-detection.html>; Purdue lecture notes: https://engineering.purdue.edu/~mrb/resources/AltLectureF/Session_21.pdf
- Stalker baseball-spin patent (radar spin outside golf): <https://patents.google.com/patent/US10935657B2/en>
- Weibel Scientific radar object-detection patent: <https://patents.google.com/patent/EP1735637B1/en>

A.8 Comparative Testing and Market Analysis

These media, vendor and retailer accounts help locate comparisons and historical context. Before using a ranking or numerical accuracy claim, inspect its reference instrument, protocol, exclusions and uncertainty; the listing itself is not an endorsement of the test.

- MyGolfSpy indoor accuracy test: <https://mygolfspy.com/news-opinion/three-of-the-most-accurate-indoor-launch-monitors-and-one-to-avoid/>
- GCQuad vs TrackMan comparison article (Golf Simulator Zone): <https://golfsimulatorzone.com/gcquad-vs-trackman-accuracy-in-2026/>
- Foresight GCQuad/GC3 comparison: <https://www.foresightsports.com/blogs/golf-tips/comparing-golf-launch-monitor-foresight-sports-gcquad-and-gc3-stand-above-the-rest>
- Bushnell Launch Pro vs GC3 vs GCQuad: <https://www.playbetter.com/blogs/golf/bushnell-launch-pro-vs-gc3-vs-gc-quad-comparison>

- Photometric vs radar explainers: <https://www.playbetter.com/blogs/golf-simulator-comparisons/photometric-vs-radar-golf-launch-monitor>,
<https://golfbays.de/en/blogs/news/radar-vs-photometric-launch-monitors-indoors>
- Indoor flight compensation:
<https://www.cerogolf.com/post/how-launch-monitors-compensate-for-indoor-ball-flight>
- TrackMan history (Golf Monthly, Tuxen profile):
<https://www.golfmonthly.com/features/from-tracking-missiles-to-tracking-golf-balls-meet-the-man-that-you-didnt-realise-changed-golf>

Appendix B

Detailed Implementation Guidance

A useful estimator must connect a reported quantity to the observations that constrain it. This appendix develops a candidate measurement chain, with explicit assumptions and checks; it does not report a validated prototype. Coordinate conventions and reference points follow Chapter 2, and hardware limits follow Chapter C. Measured sensor channels, derived physical quantities and estimates dominated by priors must remain distinguishable in the output. A plausible number, a small filter covariance or agreement with another launch monitor is insufficient evidence on its own.

B.1 Radar Signal Chain (OPS243-A)

B.1.1 Doppler Scaling and Resolution Budget

For the illustrative carrier $f_c = 24.125$ GHz, Equation (4.1) gives approximately 71.95 Hz per mph of radial speed. With uniformly spaced complex I/Q samples at $f_s = 30$ kHz, the ideal principal frequency interval is $[-f_s/2, f_s/2)$, corresponding to a radial-speed half-width of about 208.48 mph. This sampling calculation does not certify the analog passband, firmware, detection sensitivity or total-speed envelope of an OPS243-A installation [60, 57]. At 200 mph radial speed, the carrier Doppler is already about 14.39 kHz; rotational sidebands and acceleration can consume the remaining frequency margin. Confirm sample cadence, I/Q ordering, mixer sign and missing-frame behavior with the actual acquisition mode.

A 128-sample record lasts 4.267 ms and has raw Fourier spacing $f_s/128 = 234.375$ Hz, equivalent to approximately 3.26 mph. Padding it to 4096 points samples the same finite-record spectrum more densely; it does not create independent observations. Distinguish bin spacing, separation of neighboring components, single-tone estimation precision and absolute speed accuracy. All depend differently on window shape, signal-to-noise ratio, calibration and model mismatch.

At 3,000 rpm, mechanical rotation frequency is 50 Hz. An observation of 20 ms gives a 50 Hz unpadding grid; 40 ms gives 25 Hz. Neither duration guarantees that weak spin sidebands can be separated from a strong drifting carrier: sidelobes, missing harmonic orders and frequency variation also matter. Choose a window and acceptance criterion using injected signals and recorded shots over the intended operating envelope, rather than treating two rotation periods as a universal requirement.

B.1.2 Candidate Spin-Rate Estimation

US8845442 describes candidate spectral traces, harmonic relationships and order assignment [98]. The following is an engineering proposal informed by that disclosure, not its exact algorithm or a claim about current commercial firmware:

1. Preserve coherent raw I/Q with acquisition timestamps and configuration metadata. Associate a post-impact ridge with the ball using temporal continuity and independently justified gates; the strongest peak need not be the ball. Exclude saturation and unresolved aliasing before interpreting peak locations.

2. Estimate carrier frequency as a continuous function of time. For a convention with carrier phase $+2\pi \int f_d dt$, coherent demodulation uses

$$y(t) = z(t) \exp \left[-j2\pi \int_{t_*}^t \hat{f}_d(\tau) d\tau \right]. \quad (\text{B.1})$$

Here z is the recorded complex signal and t_* is a fixed phase origin. An independent frequency shift per short-time Fourier frame can recenter its peak but does not generally remove intraframe chirp; discontinuous phase corrections also introduce artifacts. Carrier-estimation error belongs in the residual uncertainty.

3. Search for harmonic families using a declared spectral or time-domain model. Cepstral or harmonic-product scores can propose candidates, but missing orders, clutter and approximately symmetric patterns can cause octave errors or competing fundamentals. Test the candidate order assignments and residuals, rather than declaring either method intrinsically robust to missing harmonics.
4. Require prespecified evidence across usable observations and check against an independent spin reference. Overlapping Fourier windows are correlated, so three consecutive frames are not three independent confirmations. Report ambiguity, detection failures and the fraction of attempted shots yielding an accepted measurement. If only a club-conditioned prior is available, label the result estimated and retain the missing-measurement flag.

Scattering contrast, viewing direction and ball pattern influence which modulation components are observable (Section 4.4.2). Log ball identity and markings rather than assigning a universal ranking to clean urethane, range or marked balls. A candidate rotation frequency does not by itself identify the full spin vector. US8845442 and US9868044 offer distinct technical approaches to study; the latter's dielectric-lens explanation and claimed processing requirements must remain separate [98, 42]. These disclosures establish neither implementation clearance nor technique-wide expiry dates (Section 7.8).

B.1.3 Club-Speed Extraction

The velocity of a rigid-head point P is $\mathbf{v}_P = \mathbf{v}_O + \boldsymbol{\omega} \times \mathbf{r}_{OP}$; radar observes its line-of-sight projection, weighted by the actual scattering response (Section 4.5 and Chapter E). The heel need not always be the slower return, and shaft returns need not occupy a separable bottom decile. Trimming the highest and lowest spectral deciles and taking a median defines a reproducible spectral statistic, not a geometric-center speed. It can become a calibrated proxy only after reference-point, timing and pose dependence have been characterized against an independent measurement.

TrackMan's documented geometric-center speed definition specifies the intended reported quantity, not a reconstruction algorithm that can be assumed for a different sensor [94]. For a candidate club track, retain the pre-impact observation interval, target-association evidence and uncertainty in projection and reference-point conversion. A smash-factor discrepancy is a diagnostic lead, not proof of toe-lock or grounds for a universal rejection threshold (Section 3.2). Validate the speed estimate before using it to validate an impact model; otherwise the same assumption can appear to confirm itself.

B.2 Angle Measurement (K-LD7) and the EKF

B.2.1 Interferometric Angles and Hardware Limits

For the declared ideal two-receiver geometry, phase difference obeys Equation (4.2). Phase calibration, wrapping, multiple targets and the antenna response affect the inverse angle estimate. The K-LD7 revision-B datasheet specifies a 6.223 mm receiver separation, FSK ranging, processed detections and a raw-data request interface; it does not justify treating the module as an arbitrary FMCW range–Doppler instrument [72]. Its maximum speed setting is 100 km/h, with a typical frame duration of 29 ms at that setting. A ball whose radial component exceeds that limit is outside the stated unambiguous speed range; combining sensors in a filter does not make an invalid channel valid. Thus a 50 ms interval cannot be assumed to contain ten to twenty independent detections from one module at that cadence. Raw transport bandwidth, settings and dropped frames must also be checked (Chapter C).

Signal-to-noise ratio can inform a calibrated observation covariance, but a universal inverse-SNR weight ignores phase ambiguity, geometry and systematic alignment error. A small mount-yaw error produces an approximately equal azimuth bias near the horizontal plane; a general three-dimensional pose error couples the angle channels. Common calibration errors correlate observations across time and must not be averaged away as independent noise.

B.2.2 Measurement Functions and Observability

For a fixed sensor at world position $\mathbf{s}_i$, let R_i map its calibrated local frame into the world frame. Choose local axes consistently with the book’s forward/up/right convention and explicitly convert any vendor-native axes. For ball state $\mathbf{x} = (\mathbf{p}, \mathbf{v})$, define

$$\mathbf{q}_i = R_i^\top (\mathbf{p} - \mathbf{s}_i), \quad \mathbf{w}_i = R_i^\top \mathbf{v}, \quad \rho_i = \|\mathbf{q}_i\|. \quad (\text{B.2})$$

A candidate observation vector is

$$h_i(\mathbf{x}) = \begin{pmatrix} \rho_i \\ \mathbf{q}_i^\top \mathbf{w}_i / \rho_i \\ \text{atan2}(q_{iz}, q_{ix}) \\ \text{atan2}(q_{iy}, \sqrt{q_{ix}^2 + q_{iz}^2}) \end{pmatrix}. \quad (\text{B.3})$$

Use only the components actually supplied by the sensor and supported by its calibration; the OPS243-A speed channel does not supply all four. Range rate requires $\rho_i > 0$, and azimuth is undefined on the local vertical axis. The elevation value approaches $\pm\pi/2$ there, but its Cartesian Jacobian is also singular; use a nonsingular local representation or exclude that geometry from an angle-based linearization. Wrap angular innovations consistently and propagate pose and timestamp uncertainty. Moving sensors require their own translational and rotational motion in the observation model.

Writing $\hat{\mathbf{u}}_i = (\mathbf{p} - \mathbf{s}_i)/\rho_i$ and $v_{ri} = \hat{\mathbf{u}}_i^\top \mathbf{v}$, the radial-speed Jacobian blocks are

$$\frac{\partial v_{ri}}{\partial \mathbf{p}} = \frac{(\mathbf{v} - v_{ri} \hat{\mathbf{u}}_i)^\top}{\rho_i}, \quad \frac{\partial v_{ri}}{\partial \mathbf{v}} = \hat{\mathbf{u}}_i^\top. \quad (\text{B.4})$$

Even with position known, a single instantaneous radial-speed observation leaves two velocity directions unconstrained. Additional times, sensor viewpoints and dynamical assumptions can add information, but nearly repeated geometry can remain poorly conditioned. Inspect local sensitivity

over the actual observation window and compare posterior uncertainty with the prior; adding a state variable does not create a measurement of it. In particular, including spin states does not establish that a short flight segment identifies spin magnitude, axis, wind and aerodynamic coefficients separately (Section 4.4.3).

B.2.3 Filtering, Smoothing, and the Impact Event

An extended Kalman filter (EKF) and an extended Rauch–Tung–Striebel smoother are approximate nonlinear estimators, based on local linearization [73]. A proposed translational model is $\dot{\mathbf{p}} = \mathbf{v}$ and $\dot{\mathbf{v}} = \mathbf{g} + \mathbf{a}_{\text{aero}}$, with a declared aerodynamic law and its parameter uncertainty. If spin is omitted from the state, uncertain spin-dependent acceleration must still be represented through nuisance parameters or justified model-error covariance. It cannot silently be treated as a known force. Choose numerical time steps and interpolation by convergence checks at the required measurement precision.

Propagate between acquisition times, update only with associated valid observations, and retain the predicted and filtered means, covariances and transition Jacobians for backward smoothing. For innovation $\boldsymbol{\nu}_k = \mathbf{y}_k - h_k(\hat{\mathbf{x}}_k^-)$, a locally independent Gaussian observation model gives

$$S_k = H_k P_k^- H_k^T + R_k, \quad d_k^2 = \boldsymbol{\nu}_k^T S_k^{-1} \boldsymbol{\nu}_k. \quad (\text{B.5})$$

Here H_k is the measurement Jacobian, P_k^- the predicted state covariance and R_k the observation-noise covariance; R_k is distinct from the sensor rotation R_i above. A prespecified chi-square gate is only an approximate screening rule under the stated model. Multipath or a misassociated club return can pass it, while a valid shot can fail a misspecified filter. Inspect residual bias, temporal correlation, uncertainty coverage and rejection rates on held-out data.

Let a raw acoustic timestamp, expressed on the common clock, include sound travel over source-to-microphone distance d at speed c_s and electronic/detection delay τ_e . The estimated acoustic source-event time is

$$\hat{t}_{\text{event}} = t_{\text{stamp}} - d/c_s - \tau_e. \quad (\text{B.6})$$

The sound source event need not coincide exactly with face separation; establish that offset independently. Estimate d from the source location and microphone pose, propagate their uncertainty, and avoid subtracting delays already removed by acquisition software. At $c_s = 343 \text{ m/s}$, one meter adds about 2.92 ms, much more than an assumed 10 μs electronic delay. To first order, 100 μs of timing uncertainty at 150 mph contributes about 6.71 mm of along-track position uncertainty before other errors are included. A post-impact flight smoother may extrapolate to the declared separation event, with increasing uncertainty outside its observed interval; it must not propagate a smooth free-flight model through the contact impulse into the pre-impact state. Clock calibration and event definition are therefore part of the launch measurement, not merely data-logging details.

B.2.4 Alignment Calibration Procedure

Use surveyed targets or a calibration artifact to identify the sensor-to-world transforms, with several positions spanning the intended observation region. Optical floor charts are useful precedents [7], but radar needs a target and motion visible to its own measurement channels. A ball rolled or swung along a surveyed line can test selected geometric relationships; rolling contact or a curved pendulum path is not automatically a known free-flight trajectory. Validate range, direction, Doppler scale and clock alignment with separate checks where possible. Record poses, calibration residuals, covariance assumptions, firmware, configuration and environmental conditions for each session.

Use independent poses or trajectories to test calibration transfer, rather than reporting the fitting residual as general accuracy.

B.3 Derived Club Data, With Priors

The forward collision problem and inverse club-state problem have different information requirements (Sections 3.1 and 4.5.2). For a forward calculation, declare contact-point incident velocity, local face normal, impact location, inertial properties and contact law. The empirical relations in Equations (3.5) and (3.9) are conditional approximations, not mutually independent physical identities [97]. Check that translation and spin remain compatible with the same impulse model and its domain of calibration.

For a local scalar launch approximation, write $b = wF + (1 - w)P$, where b , F and P are launch, face-normal and path angles in one consistent projection. For known $w \neq 0$, its inverse and sensitivities are

$$F = P + \frac{b - P}{w}, \quad (\text{B.7})$$

$$\left(\frac{\partial F}{\partial b}, \frac{\partial F}{\partial P}, \frac{\partial F}{\partial w} \right) = \left(\frac{1}{w}, -\frac{1 - w}{w}, -\frac{b - P}{w^2} \right). \quad (\text{B.8})$$

Use these derivatives with the joint input covariance and a separate allowance for model discrepancy. For illustration, $w = 0.87$ gives a launch-angle sensitivity of about 1.15, but it is not a universal amplification factor for a three-dimensional mounting error. Study-specific weights cannot be turned into a universal driver-to-wedge interpolation solely from dynamic loft (Section 3.1.1). Separate horizontal and vertical inversions also require checking the validity of the small-angle projection approximation.

A club-type prior can regularize an underdetermined estimate, but the reported face angle, loft or spin then depends on that prior. Record the prior distribution, its population and calibration range, and test sensitivity to plausible alternatives. Do not use a spin fallback estimated from club priors as an independent confirmation of the same club estimate. Unknown strike location, head inertia and local face geometry introduce gear-effect ambiguity; a fixed spin-axis allowance per millimeter of miss is not a universal uncertainty bound (Section 3.4). Where those effects cannot be bounded from evidence, report the inverse quantity as unresolved or explicitly model-dependent.

B.4 Optical Module (Phase 2) Design Parameters

The Wintriss disclosures and PiTrac provide optical concepts and implementation examples, respectively [45, 63]. They do not establish the accuracy or timing of a new camera, lens, illumination and trigger combination. The following manufactured budget illustrates what must be checked.

B.4.1 Shutter, Illumination, and Motion Blur

The Raspberry Pi Global Shutter Camera uses an IMX296 sensor with 1456×1088 output, and its documented short-exposure capability is about $30 \mu\text{s}$ with adequate light [71, 70]. Global exposure avoids row-dependent capture times; it does not remove motion during integration. The standard camera includes an infrared-cut filter, so an 850 nm illuminator requires an explicitly verified optical configuration [70]. Do not assume unmodified spectral sensitivity or an arbitrary shorter exposure from the sensor name.

For locally transverse translation at speed $v_{\perp}$ and effective illumination duration τ , object-plane blur is approximately $v_{\perp}\tau$. At 150 mph and 30 μ s, this is 2.01 mm, about 4.71% of a nominal 42.67 mm ball diameter. Pixel blur is that fraction of the observed ball diameter in pixels, with additional effects from spin, defocus and lens distortion. A shorter light pulse can determine the effective exposure only if it falls within the active integration interval and ambient light contributes negligibly. Choose pulse width from the allowable image error and measured photon budget, not from a universal “frozen motion” threshold.

B.4.2 Multiple Exposures and Trigger Timing

For same-frame strobed images at approximately constant depth, pulse-center spacing Δt produces center separation $v_{\perp}\Delta t$. At 150 mph, 300 μ s and 500 μ s give 20.1 mm and 33.5 mm, both less than the nominal ball diameter. The silhouettes overlap; an algorithm must explicitly handle that overlap or use a different capture schedule. An ideal no-overlap condition, before blur and margins, is

$$\Delta t > \frac{D}{v_{\perp}}, \quad (N-1)v_{\perp}\Delta t + D \leq L, \quad (\text{B.9})$$

where D is diameter, N the number of pulses and L the usable lateral field of view at the ball plane. The first bound is about 636 μ s for this example. For a rectangular light pulse of duration τ with pure transverse translation, replace D by $D + v_{\perp}\tau$ in both bounds to include the illuminated motion extent; additional errors still require margins. Five pulses spaced by 0.8 ms occupy about 257 mm, including one ball diameter, within an ideal 300 mm field before allowance for blur, pose, depth change and timing error. These numbers specify a test case, not a commercial capture geometry or a validated design.

Budget acoustic propagation, threshold-crossing jitter, transport delay and camera/strobe synchronization together with the event offset in Equation (B.6). Use pretrigger buffering where appropriate; an asserted microphone-module delay alone does not solve impact timing. Verify actual pulse times and integration windows electrically and with a moving target.

B.4.3 Position, Spin, and Club Pose

Circle detection is a candidate image-processing step, not proof that partially overlapping, blurred or occluded balls are correctly localized. A known-diameter depth estimate requires calibrated projection, actual ball size and a suitable silhouette model; stereo additionally requires calibrated extrinsics, correspondence and synchronization. For close views, account for the perspective silhouette of a sphere rather than assuming the far-field diameter formula is exact.

Texture registration across ball images can estimate rotation on $SO(3)$ when visible features constrain it [45, 18]. At 3,000 rpm, 400 μ s corresponds to 7.2°, but a small angle alone proves neither uniqueness nor convergence of a local search. Dimple repetition, partial visibility, lighting and image noise can leave multiple compatible rotations. Use multiple hypotheses or intervals, inspect residuals and determine which rotation components are constrained before converting an increment to a spin vector. A D-plane prior may guide a search; it must not force the answer and then be counted as independent optical evidence. Calibrated face fiducials or additional viewpoints can constrain club pose and strike location, with their own geometry and uncertainty requirements [31, 46].

B.5 Flight Model and Validation Protocol

Treat aerodynamic coefficients, spin decay, wind, ball properties and numerical integration as parts of a declared model, not universal constants inherited from a named publication. A literature

model can supply a starting hypothesis [75]; trajectory fitting can estimate parameters [51], but neither guarantees identification over a short observation arc. Carry alone is one scalar outcome and generally cannot determine a complete lift, drag and spin-decay law. Version the model and coefficient domain, verify signs and units, and test integration convergence and event detection before interpreting a physical discrepancy. Do not infer golf accuracy from solver convergence alone.

Separate calibration trajectories from held-out validation by the relevant independent units, such as session, ball batch or participant. Prespecify reference definitions, practical error limits, exclusions and missing-shot reporting (Chapters 9 and 11). Paired commercial measurements test agreement, including the reference instrument's uncertainty; they are not automatically ground truth. Use repeated-measures analysis when shots share a participant or session, and examine condition-dependent bias and spread rather than relying on correlation or one pooled mean [3].

For MLM2PRO spin comparison, Rapsodo specifies RPT printed balls; an RCT radar ball is not an interchangeable optical reference [69]. Record ball identity, device mode, firmware, failed reads and whether every reported channel was measured, derived or estimated. Log raw I/Q, available frames, timestamps and calibration/configuration identifiers so an algorithm revision can be replayed, while retaining an untouched test set for final evaluation. Replayability makes errors inspectable; independent observations and appropriate uncertainty analysis establish what the system actually measures.

Appendix C

Sensor Hardware and Integration Reference

A launch monitor connects sampled signals to estimates of ball and club motion, then connects those estimates to impact and flight models. Every connection adds assumptions: the target must be identified, the measurement time and reference frame must be known, and any missing components must be supplied by additional observations or a declared model. This appendix uses documented components as worked examples, not as an endorsed bill of materials or evidence that a particular combination achieves launch-monitor accuracy. Read the specifications below with the measurement hierarchy in Table 2.3 and the impact definitions in Chapter 3. Manufacturer specifications describe particular hardware, firmware and test conditions; the calculations here establish conditional limits, not measured performance of a completed instrument.

C.1 OmniPreSense OPS243-A (24 GHz CW Doppler)

C.1.1 Hardware and Configuration Boundaries

The OPS243 product brief distinguishes the Doppler-only OPS243-A from the Doppler/FMCW OPS243-C: the former does not provide range from its unmodulated carrier [60]. The brief lists a 24.00–24.25 GHz band, 11 dBm transmit power and a $20^\circ \times 24^\circ$ transmit beamwidth at -3 dB. The horizontal contour width at axial distance r is approximately $2r \tan(10^\circ)$: 0.353 m at 1 m and 1.763 m at 5 m. A beamwidth contour is not a hard detection boundary, and the family-level 1–100 m motion-detection claim is not a golf-ball range guarantee.

Electrical details also require the exact variant: the USB/UART version uses 3.3 V UART signaling, whereas an RS-232 version uses different levels. The brief permits a 24 V supply only for revision-C boards, otherwise 13.5 V maximum; its 1.7 W typical active-power table entry explicitly names OPS243-C [60]. Do not transfer these values indiscriminately between board revisions and sensor variants or use a headline accuracy percentage as an established golf-shot uncertainty.

AN-010 documents the default 10 ksp/s rate, selectable sample buffers, zero-padding, timestamped JSON and raw I/Q outputs [58]. For the golf example below, **S=30** sets 30 ksp/s, **S(** selects 128 acquired samples and **X=32** requests a 4096-point transform; configure the buffer before the padding multiplier. **OT**, **OM** and **OJ** select timestamps, magnitudes and JSON; **OR** supplies raw samples, while **O2** requests two reported speeds. The UART default is 19,200 baud, 8N1. Verify supported commands and read back the configuration for the installed firmware before saving it with **A!**.

C.1.2 Sampling, Grid Spacing and Identifiability

For an ideal monostatic CW observation, the magnitude of Doppler frequency satisfies $|f_D| = 2|v_r|/\lambda$, where v_r is radial velocity and $\lambda = c/f_c$. The sign depends on the receiver convention; it must be checked before transforming observations into the book’s coordinate frame. For centered complex

sampling at rate f_s , N acquired samples and an M -point transform with $M \geq N$, the nominal quantities are

$$|v_r| < \frac{\lambda f_s}{4}, \quad T = \frac{N}{f_s}, \quad \Delta v_N = \frac{\lambda f_s}{2N}, \quad \Delta v_M = \frac{\lambda f_s}{2M}. \quad (\text{C.1})$$

Here Δv_N and Δv_M are the unpadded and padded velocity-grid spacings. The first expression is a sampling limit, subject also to the analog passband and firmware; it is not a guaranteed detection speed. With $f_c = 24.125$ GHz, $c = 299,792,458$ m/s, $f_s = 30,000$ s⁻¹, $N = 128$ and $M = 4096$, these give 93.20 m/s (208.48 mph), 4.267 ms, 1.456 m/s (3.258 mph) and 0.04551 m/s (0.1018 mph), respectively.

Zero-padding samples the same finite-record spectrum on a finer grid; it adds no observed cycles. An isolated peak can be localized more finely than the unpadded bin spacing under suitable signal-to-noise and signal-model conditions, but neither spacing is a universal accuracy specification or a guaranteed separation between two resolvable targets. Windowing, unequal target strengths, acceleration and clutter change the estimation problem. This distinction explains why AN-029's nominal 0.1 mph figure cannot establish 0.1 mph measurement accuracy [57]. Likewise, a 200 Hz reporting example has 5 ms between outputs; that alone does not establish a 5 ms independent observation, capture duty cycle or processing latency.

The sensor observes $v_r = \hat{\mathbf{r}}^T \mathbf{v}$, where $\hat{\mathbf{r}}$ points from sensor to target for a positive-receding convention. Target bearing and velocity direction are different quantities: knowing the former does not determine the latter. If the angle θ between velocity and line of sight is independently known, $v = v_r / \cos \theta$; near transverse motion, small angular errors become strongly amplified. For small signed perturbations, $\delta v / v \approx \delta v_r / v_r + \tan \theta \delta \theta$, with $\delta \theta$ in radians and nonzero denominators. A fixed cosine correction cannot recover an unknown, changing trajectory.

C.1.3 Rolling Buffer and Trigger Timing

AN-027 describes a separate capture mode: **G1** stores 4096 complex samples in 32 segments of 128, with subsequent processing off the sensor [59]. A software **S!** command or 3.3 V rising edge on J3 pin 3 (**HOST_INT**) triggers completion. **S#n** selects the historical segment count; the default eight gives 1024 pre-trigger and 3072 post-trigger samples. The record spans 136.53 ms at 30 kpsps. The note illustrates a microphone gate connection, but its **S#18** example uses 250 kpsps and must not be copied as a universal golf setting.

Choose the pre-trigger history from an explicit timing budget. For source-to-microphone distance d , sound speed c_s , trigger-chain delay τ_e , desired pre-impact history τ_h and a bound τ_q on segment/timestamp uncertainty, require

$$\frac{128n}{f_s} \geq \frac{d}{c_s} + \tau_e + \tau_h + \tau_q, \quad 0 \leq n \leq 32. \quad (\text{C.2})$$

At the declared illustrative value $c_s = 343$ m/s, 2 m takes 5.831 ms, not 6.6 ms; the latter estimate in AN-027 is not adopted here. Ignoring electronics and quantization only for an arithmetic example, 2 ms of pre-impact history requires two segments at 30 kpsps, providing 8.533 ms total history. This is not a deployment recommendation: measure or conservatively bound the omitted delays and retain enough post-trigger data for the intended estimator.

C.1.4 Interpreting the Vendor Golf Example

AN-029 places the sensor about 2–3 m behind the ball and gives a configuration using 30 kpsps, 128 samples, padding to 4096, mph output, a lower speed filter, a magnitude threshold and peak

averaging [57]. Its reported 5–10 m golf-ball detection distance is vendor application guidance, not a qualification across balls and environments. Two reported peaks still require target association: neither peak order nor a plausible speed ratio proves which echo belongs to the ball or to a specified point on the clubhead. The ball and rotating club can have different line-of-sight projections and different measurement times. Before forming the smash factor of Section 3.2, establish the relevant ball/club speed definitions, timing and projection corrections. A 1.0–1.50 acceptance gate would be an implementation heuristic, not a physical theorem; record rejected shots to avoid hiding failures through selection. Radial-speed observations alone do not uniquely determine three-dimensional launch velocity or spin.

C.2 RFbeam K-LD7 (24 GHz FSK, Dual-RX Angle)

Revision B describes one transmitter and two I/Q receivers separated by 6.223 mm, with FSK range estimation and an $80^\circ \times 34^\circ$ antenna pattern [72]. Its table lists 6 dBm typical and 10 dBm maximum EIRP; the distance range starts at 0.005 m, not 0.05 m, and the nominal distance resolution is 5 cm at the 5 m range setting. These are distinct entries, not demonstrated millimeter-range golf performance. The 1° angular-resolution entry likewise does not certify one-degree golf accuracy or full-field sensitivity.

The maximum speed setting is 100 km/h (27.78 m/s or 62.14 mph), with a typical 29 ms frame duration [72]. It restricts both club and ball observations when their radial components exceed that limit; assigning the module to club speed does not remove the restriction. More transverse placement may reduce Doppler but worsens the conversion to total speed and changes visibility. Out-of-range targets can produce erroneous detections, rather than a clean indication that the limit was exceeded.

The host interface defaults to 115200 baud, 8E1; INIT can select up to 3 Mbaud. GNFD requests raw samples (RADC), spectra (RFFT), detections (PDAT), a tracked target (TDAT) and/or frame completion (DONE) [72]. Check frame counters and measured transfer time: a nominal acquisition period does not guarantee that every requested raw frame reaches the host. Calibration must also establish antenna geometry, phase offsets, target association and timing before combining this module with another sensor.

C.3 TI IWR6843 (60–64 GHz FMCW MIMO)

The IWR6843 has three transmitters, four receivers, a DSP and radar hardware acceleration; the datasheet distinguishes 12.5 Msps complex-1x sampling from 25 Msps real/complex-2x sampling [78]. The 60–64 GHz tuning range is not a statement that every configuration processes 4 GHz of useful chirp bandwidth. For an effective processed sweep bandwidth B , the nominal range-separation scale is $\Delta R = c/(2B)$, or 3.747 cm at $B = 4$ GHz [77]. Windowing and signal conditions affect separation; isolated-target range precision is a different quantity. Array geometry and the selected board determine angular coverage and aperture, so a count of twelve TX–RX channels is not twelve independent azimuth elements.

Time-division MIMO acquires the transmitter channels at different instants; velocity-induced phase must be compensated before interpreting the virtual array as an angle measurement [67]. For a uniform single-target phase-sampling model with same-transmitter inter-chirp revisit interval T_r within a burst, the principal unaliased radial-speed interval has half-width $\lambda/(4T_r)$, while coherent observation time sets a velocity-separation scale near $\lambda/(2T_{\text{obs}})$. Thus faster updates, wider unambiguous speed, more transmitter channels and finer Doppler separation compete for

acquisition time [77, 67]. Ambiguity-resolution methods can alter this design tradeoff but need their own assumptions and validation.

A 10–20 Hz output stream has 50–100 ms between reports; this is not necessarily its internal coherent dwell time. Such reporting alone does not establish coverage of a chosen 3 ms launch interval, and shortening the frame alone does not prove a working launch monitor. Specify the chirps, usable bandwidth, ADC interval, transmitter schedule, synchronization, target return strength and processing/transport latency together. Range, direction and radial velocity estimates can then constrain a trajectory model; a single range-angle-Doppler detection still lacks the tangential velocity components and does not establish spin or club delivery.

C.4 Raspberry Pi Global Shutter Camera (Optical Module)

The IMX296 camera provides 1456×1088 pixels of $3.45 \mu\text{m}$ pitch, full-resolution operation near 60 fps and short exposures near $30 \mu\text{s}$, subject to operating conditions [71]. Its XTR trigger is a **1.8 V input**; the documented exposure is the low pulse width plus $14.26 \mu\text{s}$. Shared pulses can synchronize cameras, but accepted pulse rates and actual frame timing must be verified. Raspberry Pi documents enabling the IMX296 external-trigger mode through

```
echo 1 | sudo tee /sys/module/imx296/parameters/trigger_mode
```

Use the documented procedure for the installed driver version. Boards with Q2 fitted require the documented R11 modification; use suitable level conversion and board-specific instructions [70].

Exposure duration controls motion blur; frame interval controls temporal sampling. At 60 fps, consecutive frames are 16.67 ms apart: a ball moving at 70 m/s travels 1.167 m between them, but only 2.1 mm during a $30 \mu\text{s}$ exposure. Whether that yields usable images depends on field of view, magnification, illumination and trigger timing. It cannot supply two consecutive frame samples within a 3 ms interval, but it can record flight over a larger observation region. Multiple flashes within one exposure, as discussed in Section B.4, are one possible architecture; a faster camera or a different observation window is another. None alone establishes spin observability without resolvable surface features, timing and calibrated geometry.

C.5 Simulator Integration: GSPro Open Connect

GSPro Open Connect v1 is a documented JSON-over-TCP interface; its sample endpoint is localhost port 921, with the launch monitor acting as client [36]. It is an integration example, not an assertion that no other documented protocol exists. A shot supplies device identity, shot number, API version, data-presence options and ball fields including **Speed** (mph), **VLA/HLA** (degrees) and spin data. The specification accepts **TotalSpin/SpinAxis** or **BackSpin/SideSpin**; it also describes optional club fields, readiness/heartbeat messages and a player-information response [36].

Numeric field names do not preserve the scientific meaning of a measurement automatically. Verify units, angle signs, handedness, reference points and event times against Chapter 2; avoid labeling a signed spin-axis value as a draw independently of handedness and convention. Two spin parameters also do not encode an unrestricted three-component spin vector; any mapping must state its representation assumptions. The documented schema supplies no general per-field uncertainty or measurement-provenance contract. Keep a separate, shot-linked record of raw evidence, calibration, inferred versus observed quantities and unavailable values, as motivated by Chapter 10. Do not replace unavailable spin with an apparently measured zero or silently present a prior-driven estimate as an observation simply to fill a required field. An accepted simulator message confirms transport compatibility, not measurement validity.

C.6 Equipment Rules and Model Parameters

Equipment conformance limits are constraints, not measured properties of every conforming club or ball. The ball rules specify mass at most 45.93 g and diameter at least 42.67 mm [81]. Using these limiting values in Equation (8.1) is a declared modeling choice; it does not identify a particular ball's mass, diameter, inertia or aerodynamic coefficients.

The driver spring-effect test uses a characteristic-time limit of 239 μs with an 18 μs test tolerance [107]. This pendulum-test quantity is not the club-ball contact duration or a universal conversion to the restitution coefficient e in Equation (3.8). That coefficient belongs to the declared impact model, incident speed, impact location and club/ball pairing. The wood-head volume limit is 460 cm^3 with a 10 cm^3 tolerance; the MOI limit is 5900 g cm^2 with a 100 g cm^2 test tolerance about the vertical CG axis at a 60° lie angle [80]. This is one specified inertia component, not a bound automatically transferable to every axis in Equation (3.10).

Status checked October 4, 2026: the December 2023 announcement described changing the ODS test from 120 mph/10°/2520 rpm to 125 mph/11°/2200 rpm, originally beginning in 2028 [12]. The June 17, 2026 joint statement superseded that timetable: no ODS testing change before January 2030 while alternative approaches are evaluated [108]. Do not present the earlier 2028 schedule as current or assume the eventual conditions are settled. The 317 yd limit plus 3 yd tolerance concerns the specified overall-distance test, not a universal maximum carry for a conforming ball [81]. A flight-model comparison requires the actual launch conditions, atmosphere, aerodynamic characterization and prescribed bounce/roll treatment; merely keeping an arbitrary simulation below 320 yd neither validates it nor certifies ball conformance.

C.7 Detection Theory: CFAR Selection

Constant-false-alarm-rate (CFAR) detection adapts a threshold to an estimate of local background power. A fixed SNR cutoff and a masked region around zero Doppler do not, by themselves, specify a cell-averaging CFAR detector. CA-CFAR uses the mean of training-cell powers, excluding guard cells around the cell under test; OS-CFAR uses a selected order statistic [49, 50]. The training region, guard width, rank where applicable, noise model and target false-alarm probability are part of the algorithm.

For an illustrative square-law CA detector, suppose the test-cell noise and K training-cell powers are independent exponential variables with common mean μ . Let $\bar{P}$ be their training mean and declare a detection when $P_{\text{CUT}} > \alpha \bar{P}$. Conditioning on the training samples gives

$$P_{\text{FA}} = \mathbb{E}\left[e^{-\alpha \bar{P}/\mu}\right] = \left(1 + \frac{\alpha}{K}\right)^{-K}, \quad \alpha = K \left(P_{\text{FA}}^{-1/K} - 1\right). \quad (\text{C.3})$$

The product follows from the exponential Laplace transform for each independent training cell. It is a conditional model calculation, not a false-alarm guarantee for real golf data: windowing and zero-padding can correlate spectral samples, while clutter and target leakage can invalidate the common-background model. A mask specified only as a number of padded bins also changes its physical velocity width when the FFT configuration changes.

OS-CFAR can reduce the effect of some high-power contaminated training cells, with performance depending on the chosen rank and contamination pattern; it is not universally superior for a club followed by a ball [50]. Their closeness in physical space or time does not establish closeness in the processed range-Doppler cells. Guard cells cannot separate unresolved targets, and neither a universal 0.5–1 dB loss nor a preferred detector follows from the shot geometry alone. Compare

candidate detectors at matched false-alarm targets on declared single-target, multiple-target and clutter cases, including nondetections and mistaken associations.

Design implication

The useful design unit is the complete observation chain: sensor geometry, acquisition window, synchronization, estimator, uncertainty record and downstream model. A finer reported number does not create a new physical observation; a plausible impact or flight result does not independently validate the measurements used to produce it. Preserve these distinctions when relating measured ball flight back to club delivery and, further upstream, to the golfer's swing.

Appendix D

Patent Portfolio Compendium

This appendix organizes selected launch-monitor patent publications by technical topic and named organization. It includes US grants, published US applications and explicitly identified foreign or PCT documents. The links provide a route to the disclosures; they do not establish an exhaustive portfolio search or a commercial product’s implementation. TrackMan’s marking notice lists 47 US numbers at the October 1, 2026 check [90]; Uneekor’s product-specific marking list expressly notes that other products and applications may exist outside its table [104].

The priority column records the indexed priority year of each row’s leading document. It is distinct from that application’s filing year and the document’s publication year, and it does not assign a common priority to every related number in the notes. A claim’s entitlement to priority and a patent’s term require separate analysis. Descriptions identify technical subject matter rather than the complete limitations of every claim. Neither a matching title nor a shared inventor proves common ownership, a continuation relationship or use in a named product. *Also* denotes another technical reference, not identical claim scope or verified family membership. Publication of an application is a document type, not an assertion that prosecution remains pending.

Two attribution corrections surfaced by this sweep are worth flagging prominently: (i) the widely cited “radar + image data 3D tracking” publications US10596416 / US11697046 / US12128275 identify **Topgolf Sweden AB (Toptracer)**, not TrackMan; and (ii) Full Swing’s launch-monitor publication US2020/0147470A1 corresponds to the grant US11311789B2. The Garmin and USGA entries below separate the disclosed mechanism from unsupported product and company-portfolio attributions.

D.1 TrackMan A/S (Incl. Interactive Sports Games A/S)

Radar Fundamentals and Target-Line Deviation (2004–2011)

Patent	Subject	Prio.	Notes
US8085188B2	Deviation of launched projectile vs. image-designated target direction	2004	
US8912945B2	Radar measurements transformed to a launch-and-target coordinate frame	2004	
US9857459B2	Target-direction deviation using co-rotated camera and radar	2004	
US10473778B2	Radar trajectory correlated with an image feature and displayed	2004	
US10690764B2	Radar trajectory mapped into launch-and-target coordinates	2004	

Patent	Subject	Prio.	Notes
US9958527B2	Direction-of-arrival sensor: extra RX antenna resolves monopulse phase ambiguity	2011	the sparse-array geometry of Section 4.2

Spin Rate and Spin Axis

Patent	Subject	Prio.	Notes
US8845442B2	Spin rate via harmonic sidebands; model-dependent axis from flight curvature	2005	Also EP1698380B1
US9645235B2	Unmarked-ball spin frequency from radar harmonic spacing	2005	
US10393870B2	Radar harmonic spin estimation without ball-marking information	2005	
US10962635B2	Spin frequency from radar modulation harmonic spacing	2005	
US11143754B2	Model-dependent axis from trajectory accelerations and spin frequency	2005	
US10850179B2	Projected spin axis from multi-receiver Doppler components	2018	
US11446546B2	Spin-axis projection from angular positions of radar frequency components	2018	
US11938375B2	Spin-axis component from receiver-pair phase differences	2018	
US11673029B2	Embedded radar markers with a great-circle proximity constraint	2019	
US12179068B2	Interior radar-reflective strips with great-circle coverage	2019	
US12042698B2	Toppling frequency of non-spherical rotating objects	2018	

Club Impact From Image Features

Patent	Subject	Prio.	Notes
US10953303B2	Impact timing and location from tracked club-image points	2017	
US11439886B2	Image-based impact geometry; single-imager embodiment with optional shaft marker	2017	
US11612801B2	Image-tracked line angle evaluated at impact time	2017	

Patent	Subject	Prio.	Notes
US12263393B2	Tracked fix-line angle (e.g., shaft); known geometry for 3D reconstruction	2017	

Radar+Camera Fusion, Calibration, Tracer, Range, Short Game

Patent	Subject	Prio.	Notes
US10052542B2	Coordinating radar + image data; overlay	2004	
US10471328B2	Radar/image trajectory and target-line overlay	2004	
US10989791B2	Fused radar range/rate + imager angles track	2016	
US11828867B2	Radar range and image angles combined with sensor separation geometry	2016	
US11619708B2	Calibration using radar tracks and camera bearings augmented by radar range	2020	
US12517218B2	Sensor calibration by weighted 3D track-residual minimization	2020	
US11748985B2	Master clock; composite multi-moment images	2019	
US12067775B2	Clock-referenced composite imaging	2019	
US12586211B2	Motion-triggered changes to imager operating parameters	2023	
US9855481B2	Radar trajectory projected into sports-video images	2009	Also US10315093B2, US10441863B2, US11135495B2, US11291902B2
US10379214B2	Launch-location association; one- and two-radar embodiments	2016	also US11086005B2, US11921190B2, US12618962B2
US11452911B2	Imager/secondary-sensor correspondence; range-radar embodiment	2019	also US11986698B2
US12036465B2	Player/shot association from motion-sensor time and location	2021	
US12186643B2	Camera line-of-sight and terrain model for bouncing/rolling ball position	2021	
US12702912B2	Robotic camera calibration and terrain-based ball localization	2021	USPTO Gazette
US10444339B2	Bounce/slide/roll transitions from velocity or acceleration changes	2016	also US11079483B2, US11619731B2, US11946997B2
US11285367B2	Strategy simulation from player capability	2018	also US12109473B2

Patent	Subject	Prio.	Notes
US11951372B2	Model-based shot-improvement estimates; mishit-filtering disclosure	2020	also US12539454B2
US12616891B2	Automated ball/strike, biometric strike zone	2021	Baseball

D.2 Topgolf Sweden AB (Toptracer / Protracer)

The publications in this section identify Topgolf Sweden AB and include camera, radar and combined tracking methods.

Patent	Subject	Prio.	Notes
US8077917B2	Enhancing sports video with trajectory imagery	2005	
US10596416B2	3D tracking: radar + image data	2017	also US11697046B2, US12128275B2
US10898757B1	3D trace from image motion and polynomial/exponential radar-speed fits	2020	also US11504582B2, US11883716B2, US12330020B2
US11335013B2	Motion-filtered image sequences and timestamp-based virtual-position interpolation	2020	also US11557044B2, US12322122B2
US11644562B2	Backward trajectory extrapolation with error-gated launch-origin association	2020	Also US11771957B1, US12121771B2
US11513208B2	Radar spin from radial-velocity variations about center velocity	2020	Also US12105184B2
US12544624B2	Spin-candidate selection from observed trajectory and probable club	2023	Model-dependent inference
US11964188B2	Position-dependent trajectory error estimates and sensor/shot placement guidance	2021	
US11995846B2	Multi-hypothesis tracking with pruning and progressively richer motion models	2020	also US12361570B2
US11815618B2	Co-located FMCW radar coordination through sweep-time and frequency offsets	2022	also US12253622B2
US12206977B2	Predictive pan/tilt/zoom control from launch data and updated 3D tracking	2021	
US12298326B2	Wind estimation from observed/model acceleration differences across ball trajectories	2022	
US12594460B2	Scored blob traces and calibration data carried in a video audio channel	2023	

D.3 FlightScope / EDH (Henri Johnson)

Patent	Subject	Prio.	Notes
WO2003032006A1	Path speed from directional velocity and angular deviation	2001	PCT publication
US8189857B2	Impact-mark localization from before/after playing-surface images	2007	
US9036864B2	Bounce localization from composite long-exposure image envelopes	2011	
US9868044B2	Spin rate from periodic phase-demodulated radar features	2013	
US10775492B2	Spin-axis inference from receiver-pair delays and separations	2013	
US10338209B2	Camera/radar calibration and tracking through missing data	2015	also US11016188B2
US11573082B2	Effective-altitude estimation by matching environmental air density	2019	
US12528005B2	Weather-based range prediction, club selector	2023	
US20160306036A1	Putting-green tracking	2013	Published application
US20180239012A1	Antenna with boresight optical system	2013	Published application

D.4 Full Swing Golf

Patent	Subject	Prio.	Notes
US11311789B2	CW/FMCW radar and non-uniform antenna array; grant of US2020/0147470A1	2018	Also US11844990B2
US11875517B2	Frame-difference tracking and calibrated golf-surface impact location	2020	also US12354282B2
US8758103B2	Fixed-plane crossing and image-based translation/rotation; IR embodiment	2009	also US9616346B2, US11033826B2
US8926416B2	Strobed-image spin with separate two-plane timing/position sensing	2007	also US10058733B2
US8414408B2	Ball-permeable screen, ball return	2009	also US8834284B2

Caution: US10605910B2/US11086008B2 (Alphawave Golf) and US11565166B2 (individual) surface in “Full Swing” text searches but are unrelated assignees.

Historical Light-Curtain Method: US5333874A (indexed priority 1992) describes two receiver planes and a post-screen rebound measurement used with stored spin values. Kiraly and Wintriss appear among the inventors; shared inventorship does not establish Foresight ownership. The index lists an individual original assignee and Full Swing as current assignee, rather than a Foresight portfolio entry. This is a rebound-based inference, not direct image tracking of a rotating ball surface.

D.5 Garmin

US11351436B2 — “Hybrid golf launch monitor” (indexed priority 2019): two laterally separated cameras whose capture timing depends on clubhead speed and a swing trigger. Claim 1 describes the two-camera arrangement; claim 9 adds a radar-based trigger and clubhead speed estimate. This disclosure does not establish that the Approach R10 implements the claimed arrangement. For scope comparison, other Garmin disclosures are US8647214B2 (2008, associating inertial swing data with ball start/rest positions) and US7467060B2 (2006, stride-based estimation for cyclical motion). The latter’s first claim uses acceleration and stride duration; it should not be presented as a ball-flight or golf-swing reconstruction method. These entries do not establish the completeness of Garmin’s portfolio.

D.6 Rapsodo Pte. Ltd.

Patent	Subject	Prio.	Notes
US9955126B2	Radar-speed-based camera triggering for a projectile entering the image region	2015	
US11170513B2	Surface-feature template matching for ball-spin estimation	2016	
US11747461B2	Radar-track prediction; camera/radar association disclosed	2018	
US20210299540A1	Scanned-club model with camera/radar swing representations	2018	Published application
US20230065614A1	Neural-network spin estimation from radar features	2021	Published application
US20230364468A1	Neural estimation from separate ball-image and swing-radar representations	2021	also club-side US20230070986A1
US12169941B1	Target-plane crossing localization	2024	
US12586248B2	Camera/radar depth inference using radial speed, its slope and zero-speed timing	2024	also US12548194B2
US12158517B1	Imager triggering from a two-tone radar phase-difference zero crossing	2024	

D.7 Camera Vendors: Foresight/Wintriss, Creatz/Uneekor, Golfzon

Foresight Sports, Wintriss and WAWGD Disclosures

Patent	Subject	Prio.	Notes
US7292711B2	Image-based ball motion and surface-feature spin disclosure	2002	
US7324663B2	Image rotation/correlation search and alignment correction	2002	
US7497780B2	Integrated trigger, capture, calculation and display	2006	
US7540500B2	Foldable monitor housing	2006	
US7641565B2	Ball-placement detection / auto-arm	2006	
US8951138B2	Club path and face orientation around impact	2012	
US9737757B1	Alignment-stick target alignment	2016	
US10639537B2	Associating observed flight/landing tracks with launch-monitor predictions	2018	

Creatz Inc. (Uneekor)

Patent	Subject	Prio.	Notes
US9448067B2	Multi-camera (unsynchronized) trajectory via projection-plane intersection	2011	
US9605960B2	Single-camera trajectory from an image-defined plane and image-size database	2011	
US9752875B2	Iterative exposure/gain control using reference regions of different reflectance	2011	
US10247553B2	Start sensor: ball state at first movement	2011	
US10587797B2	Ball-image brightness compensation for spin marks	2016	
US10776929B2	Ball/camera-position and club-dependent imaging region; predicted-motion embodiment	2016	
US11191998B2	Speed/angle-based spin estimates correct or replace mark-derived spin	2018	
US12008770B2	Dimple-position comparison with lighting correction for spin	2020	

Uneekor's marking page also lists the four licensed Wintriss/Foresight patents (US7497780, US7292711, US7641565, US7324663) [104, 6].

Golfzon Co., Ltd. (Sensing Core)

Patent	Subject	Prio.	Notes
US9242158B2	Two-stage sensing → simulation (latency hiding)	2011	
US9333409B2	Ball-candidate 2D trajectories used to derive 3D motion	2011	Also US9333412B2
US9162132B2	Image-based ball recognition and shot detection on a hitting area	2011	
US9514379B2	Difference images and estimated club trajectory inform ball-spin estimation	2011	
US10045008B2	Alternating stereo capture with estimated intermediate images	2011	
US11364428B2	Spin fit by trajectory iteration vs. observed positions	2017	
US12002222B2	Database-derived loft/backspin and motion-vector spin-axis inference	2016	
US12599828B2	Marker-constellation spin between frames	2021	
US12605593B2	Next-frame image-search region from a projected 3D motion bound	2021	
US12478849B2	Single-camera putting-mat sensing	2020	
US20230347209A1	Stereo impact position on club face	2020	Published application

D.8 Acushnet (Titleist) — Selected Optical Disclosures

Patent	Subject	Prio.	Notes
US4136387A	Enhanced-point optical estimates of translation and rotation	1977	
US5471383A	Shutterable cameras + retroreflective markers	1992	
US5501463A	Stereo + club/ball dots: face, path, contact location	1992	
US6241622B1	Portable camera/lighting system and trajectory computation	1998	Also US6533674B1, US6616543B1, US7086955B2
US6500073B1	Image-derived launch conditions and computed trajectory	1992	
US6758759B2	Separate club and ball camera arrangements	2001	
US7143639B2	Integrated club/ball cameras, illumination and trigger layout	2004	Also US7395696B2, US8500568B2, US8556267B2

Patent	Subject	Prio.	Notes
US10668350B2	3D-image reconstruction; light-field camera in claim 1	2017	

D.9 Others and Foundational Radar Art

United States Golf Association: US6186002B1 (indexed priority 1998) is assigned in the publication record to the USGA, not Acushnet. It describes fitting lift and drag coefficients from launches with specified initial conditions and timed ballistic light-screen crossings, then relating those coefficients to Reynolds number and spin ratio. It is a useful experimental-modeling reference, not evidence that arbitrary launch-monitor outputs uniquely identify the aerodynamic coefficients.

AccuSport: US7209576B2 (indexed priority 2002; original assignee AccuSport International Inc.) describes golf-ball image processing, including marked-surface spin estimation and image calibration.

SkyTrak / SkyHawke: US12515116B2 (2023, swing-tag + launch-monitor fusion display) is one relevant disclosure. This catalog does not establish the complete SkyHawke portfolio or the patent marking of every SkyTrak model. No verified entries for **Zelocity**, **GolfTek**, **Ernest Sports** or **ProTee United** are supplied here; that search limitation establishes neither the absence of patents nor the technical ancestry of their products. **Voice Caddie (Ucomm):** US10338212B2 (2014) specifies in claim 1 club-speed calculation from time-domain processing and ball-speed calculation from frequency-domain processing of an output generated from a mixed Doppler return. **Sports Sensors (Dilz):** US6079269A (1997) describes a short-range sports-object speed sensor, including a ball-catching application; it should not be labeled the Swing Speed Radar product patent. US6898971B2 (2000) addresses short-range speed measurement, including clubhead speed; US8007367B2 (2005) explicitly addresses swing duration and clubhead speed. These are different measurement tasks.

Weibel Scientific: EP1735637B1 (2004) describes simultaneous CW and FMCW tracking with three receivers, phase-based angles and maintained track records. Calling it only a CW-speed method omits a material part of the arrangement. **Applied Concepts:** US10935657B2 (2019) estimates projectile spin from radar spectral shoulder frequencies adjacent to a peak. Its relevance extends beyond golf; the disclosure is not evidence of a particular product's implemented accuracy.

Design implication

The catalog compares observations and assumptions behind similarly named outputs. Radar periodicity, image rotation and trajectory fitting need different calibration and ambiguity tests; a grant establishes no accuracy benchmark. For each implementation, retain the measured signals and model dependencies when reporting derived or estimated quantities. The rights boundaries in Section 7.8 apply throughout this appendix.

Appendix E

Radar Clubhead Kinematics: Motion and Observability

A useful club-motion model must connect what the sensor records to what a golfer wants to know: which point on the head is moving, how the face is turning, and how confidently those quantities are known near impact. Screw theory provides a compact representation of rigid motion, but choosing that representation does not make the motion observable. This appendix develops the measurement equations, exposes an exact ambiguity in instantaneous single-origin Doppler data, and specifies what an implementer must add and validate before reporting club-delivery parameters. The estimator described below is a research design, not a demonstrated IWR6843 golf instrument. The point-tracking framework in Section B.2 is a starting point; a rigid-body estimator also needs orientation, a calibrated head reference, and an explicit account of unresolved motion.

E.1 Context: Definitions Are Not an Estimator Specification

TrackMan defines club speed at the head’s geometric center immediately before first contact, while its path and attack-angle definitions use maximum compression as their time reference [94, 92]. These definitions specify intended outputs, not the complete sensing and estimation algorithm. The manufacturer’s discussion of a three-dimensional club silhouette and its patents describe additional information beyond one list of radial velocities; they do not establish that the particular Doppler-only solve below reconstructs a head pose [87, 100]. Nor does an algorithm’s appearance in a patent establish its implementation or independently measured accuracy in every product version. This appendix makes no claim about an undisclosed vendor state representation or the novelty or legal scope of the proposed design.

Instantaneous screw axes also have a distinct use in biomechanics. Vena et al. studied optically reconstructed pelvis, shoulder, and left-arm motion: Part 1 reports predominantly rotational contributions to marker velocity in those segments, and Part 2 reports the expected segment sequence in two of five subjects [109, 110]. Those observations neither validate a radar clubhead estimator nor establish a universal axis near the hands. The publisher abstracts and the visible Part 1 mathematical appendix were checked for this review; the subscription main texts were not accessed. The robotics conventions below are independently supported by the authors’ explanation of body and spatial twists [48]; Murray, Li, and Sastry remain a general reference [52].

E.2 Rigid Motion With Declared Frames and Reference Points

Assume a rigid clubhead over the interval being modeled, a stationary radar origin, and a right-handed Cartesian sensor frame. This is a head model: shaft bending and deformation during contact require separate states or an explicitly bounded approximation. Let O be a declared body-fixed reference point, $\mathbf{p}_O$ its sensor-frame position, $\mathbf{v}_O = \dot{\mathbf{p}}_O$ its velocity, and $\boldsymbol{\omega}$ the head’s angular velocity

expressed in the same frame. Use linear-first ordering throughout:

$$\xi_O = \begin{bmatrix} \mathbf{v}_O \\ \boldsymbol{\omega} \end{bmatrix} \in \mathbb{R}^6. \quad (\text{E.1})$$

This six-component velocity description becomes a body or spatial Lie-algebra twist only after the reference and expression frame are specified below. For a body-fixed point whose instantaneous sensor-frame offset from O is $\mathbf{r}$,

$$\mathbf{v}(\mathbf{r}) = \mathbf{v}_O + \boldsymbol{\omega} \times \mathbf{r}. \quad (\text{E.2})$$

Moving the reference to another body-fixed point Q therefore changes the linear component to $\mathbf{v}_Q = \mathbf{v}_O + \boldsymbol{\omega} \times (\mathbf{p}_Q - \mathbf{p}_O)$. The physical velocity field is unchanged; its six numerical coordinates are not invariant to changes of reference or frame.

For $\|\boldsymbol{\omega}\| > 0$, the instantaneous screw axis is the line

$$\begin{aligned} \mathbf{p}_{\text{axis}}(a) &= \mathbf{p}_O + \mathbf{r}_{\text{ISA}} + a\hat{\boldsymbol{\omega}}, & \mathbf{r}_{\text{ISA}} &= \frac{\boldsymbol{\omega} \times \mathbf{v}_O}{\|\boldsymbol{\omega}\|^2}, \\ h &= \frac{\boldsymbol{\omega} \cdot \mathbf{v}_O}{\|\boldsymbol{\omega}\|^2}, & \hat{\boldsymbol{\omega}} &= \frac{\boldsymbol{\omega}}{\|\boldsymbol{\omega}\|}. \end{aligned} \quad (\text{E.3})$$

Here a is distance along the axis and h is signed axial translation per radian. Substitution into Equation (E.2) gives velocity $h\boldsymbol{\omega}$ at every point on the axis, which verifies the construction. Pure translation has no unique finite axis, rest has no preferred axis, and near-zero angular speed makes the axis and pitch numerically sensitive. An instantaneous axis is not a uniquely defined swing plane: its moving line need not remain normal to one plane, and a head-point trajectory need not be planar. A plane estimate needs its own declared point trajectory, fitting interval, residual, and uncertainty. Low pitch or an axis near the grip can be hypotheses to test; neither is an assumption that the cited segment study licenses for every club swing.

E.2.1 Body and Spatial Twists for Pose Integration

Let $T = \begin{bmatrix} R & \mathbf{p}_O \\ 0 & 1 \end{bmatrix}$ map body coordinates to the sensor frame, with $R \in SO(3)$. For a linear-first pair $V = (\mathbf{v}, \boldsymbol{\omega})$, define $V^\wedge = \begin{bmatrix} [\boldsymbol{\omega}]_\times & \mathbf{v} \\ 0 & 0 \end{bmatrix}$, where $[\mathbf{a}]_\times \mathbf{b} = \mathbf{a} \times \mathbf{b}$. Then

$$\begin{aligned} V_b &= (R^\top \mathbf{v}_O, R^\top \boldsymbol{\omega}), & V_b^\wedge &= T^{-1} \dot{T}, \\ V_s &= (\mathbf{v}_O - \boldsymbol{\omega} \times \mathbf{p}_O, \boldsymbol{\omega}), & V_s^\wedge &= \dot{T} T^{-1}. \end{aligned} \quad (\text{E.4})$$

These are related by the adjoint of T [48]. A constant body-twist step uses $T_{k+1} = T_k \exp(V_{b,k}^\wedge \Delta t)$; a constant spatial-twist step uses $T_{k+1} = \exp(V_{s,k}^\wedge \Delta t) T_k$. For varying motion these are integration approximations; the increments must be composed in time order. Inserting sensor-frame $\mathbf{v}_O$ directly into the right-multiplying body update generally gives the wrong trajectory. Likewise, a logarithm of two measured poses describes a finite interframe displacement; dividing it by the interval is not automatically an unaliased instantaneous angular velocity.

E.3 The Measurement Model and Its Exact Ambiguity

First consider an ideal monostatic measurement represented by a single origin: a localized scattering point $\mathbf{p}_i \neq 0$ has sight line $\hat{\mathbf{u}}_i = \mathbf{p}_i / \|\mathbf{p}_i\|$, offset $\mathbf{r}_i = \mathbf{p}_i - \mathbf{p}_O$, and radial velocity z_i positive away

from the radar. Assume its Doppler return represents the instantaneous velocity of head material there. The conditional measurement equation is

$$z_i = \hat{\mathbf{u}}_i^\top \mathbf{v}_O + (\mathbf{r}_i \times \hat{\mathbf{u}}_i)^\top \boldsymbol{\omega} + \epsilon_i. \quad (\text{E.5})$$

With the ordering in Equation (E.1),

$$\mathbf{z} = A\boldsymbol{\xi}_O + \boldsymbol{\epsilon}, \quad A_i = [\hat{\mathbf{u}}_i^\top \mid (\mathbf{r}_i \times \hat{\mathbf{u}}_i)^\top]. \quad (\text{E.6})$$

Linearity holds conditional on the positions, associations, and rigid-motion model. It does not say that six independent unknowns can be recovered. Because $\mathbf{p}_i \times \hat{\mathbf{u}}_i = 0$, the same equation factors as

$$\begin{aligned} \mathbf{z} &= U(\mathbf{v}_O - \boldsymbol{\omega} \times \mathbf{p}_O) + \boldsymbol{\epsilon}, \\ A &= U[I \mid [\mathbf{p}_O]_\times], \quad \text{rank}(A) \leq 3, \end{aligned} \quad (\text{E.7})$$

where row i of U is $\hat{\mathbf{u}}_i^\top$. For any $\delta\boldsymbol{\omega}$, the perturbation $\delta\mathbf{v}_O = \delta\boldsymbol{\omega} \times \mathbf{p}_O$ leaves every predicted Doppler value unchanged. This is an exact three-dimensional null family even with arbitrarily many noncoplanar localized detections and noise-free measurements. Nearly parallel sight lines can further weaken the remaining three combinations; they do not explain away the structural nullspace. The observed combination is the translational component of V_s , not necessarily the speed of the head center.

Key point

One origin's instantaneous Doppler rows cannot determine the complete club velocity field under this model. Better signal-to-noise ratio or more rows at that same origin cannot remove this ambiguity. Range and angle histories, known feature geometry, independent orientation measurements, or sufficiently informative additional sensor geometry can supply information outside these rows; their contribution must appear in the actual measurement model.

For example, known and correctly associated noncollinear body features with resolved three-dimensional positions can constrain pose; their time history can then constrain motion. Unlabeled specular detections are not automatically such features. Two separated monostatic stations can generically raise the Doppler-only rank to five, but retain rotation about their baseline as a null direction with its corresponding translational change. Three noncollinear station origins can generically reach six with sufficient point/sight-line diversity. These are geometric possibilities, not guarantees of adequate precision. A multiple-input multiple-output (MIMO) array also requires its actual transmit/receive path model: a bistatic path derivative involves both ray directions. Twelve virtual array channels are not twelve independent monostatic stations, and the single-origin bound is not a theorem that every near-field, coherent, or feature-tracking radar architecture is limited to three observable motion coordinates.

E.3.1 What Least Squares and Covariance Actually Provide

For fixed A and zero-mean measurement error with known positive-definite covariance R_z , generalized least squares minimizes $(\mathbf{z} - A\boldsymbol{\xi})^\top R_z^{-1}(\mathbf{z} - A\boldsymbol{\xi})$. If A has full column rank, its estimator covariance is $(A^\top R_z^{-1}A)^{-1}$ under that model. This inverse does not exist for the six-unknown single-origin problem. A pseudoinverse selects a convention for the unobserved components; a zero eigenvalue in a pseudoinverse covariance is not evidence of zero uncertainty there. Report the identifiable combinations and the remaining nullspace, or explicitly state the extra observations or priors that define a full posterior.

Inspect singular values after noise whitening and declared parameter scaling, for example $R_z^{-1/2}AD$ with D converting dimensionless unknowns into chosen linear- and angular-velocity scales. Raw singular values mix meters per second with radians per second, so an unexplained threshold is not a physical observability criterion. Uncertain positions make A uncertain; position–Doppler correlations, calibration bias, ambiguous associations, and model discrepancy require joint propagation or an appropriate errors-in-variables model. Robust losses such as Huber or Cauchy generally require iterative fitting and their own uncertainty treatment. They do not turn an incorrect scattering model into a calibrated rigid-body measurement.

E.4 What Representative Hardware Adds

E.4.1 OPS243-A: Radial-Speed Information

The manufacturer’s OPS243 product brief distinguishes the Doppler-only A model from the range-capable C model [60]. Without per-return range and direction, the A output alone cannot assemble the located rows above. A Doppler spectrum is a scattering-amplitude-weighted signal, not a uniform sample of clubhead material velocities. Its peak need not always be the toe, its midpoint need not be the geometric-center speed, and its width is not a uniquely calibrated angular speed. For a stated example of 30 kps, 128 samples, and 24 GHz, $\Delta v = cf_s/(2f_cN) \approx 1.46$ m/s, or 3.27 mph, before accounting for the spectral window. Zero-padding samples the spectrum more finely but does not add the observation time needed to resolve closer velocities. A percentile or spectral-width feature may be useful after empirical calibration across aspect, club, speed, and clutter; it should retain that provenance.

E.4.2 K-LD7: Geometry and Operating Envelope

The K-LD7 Revision B datasheet specifies range, radial speed, and one angular coordinate, with a maximum speed setting of 100 km/h and a 29 ms cycle at that setting [72]. The speed limit is approximately 27.8 m/s, not a limit on total three-dimensional speed independent of viewing direction. A head moving at 50 m/s travels 1.45 m during that cycle; its radial component may alias even though another orientation does not. A single angular coordinate also does not localize arbitrary three-dimensional scatterers. Neither these specifications nor a frame’s angle output establishes that high-speed ball returns remain valid angle measurements after Doppler aliasing. Qualification would need the chosen geometry, waveform, association, and motion envelope.

E.4.3 IWR6843: A Coupled Waveform Budget

The IWR6843 has three transmitters, four receivers, a 60–64 GHz band, and a maximum specified ramp rate of 250 MHz/μs in Revision F [78]. Array angle estimation and multiplexing depend on antenna geometry and processing; they are not generically synonymous with monopulse [67]. Let T_c be the start-to-start chirp interval, n_T the number of cyclic time-division-multiplexed transmitters, and N the number of repeated samples *per transmitter*. For uniform slow-time sampling, wavelength λ , and a conventional unambiguous Doppler FFT,

$$T_r = n_T T_c, \quad |v_r| < \frac{\lambda}{4T_r}, \quad \Delta v = \frac{\lambda}{2NT_r}, \quad T_{\text{obs}} \approx NT_r. \quad (\text{E.8})$$

These follow from the round-trip phase increment $4\pi v_r T_r / \lambda$ and the sampled Fourier frequency grid [77, 67]. At the illustrative $\lambda = 5$ mm, $T_c = 25$ μs, and $N = 128$:

Transmit Schedule	Speed Limit (m/s)	Δv (m/s)	T_{obs} (ms)
One transmitter	50.0	0.781	3.2
Three-transmitter TDM	16.7	0.260	9.6

The table is arithmetic, not a validated firmware configuration; frame idle time and processing can lower the output rate further. At the sampling boundary the signed velocity is ambiguous, so a 50 m/s limit provides no margin for a 50 m/s return. For three-transmitter TDM to exceed that unambiguous radial speed at this wavelength requires $T_c < 8.33 \mu\text{s}$. Sweeping 4 GHz already needs at least 16 μs at the stated maximum ramp rate, before settling and idle time. Thus those requirements cannot be combined in that simple waveform; ambiguity-resolution schemes or other multiplexing designs need their own analysis and validation.

Nominal range resolution is $c/(2B) \approx 3.75 \text{ cm}$ for an effective sampled bandwidth $B = 4 \text{ GHz}$ [77]. It does not guarantee three or four distinct returns from a head: projected radial extent, reflectivity, angular resolution, windowing, and detection thresholds also matter. Even over 3.2 ms, a 50 m/s radial return migrates 16 cm, more than four such bins. Acceleration, changing aspect, range migration, and inter-transmitter phase changes must therefore be included in the signal model or shown to be negligible for a bounded operating case. TI specifically describes motion-phase correction before TDM angle estimation [67]. Select sampled bandwidth, slope, ADC start/sample time, IF bandwidth, settling, transmitter schedule, output timing, and processing load together [77]; a bandwidth, frame rate, and velocity ceiling cannot be independently promised.

E.5 An Estimation Recipe With Explicit Acceptance Conditions

Step 1 — Preserve Geometry, Timing, and Detection Uncertainty

Retain the raw signal or sufficient diagnostics to check aliasing, motion compensation, and false detections. Choose CFAR training/guard cells and any order-statistic variant against recorded clutter and false-alarm requirements; no detector is universally preferable for every impact scene. For a declared right-handed sensor frame with range ρ , azimuth θ , and elevation ϕ , convert spherical coordinates to Cartesian positions:

$$\mathbf{p} = \rho \begin{bmatrix} \cos \phi \cos \theta & \cos \phi \sin \theta & \sin \phi \end{bmatrix}^T. \quad (\text{E.9})$$

Propagate the full range/angle covariance through this mapping and preserve relevant correlations with Doppler. A reported FFT bin spacing, a precision estimate, and demonstrated accuracy are different quantities. Assign timestamps to the acquisition intervals; a frame output timestamp is not an instantaneous sample of all returns.

Step 2 — Associate Returns With a Head Model

Use range, radial velocity, geometry, temporal continuity, and calibrated uncertainties jointly to distinguish head, shaft, ball, and body returns. A shaft point is not invariably slower in radial speed or nearer the sensor than the head, and speed alone cannot guarantee removal of body clutter. An SNR-weighted detection centroid can migrate as aspect changes; its derivative includes that migration and is not automatically the velocity of a body-fixed point. Choose O through a calibrated geometric model or identified body feature, and distinguish material-point velocity from apparent glint motion. If correspondence or head geometry is unresolved, retain multiple hypotheses or withhold outputs that depend on it.

Step 3 — Fit the Identifiable Quantities First

Use Equation (E.7) to fit the three-component combination $\mathbf{b} = \mathbf{v}_O - \boldsymbol{\omega} \times \mathbf{p}_O$ when the weighted sight-line matrix supports it. A smaller observable subspace may be all the data justify. Record residuals, scaled singular values, associations, and the acquisition interval. Do not publish an arbitrary six-component pseudoinverse solution as a fully measured twist. A full solve needs additional, explicitly modeled observations, such as calibrated feature-position histories or independent orientation information.

Step 4 — Establish Temporal Observability

A changing sight line is not a proof that smoothing recovers six arbitrary motion functions. For Doppler-only histories, even a constant rotation of every head position and velocity about the sensor preserves all radial histories while changing the target-relative motion; both trajectories can be smooth. Angle or feature-position observations may distinguish them, which is precisely why those observations must be included rather than used only as fixed coefficients in a velocity solve. Analyze the linearized multi-time measurement model with the chosen state transitions, geometry, calibration states, and priors. Report structural nullspaces separately from poor conditioning and from uncertainties controlled chiefly by a prior. Use withheld trajectories and deliberate near-degeneracies to test the claimed operating region.

Step 5 — Estimate Pose and Motion Consistently

An $SE(3)$ filter or batch smoother can combine pose, velocity, calibration, and feature observations using Equation (E.4). Feed raw residuals where practical, so the estimator retains their actual information and correlations; a per-frame diagnostic fit can still be useful. A backward smoother uses future measurements and must be labeled offline if they are unavailable at the desired reporting time. Select process uncertainty against observed dynamics, not an unsupported universal acceleration of one mile per hour per millisecond. Low-pitch or grip-proximity penalties are optional model assumptions whose sensitivity must be reported; they cannot resolve an unobservable quantity through measurement alone and become singular near zero angular speed if written using Equation (E.3) without safeguards. A prior that is weak in observed directions may completely determine an unobserved one.

Step 6 — Define the Impact Event and Model Boundary

Synchronize the radar, any optical reference, and the contact-event measurement, including sound-propagation and trigger latency where an acoustic trigger is used. Distinguish the last resolved pre-contact state, first contact, maximum compression, and separation. A prediction from the last pre-contact sample to another event is an extrapolated estimate with event-time and model uncertainty. The manufacturer's historical club-data explanation explicitly discusses using pre-impact data for an event-time definition [87]; the chosen product definition must accompany any comparison. Do not assume that every head is exactly rigid until a universal 500 μs switch, or extend a pre-contact process model through collision and turf contact without validation. Fit pre- and post-contact intervals separately if the purpose is to study collision-induced changes.

Step 7 — Report Only Supported Delivery Quantities

Let the target frame be right-handed with x down the target line, y to the golfer's left when looking down that line, and z upward. For a calibrated body-fixed point with body offset $\mathbf{r}_{gc}^b$,

first compute $\mathbf{v}_{gc} = \mathbf{v}_O + \boldsymbol{\omega} \times R\mathbf{r}_{gc}^b$ and express it in that target frame. Then speed is $\|\mathbf{v}_{gc}\|$, path positive to the right is $\text{atan2}(-v_y, v_x)$, and attack angle positive upward is $\text{atan2}(v_z, \sqrt{v_x^2 + v_y^2})$. Path is undefined for zero horizontal velocity, and the velocity direction is undefined at rest. A calibrated geometric-center offset, an anatomical grip point, and a changing radar centroid are not interchangeable.

Face closure needs a definition and orientation information. For a known target-frame unit face normal $\mathbf{n}$, rigid rotation gives $\dot{\mathbf{n}} = \boldsymbol{\omega} \times \mathbf{n}$. Its mathematical horizontal azimuth $\gamma = \text{atan2}(n_y, n_x)$ satisfies

$$\dot{\gamma} = \frac{n_x \dot{n}_y - n_y \dot{n}_x}{n_x^2 + n_y^2}, \quad n_x^2 + n_y^2 > 0. \quad (\text{E.10})$$

This generally differs from $\boldsymbol{\omega}$ projected onto the shaft axis. A right-positive face-angle convention reverses this azimuth sign; a label such as closing also needs the handedness and stated convention. A full head frame requires more orientation information than one normal, and face angle at the contact location may additionally require head geometry and impact location. Thus closure rate is not a free Doppler-only output or a parameter whose product availability can be declared universally exclusive.

Estimate swing plane and swing direction only through a specified trajectory-plane fit, with its residual and fit window. A low point requires an observed or modeled trajectory with $v_z = 0$ and a local height minimum; a vertical-velocity zero alone can be a maximum. A difference between pre- and post-impact angular velocities is a net kinematic change. Attributing it to gear effect requires an impact/contact model, timing, inertia, and consideration of continuing shaft/grip and other contact loads.

Propagate joint state, calibration, association, timing, and model uncertainty into each output and disclose unresolved directions or dependence on priors. Keep the book's distinction between sensor observations, quantities derived from them, and model-based estimates; merely changing the state representation promotes no output into a measured category. Kinematics alone also identifies neither golfer torques nor a control strategy: a hand-axis prior must not be presented as evidence that the golfer actually produced that motion by the assumed mechanism.

E.6 Validation That Can Falsify the Proposed Estimator

The following is a proposed qualification sequence, not a report of completed hardware or human testing. Use the release and evidence discipline of Chapter 11.

1. **Analytic and Synthetic Checks.** Verify the exact null family, frame conversions, pure-translation limit, and waveform arithmetic before fitting realistic data. Simulations should include noise, glint changes, aliasing, missed detections, correlated errors, and deliberately unobservable cases, and should test empirical interval coverage as well as point error. Recovering truth in a simulation generated with the same restrictive prior cannot establish that the prior holds for golfers.
2. **Independent Rig Reference.** Use synchronized calibrated encoders or optical tracking for pose and velocity on a rigid-motion rig. A fixed-axis pendulum supplies a known axis only if the bearings, mounting, head rigidity, and coordinate calibration support that model; its angular history is not known merely because it is called a pendulum. Test translations, multiple axes, low angular speeds, varying aspects, and moving-pivot cases as well as the easy planar case.

3. **Independent Cross-System Comparison.** Match the head point, event time, coordinate convention, club, environment, and uncertainty of the reference. A commercial monitor is a comparator, not automatically ground truth; unavailable or model-derived comparator outputs cannot validate an independent direct measurement. Report bias, limits of agreement, repeatability, missing-output rates, and uncertainty coverage against predefined requirements, including difficult cases.
4. **Mechanics Consistency With Stated Assumptions.** Check geometric identities only where their planar or rigid-body assumptions apply. An identity satisfied by construction is a software consistency check, not independent physical validation. Do not discard shots merely because a simplified smash-factor or swing-plane model is violated: inspect reference-point mismatch, contact, alignment, and model inadequacy, and preserve the rejection rule and excluded cases.

Design implication

The value of a screw representation is a common account of rigid motion and point transport. Its practical benefit depends on calibrated geometry, sufficient observations, consistent timing, and validated uncertainty. Declare what the data constrain before reporting a complete club state, and keep the inferred motion separate from claims about the forces or intentions that produced the swing.

Bibliography

Peer-Reviewed Literature and Books

- [1] P. W. Bearman and J. K. Harvey. “Golf Ball Aerodynamics”. In: *Aeronautical Quarterly* 27.2 (1976), pp. 112–122. URL: <https://www.cambridge.org/core/journals/aeronautical-quarterly/article/abs/golf-ball-aerodynamics/67FE0903DB1CC12001F1ED1B1261C4B9> (visited on 07/31/2026).
- [2] C. Bishop, J. Wells, A. Ehlert, A. Turner, D. Coughlan, N. Sachs, et al. “Trackman 4: Within and between-session reliability and inter-relationships of launch monitor metrics during indoor testing in high-level golfers”. In: *Journal of Sports Sciences* 41.23 (2023). Published online 8 February 2024, pp. 2138–2143. DOI: 10.1080/02640414.2024.2314864. URL: <https://www.tandfonline.com/doi/full/10.1080/02640414.2024.2314864> (visited on 10/04/2026).
- [3] J. M. Bland and D. G. Altman. “Agreement Between Methods of Measurement with Multiple Observations Per Individual”. In: *Journal of Biopharmaceutical Statistics* 17.4 (2007), pp. 571–582. DOI: 10.1080/10543400701329422.
- [4] A. Bliss and B. L. Langdown. “Comparing indoor tracking of golf ball and club metrics: Consistency and absolute agreement of the Flightscope Mevo+ and Trackman 4 launch monitors”. In: *JSAMS Plus* 7 (2026). Published online 18 December 2025, p. 100128. DOI: 10.1016/j.jsampl.2025.100128. URL: <https://oro.open.ac.uk/107777/1/1-s2.0-S2772696725000420-main.pdf> (visited on 10/04/2026).
- [8] V. C. Chen. *The Micro-Doppler Effect in Radar*. 2nd ed. Standard treatment of Doppler signatures of rotating and vibrating rigid bodies. Artech House, 2019.
- [9] A. Cochran and J. Stobbs. *Search for the Perfect Swing*. Golf Society of Great Britain, 1968. URL: <https://archive.org/details/searchforperfect0000coch> (visited on 07/31/2026).
- [10] R. Cross. *Golf physics: oblique impact experiments*. See also Cross and Nathan, *Scattering of a ball by a bat in the limit of small deformation*, <https://arxiv.org/pdf/1610.03464>. University of Sydney. URL: <https://www.physics.usyd.edu.au/~cross/GOLF/GOLF.htm> (visited on 07/31/2026).
- [37] E. Henrikson, P. Wood, C. Broadie, and T. Nuttall. “The Role of Friction and Tangential Compliance on the Resultant Launch Angle of a Golf Ball”. In: *Proceedings* 49.1 (2020), p. 27. DOI: 10.3390/proceedings2020049027.
- [44] T. P. Jorgensen. *The Physics of Golf*. 2nd ed. Origin of the D-plane model. Springer / AIP Press, 1999.
- [47] R. J. Leach, S. E. Forrester, A. C. Mears, and J. R. Roberts. “How valid and accurate are measurements of golf impact parameters obtained using commercially available radar and stereoscopic optical launch monitors?” In: *Measurement* 112 (2017). Open access mirror: https://repository.lboro.ac.uk/articles/journal_contribution/9562799, pp. 125–136. DOI: 10.1016/j.measurement.2017.08.009. URL: <https://www.sciencedirect.com/science/article/abs/pii/S0263224117305079> (visited on 07/31/2026).

- [48] K. M. Lynch and F. C. Park. *Modern Robotics: Twists, Part 2 of 2*. Author video transcript accompanying the robotics textbook. URL: <https://modernrobotics.northwestern.edu/nu-gm-book-resource/3-3-2-twists-part-2-of-2/> (visited on 09/30/2026).
- [52] R. M. Murray, Z. Li, and S. S. Sastry. *A Mathematical Introduction to Robotic Manipulation*. Standard reference for twists, screws, and the geometry of $SE(3)$. The author archive reports that the publisher withdrew the online book PDF in January 2020; https://www.cds.caltech.edu/~murray/mlswiki/index.php?title=Main_Page. CRC Press, 1994.
- [61] A. R. Penner. “The physics of golf”. In: *Reports on Progress in Physics* 66.2 (2003), pp. 131–171. DOI: 10.1088/0034-4885/66/2/202. URL: <https://iopscience.iop.org/article/10.1088/0034-4885/66/2/202> (visited on 07/31/2026).
- [66] S. J. Quintavalla. “A generally applicable model for the aerodynamic behavior of golf balls”. In: *Science and Golf IV: Proceedings of the World Scientific Congress of Golf*. Chapter 30; linked publisher preview verifies the contents entry, not the full coefficient model. 2002, pp. 341–348. URL: https://api.pageplace.de/preview/DT0400.9781135646929_A23806389/preview-9781135646929_A23806389.pdf (visited on 10/04/2026).
- [73] S. Särkkä and L. Svensson. *Bayesian Filtering and Smoothing*. 2nd ed. Cambridge University Press, 2023. URL: https://users.aalto.fi/~ssarkka/pub/bfs_book_2023_online.pdf (visited on 10/04/2026).
- [75] A. J. Smits and D. R. Smith. “A new aerodynamic model of a golf ball in flight”. In: *Science and Golf II: Proceedings of the World Scientific Congress of Golf*. 1994. URL: <https://www.researchgate.net/publication/284037213> (visited on 07/31/2026).
- [109] A. Vena, D. Budney, T. Forest, and J. P. Carey. “Three-dimensional kinematic analysis of the golf swing using instantaneous screw axis theory, Part 1: methodology and verification”. In: *Sports Engineering* 13 (2011), pp. 105–123. DOI: 10.1007/s12283-010-0058-8. URL: <https://link.springer.com/article/10.1007/s12283-010-0058-8> (visited on 09/30/2026).
- [110] A. Vena, D. Budney, T. Forest, and J. P. Carey. “Three-dimensional kinematic analysis of the golf swing using instantaneous screw axis theory, Part 2: golf swing kinematic sequence”. In: *Sports Engineering* 13.3 (2011), pp. 125–133. DOI: 10.1007/s12283-010-0059-7. URL: <https://link.springer.com/article/10.1007/s12283-010-0059-7> (visited on 08/01/2026).
- [112] P. Wood, E. Henrikson, and C. Broadie. “The Influence of Face Angle and Club Path on the Resultant Launch Angle of a Golf Ball”. In: *Proceedings* 2.6 (2018), p. 249. DOI: 10.3390/proceedings2060249. URL: <https://www.mdpi.com/2504-3900/2/6/249> (visited on 10/01/2026).

Patents, Portfolios, and IP Records

- [6] Business Wire. *Foresight Sports and Uneekor Agree on License of Foresight Sports Patents Related to Portable Golf Launch Monitor Screen Technology*. 2024. URL: <https://www.businesswire.com/news/home/20240930838753/en/> (visited on 07/31/2026).
- [11] S. DeLeon, S. Nicora, and K. Wang. “Launch monitor”. US 11,311,789 B2. I. Full-Swing Golf. Granted from US 2020/0147470 A1; CW + FMCW dual-mode 24 GHz radar with non-uniform array; expires ~2039. Continuation US 11,844,990 B2. 2022. URL: <https://patents.google.com/patent/US11311789B2/en> (visited on 07/31/2026).

- [30] W. Gobush. “Method and apparatus to determine golf ball trajectory and flight”. US 6,500,073 B1. A. Company. Expired. 2002. URL: <https://patents.google.com/patent/US6500073B1/en> (visited on 07/31/2026).
- [31] W. Gobush, D. Pelletier, and C. Days. “Method and apparatus to determine object striking instrument movement conditions”. US 5,501,463 A. A. Company. Priority 1992; expired. 1996. URL: <https://patents.google.com/patent/US5501463A/en> (visited on 07/31/2026).
- [32] Golf Business News. *FlightScope wins patent infringement case against TrackMan in Germany*. 2022. URL: <https://golfbusinessnews.com/news/innovation-centre/flightscope-wins-patent-infringement-case-against-trackman-in-germany/> (visited on 07/31/2026).
- [40] H. Johnson. “Systems to track a moving sports object”. US 10,338,209 B2. E. U. L. (FlightScope). Fusion Tracking; priority 2015, expires ~2037. Continuation US 11,016,188 B2. 2019. URL: <https://patents.google.com/patent/US10338209B2/en> (visited on 07/31/2026).
- [41] H. Johnson. “Golf ball spin axis measurement”. US 10,775,492 B2. E. U. L. (FlightScope). Priority 2013; expires ~2035. 2020. URL: <https://patents.google.com/patent/US10775492B2/en> (visited on 07/31/2026).
- [42] H. Johnson, T. Johnson, and R. W. Rust. “Ball spin rate measurement”. US 9,868,044 B2. E. U. L. (FlightScope). Priority 2013; expires ~2034. Dielectric-lens phase demodulation. 2018. URL: <https://patents.google.com/patent/US9868044B2/en> (visited on 07/31/2026).
- [45] C. M. Kiraly and G. V. Wintriss. “Flight parameter measurement system”. US 7,292,711 B2. W. E. C. (W. N. / F. Sports). Priority 2002; **expired April 2025**. Sibling US 7,324,663 B2 expired August 2025. 2007. URL: <https://patents.google.com/patent/US7292711B2/en> (visited on 07/31/2026).
- [46] “Launch monitor system and a method for use thereof”. US 6,758,759 B2. A. Company. Priority 2001; expired 2022. Portable four-camera family: US 7,143,639 B2 (expired 2024), US 8,500,568 B2, US 8,556,267 B2; 3D imaging US 10,668,350 B2. 2004. URL: <https://patents.google.com/patent/US6758759B2/en> (visited on 07/31/2026).
- [51] “Method for determining coefficients of lift and drag of a golf ball”. US 6,186,002 B1. A. Company. Trajectory-fitting template for aerodynamic model calibration. 2001. URL: <https://patents.google.com/patent/US6186002B1/en> (visited on 07/31/2026).
- [76] “System and method for determining three-dimensional positional coordinates of a ball”. US 12,186,643 B2. T. A/S. Related sparse-array angle work: US 9,958,527 B2. 2025. URL: <https://patents.google.com/patent/US12186643B2/en> (visited on 07/31/2026).
- [82] “Timing determination in camera-based motion measurement”. US 12,401,909. Documents Gabor-filter dimple enhancement for spin tracking. 2025. URL: <https://patents.google.com/patent/US12401909B2/en> (visited on 07/31/2026).
- [90] TrackMan A/S. *Patents*. Virtual patent marking notice; not an exhaustive portfolio search. URL: <https://www.trackman.com/legal/patents> (visited on 10/01/2026).
- [98] F. Tuxen. “Determination of spin parameters of a sports ball”. US 8,845,442 B2. T. A/S. Priority 2005; PTA expiry ~2029. Continuation US 10,393,870 B2 (expires Dec. 2026); European sibling EP 1 698 380. 2014. URL: <https://patents.google.com/patent/US8845442B2/en> (visited on 07/31/2026).
- [99] F. Tuxen. “Method and a sensor for determining a direction-of-arrival of impingent radiation”. US 9,958,527 B2. TrackMan A/S. 2018. URL: <https://patents.google.com/patent/US9958527B2/en> (visited on 10/02/2026).

- [100] F. Tuxen. “System and method for determining a spin axis of a sports ball”. US 10,850,179 B2. T. A/S. Continuations US 11,446,546 B2 and US 11,938,375 B2. 2020. URL: <https://patents.google.com/patent/US10850179B2/en> (visited on 07/31/2026).
- [104] Uneekor, Inc. *Patents*. Product-specific virtual patent marking notice. URL: <https://uneekor.com/en-us/legal/patents> (visited on 10/01/2026).
- [111] “Virtual golf simulation apparatus and method”. US 9,242,158 B2. L. Golfzon Co. Two-stage latency-hiding simulation; to ~2032. 2016. URL: <https://patents.google.com/patent/US9242158B2/en> (visited on 07/31/2026).

Manufacturer Technical Documentation

- [5] Bushnell Golf. *Launch Pro: Features and Data Points*. URL: <https://www.bushnellgolf.com/products/launch-pro> (visited on 10/04/2026).
- [14] FlightScope. *Face Impact Location Overview*. URL: <https://flightscope.com/blogs/blog/s/face-impact-location-overview> (visited on 10/04/2026).
- [15] FlightScope. *FlightScope X3C: Indoor Setup Requirements and Fusion Tracking*. URL: <https://flightscope.com/products/flightscope-x3c> (visited on 10/02/2026).
- [16] FlightScope. *Frequently Asked Questions: Mevo+ Camera and Software Packages*. URL: <https://flightscope.com/pages/faq> (visited on 10/04/2026).
- [17] Foresight Sports. *Club Head Data: Measurements and Definitions*. Updated date; lie and closure-rate conventions. Dec. 8, 2025. URL: <https://help.foresightsports.com/hc/en-us/articles/47214673873811-Club-Head-Data-Measurements-Definitions> (visited on 10/04/2026).
- [18] Foresight Sports. *Ball and club data (spherical correlation spin measurement)*. URL: <https://foresightsports.eu/ball-club-data/> (visited on 07/31/2026).
- [19] Foresight Sports. *GC3: Launch Monitor and Data Features*. URL: <https://www.foresightsports.com/collections/launch-monitors/products/gc3> (visited on 10/04/2026).
- [20] Foresight Sports. *GCQuad launch monitor*. URL: <https://www.foresightsports.com/products/gcquad-launch-monitor> (visited on 07/31/2026).
- [21] Foresight Sports. *GCQuad: User Guide*. URL: https://support.foresightsports.com/sites/default/files/files/GCQuad_User_Manual.pdf (visited on 10/04/2026).
- [22] Foresight Sports. *HMT head measurement technology*. Fiducial dot-count to data-tier mapping. URL: <https://www.foresightsports.com/hmt-head-measurement> (visited on 07/31/2026).
- [23] Foresight Sports. *How to apply and maintain club markers*. URL: <https://help.foresightsports.com/hc/en-us/articles/4408197030035> (visited on 07/31/2026).
- [24] Full Swing Golf. *KIT launch monitor technology*. URL: <https://www.fullswinggolf.com/kit-launch-monitor-technology/> (visited on 07/31/2026).
- [25] Garmin Ltd. *Accuracy of Approach R10 Radar*. URL: <https://support.garmin.com/id-ID/?faq=kj37Cgzvwm98hC9WPrIQm5> (visited on 10/02/2026).
- [26] Garmin Ltd. *Approach R50: Applying Club Stickers*. URL: <https://www8.garmin.com/manuals-apac/webhelp/approachr50/EN-SG/GUID-031570F0-3067-4F21-9030-307947280EAD-8177.html> (visited on 10/04/2026).

- [27] Garmin Ltd. *Approach R50: Factors Affecting Shot Detection and Accuracy*. URL: <https://support.garmin.com/en-US/?faq=4RsZFm0Ahy00Itfb8rJ7H7> (visited on 10/04/2026).
- [28] Garmin Ltd. *Approach R50: Owner's Manual*. URL: https://www8.garmin.com/manuals/webhelp/GUID-D31AA1A2-C026-46EB-9699-FA94F1E8AF0B/EN-US/Approach_R50_OM_EN-US.pdf (visited on 10/04/2026).
- [29] Garmin Ltd. *Using a Titleist RCT Ball With the Approach R10*. URL: <https://support.garmin.com/fr-CA/?faq=COFCXdRAJv2m8r9MBtLrf8> (visited on 10/02/2026).
- [65] ProTee United. *The Power of VX: Features and Specifications*. URL: <https://proteegroup.com/the-power-of-vx/> (visited on 10/04/2026).
- [68] Rapsodo. *Frequently Asked Golf Questions: MLM2PRO Device and Data*. URL: <https://rapsodo.com/pages/frequently-asked-golf-questions-faq> (visited on 10/04/2026).
- [69] Rapsodo. *Titleist 2025 PRO V1x Golf Balls with RPT*. URL: <https://rapsodo.com/products/titleist-2025-pro-v1x-golf-balls-w-rpt> (visited on 10/02/2026).
- [74] SkyTrak. *What does the SkyTrak ST+ launch monitor measure?* URL: <https://www.skytrakgolf.com/pages/what-does-the-skytrak-st-launch-monitor-measure> (visited on 07/31/2026).
- [79] The Golf Wire. *FlightScope X3: multi-frequency radar with direct distance measurement and Fusion Tracking*. URL: <https://thegolfwire.com/326713-2/> (visited on 07/31/2026).
- [83] TrackMan A/S. *TrackMan News #5: Ball Flight Laws*. The D-plane and the 85/15 face/path weighting. 2009. URL: <https://www.yumpu.com/en/document/view/11652756/trackman-ball-flight-laws> (visited on 07/31/2026).
- [84] TrackMan A/S. *What Is Low Point?* May 30, 2024. URL: <https://www.trackman.com/blog/low-point> (visited on 10/04/2026).
- [85] TrackMan A/S. *Parameters: Carry (Tee to Green)*. Updated date; carry-flat convention. Aug. 7, 2025. URL: <https://support.trackmangolf.com/hc/en-us/articles/39726543090971-Parameters-Carry-Tee-to-Green> (visited on 10/04/2026).
- [86] TrackMan A/S. *Practice: Trackman Data Parameter Definitions*. Updated date; full-swing definitions consulted. Nov. 27, 2025. URL: <https://support.trackmangolf.com/hc/en-us/articles/5089892383515-Practice-Trackman-Data-Parameter-Definitions> (visited on 10/04/2026).
- [87] TrackMan A/S. *Club data definitions*. URL: <https://www.trackman.com/blog/golf/club-data-definitions> (visited on 07/31/2026).
- [88] TrackMan A/S. *Launch monitor technical specifications (TrackMan 4, TrackMan iO)*. URL: <https://www.trackman.com/golf/launch-monitors/tech-specs> (visited on 07/31/2026).
- [89] TrackMan A/S. *Patent: measuring the spin of a sports ball*. URL: <https://blog.trackmangolf.com/patent-measuring-the-spin-of-a-sports-ball/> (visited on 07/31/2026).
- [91] TrackMan A/S. *Spin loft and spin rate*. URL: <https://www.trackman.com/blog/golf/spin-loft> (visited on 07/31/2026).
- [92] TrackMan A/S. *TrackMan parameters explained*. URL: <https://www.trackman.com/blog/golf/40-trackman-parameters> (visited on 07/31/2026).
- [93] TrackMan A/S. *Two radars, one camera, zero doubt*. Optically Enhanced Radar Tracking (OERT). URL: <https://www.trackman.com/blog/golf/two-radars-one-camera-zero-doubt> (visited on 07/31/2026).

- [94] TrackMan A/S. *What is club speed?* Geometric-center reference-point definition. URL: <https://www.trackman.com/blog/golf/what-is-club-speed> (visited on 07/31/2026).
- [101] F. Tuxen. *How Trackman Combines Radar and Cameras for Superior Indoor Accuracy*. Vendor account; no independent accuracy validation implied. Sept. 10, 2026. URL: <https://www.trackman.com/blog/radar-and-cameras-for-superior-indoor-golf-accuracy> (visited on 10/04/2026).
- [102] Uneekor, Inc. *EYE MINI: Product Specifications*. URL: <https://uneekor.com/en-us/products/eye-mini-standalone> (visited on 10/04/2026).
- [103] Uneekor, Inc. *EYE XO2: User Manual*. URL: https://download.uneekor.com/docs/EYEXO2_Manual.pdf (visited on 10/04/2026).
- [105] Uneekor, Inc. *Photometric vs. Doppler: which launch monitor technology delivers the most accurate golf data*. Describes Dimple Optix markerless spin. URL: <https://uneekor.com/blogs/blog/photometric-vs.-doppler:-which-launch-monitor-technology-delivers-the-most-accurate-golf-data> (visited on 07/31/2026).
- [106] Uneekor, Inc. *What Are the Technical Specifications for Uneekor EYE MINI?* URL: <https://uneekor.com/en-us/help-center/general-product-information/what-are-the-technical-specifications-for-uneekor-eye-mini> (visited on 10/04/2026).

Datasheets, Protocols, and Standards

- [12] J. Driscoll. *Revised Golf Ball Testing Conditions to Take Effect in 2028*. Historical announcement; implementation timetable superseded by the June 2026 joint statement. Dec. 6, 2023. URL: <https://www.usga.org/content/usga/home-page/articles/2023/12/revised-golf-ball-testing-conditions-to-take-effect-in-2028.html> (visited on 10/04/2026).
- [13] Federal Communications Commission. *Equipment authorization filing SFX-TMAN₄ (TrackMan 4)*. TrackMan iO: SFX-TMB0201. URL: <https://fccid.io/SFX-TMAN4> (visited on 07/31/2026).
- [33] Golf Simulator Forum. *FlightScope Mevo+ teardown and FCC filings (QXP-LJ361, QXP-A7310)*. URL: <https://golfsimulatorforum.com/forum/flightscope/277362-mevo-teardown> (visited on 07/31/2026).
- [36] GSPro. *GSPro Open Connect API v1 specification*. Reference client: <https://github.com/travislang/gspro-garmin-connect-v2>. URL: <https://gsprogolf.com/GSProConnectV1.html> (visited on 07/31/2026).
- [43] Joint Committee for Guides in Metrology. *International Vocabulary of Metrology—Basic and General Concepts and Associated Terms (VIM)*. 3rd ed. JCGM 200:2012. Annotated online edition; definitions 2.15, 2.20, 2.21, and 2.39 consulted. 2012. URL: <https://jcgmbipm.org/vim/en/> (visited on 10/03/2026).
- [49] MathWorks. *Constant false alarm rate (CFAR) detection*. CA/GO/SO/OS variants and false-alarm mathematics. URL: <https://www.mathworks.com/help/phased/ug/constant-false-alarm-rate-cfar-detection.html> (visited on 07/31/2026).
- [50] MathWorks. *Constant False-Alarm Rate (CFAR) Detectors*. URL: <https://www.mathworks.com/help/phased/ug/constant-false-alarm-rate-cfar-detectors.html> (visited on 10/04/2026).

- [57] OmniPreSense Corp. *AN-029: OPS24x for Sports Applications*. Documents the golf configuration and cites the OpenFlight project as its reference implementation. 2026. URL: https://omnipresense.com/wp-content/uploads/2026/06/AN-029-A_OPS243-for-Sports_260621.pdf (visited on 07/31/2026).
- [58] OmniPreSense Corp. *AN-010: API Interface Specification*. URL: https://omnipresense.com/wp-content/uploads/2025/10/AN-010-AD_API_Interface.pdf (visited on 07/31/2026).
- [59] OmniPreSense Corp. *AN-027: Rolling Buffer Mode*. URL: https://omnipresense.com/wp-content/uploads/2025/06/AN-027-A_Rolling-Buffer-1.pdf (visited on 07/31/2026).
- [60] OmniPreSense Corp. *OPS243 Doppler Radar Sensor Product Brief*. OPS-PB-004-F. URL: https://omnipresense.com/wp-content/uploads/2024/01/OPS243-Product-Brief_004-F.pdf (visited on 07/31/2026).
- [67] S. Rao. *MIMO Radar*. SWRA554A. 2018. URL: <https://www.ti.com/lit/an/swra554a/swra554a.pdf> (visited on 09/30/2026).
- [70] Raspberry Pi Ltd. *Raspberry Pi Global Shutter Camera documentation (IMX296, XTR external trigger)*. URL: <https://www.raspberrypi.com/documentation/accessories/camera.html> (visited on 07/31/2026).
- [71] Raspberry Pi Ltd. *Raspberry Pi Global Shutter Camera: Specifications*. URL: <https://www.raspberrypi.com/products/raspberry-pi-global-shutter-camera/> (visited on 10/04/2026).
- [72] RFbeam Microwave GmbH. *K-LD7 Radar Transceiver Datasheet*. Rev. B. 2021. URL: <https://rfbeam.ch/product/k-ld7-radar-transceiver/> (visited on 09/30/2026).
- [77] Texas Instruments. *Programming Chirp Parameters in TI Radar Devices*. SWRA553A. 2020. URL: <https://www.ti.com/lit/an/swra553a/swra553a.pdf> (visited on 09/30/2026).
- [78] Texas Instruments. *IWR6843, IWR6443 Single-Chip 60 to 64 GHz mmWave Sensor*. SWRS219F. EVM documentation: <https://www.ti.com/tool/IWR6843ISK>; sensor placement app note SWRA758. 2025. URL: <https://www.ti.com/lit/ds/symlink/iwr6843.pdf> (visited on 09/30/2026).
- [80] The R&A and USGA. *Equipment Rules: Part 2, Conformance of Clubs*. URL: <https://www.randa.org/roe/the-rules-of-equipment/part-2-conformance-of-clubs> (visited on 10/04/2026).
- [81] The R&A and USGA. *Equipment Rules: Part 4, Conformance of Balls*. URL: <https://www.randa.org/roe/the-rules-of-equipment/part-4-conformance-of-balls> (visited on 10/04/2026).
- [107] USGA. *Driver Check Testing Program: Notice to Manufacturers*. Mar. 13, 2008. URL: <https://www.usga.org/content/dam/usga/pdf/Equipment/Archived-Notices/3-13-08%20Driver%20Check%20Test%20Program.pdf> (visited on 10/04/2026).
- [108] USGA, The R&A, PGA TOUR and DP World Tour. *Joint Statement on the Overall Distance Standard*. No ODS testing change before January 2030 while alternatives are evaluated. June 17, 2026. URL: <https://mediacenter.usga.org/2026-Joint-Statement-USGA-RandA-PGA-Tour-DP-World-Tour> (visited on 10/04/2026).

Independent Testing, Engineering References, and Open Source

- [7] Carl's Place. *Calibrating Uneekor launch monitors*. URL: <https://www.carlofet.com/blog/calibrating-uneekor-launch-monitors> (visited on 07/31/2026).

- [34] Golf Simulator Zone. *GCQuad vs TrackMan accuracy: robot testing summary*. URL: <https://golfsimulatorzone.com/gcquad-vs-trackman-accuracy-in-2026/> (visited on 07/31/2026).
- [35] GolfBays. *Radar vs photometric launch monitors indoors*. URL: <https://golfbays.de/en/blogs/news/radar-vs-photometric-launch-monitors-indoors> (visited on 07/31/2026).
- [38] Home Performance Lab. *TrackMan simulator data: measured vs. inferred indoors*. URL: <https://homeperformancelab.com/trackman-simulator-data/> (visited on 07/31/2026).
- [39] *How does TrackMan/FlightScope measure the club face angle?* Contains TrackMan statements on derived face data. Brian Manzella Golf Forum. URL: <https://forum.brianmanzellagolf.com/threads/how-does-trackman-flightscope-measure-the-club-face-angle.16591/> (visited on 07/31/2026).
- [53] MyGolfSimulator. *Garmin R10 accuracy: measured vs. calculated parameters*. URL: <https://mygolfsimulator.com/garmin-r10-accuracy/> (visited on 07/31/2026).
- [54] MyGolfSimulator. *Garmin R10 data and radar architecture*. URL: <https://mygolfsimulator.com/garmin-r10-data/> (visited on 07/31/2026).
- [55] MyGolfSpy. *Rapsodo MLM2PRO takes spin measurement to pro level*. URL: <https://mygolfspy.com/news-opinion/rapsodo-mlm2pro-takes-spin-measurement-to-pro-level/> (visited on 07/31/2026).
- [56] MyGolfSpy. *Three of the most accurate indoor launch monitors — and one to avoid*. URL: <https://mygolfspy.com/news-opinion/three-of-the-most-accurate-indoor-launch-monitors-and-one-to-avoid/> (visited on 07/31/2026).
- [62] Perfect Golf Swing Review. *Ball flight laws*. URL: <https://www.perfectgolfswingreview.net/ballflight.htm> (visited on 07/31/2026).
- [63] PiTrac contributors. *PiTrac — the DIY golf launch monitor*. Software and model licensing should be checked separately; documentation <https://docs.pitrac.org/>; design logs at Hackaday.io project 195042. URL: <https://github.com/PiTracLM/PiTrac> (visited on 07/31/2026).
- [64] PlayBetter. *Bushnell Launch Pro vs GC3 vs GCQuad comparison*. URL: <https://www.playbetter.com/blogs/golf/bushnell-launch-pro-vs-gc3-vs-gc-quad-comparison> (visited on 07/31/2026).
- [95] D. Tutelman. *Gear effect*. Companion: spin decay, <https://www.tutelman.com/golf/ballflight/spinDecay.php>. URL: <https://www.tutelman.com/golf/ballflight/gearEffect1.php> (visited on 07/31/2026).
- [96] D. Tutelman. *Smash factor*. URL: <https://www.tutelman.com/golf/ballflight/smashfactor.php> (visited on 07/31/2026).
- [97] D. Tutelman. *Three-dimensional launch conditions from impact conditions*. URL: <https://www.tutelman.com/golf/ballflight/3dlaunch.php> (visited on 07/31/2026).